

4 – Prototype

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4.1 Justification of Prototype

#	Area of System	Feature	Included / Omitted	Justification
1	Login	Username and password validation	Omitted	This would complicate the testing process as the username and password would have to be entered every start-up of the system. Additionally, validation of usernames and password is not essential.
2		Login facilities	Included	The ability to log the attendance of, and store information about, multiple users is one of the main requirements of the system. In addition, when the system is used in practice it's important that multiple users are able to use it. Therefore, a login system is essential because it allows multiple different user accounts to log in. This means that, the features of the system that rely on multiple user accounts can be fully prototyped and tested.
3		Automatic attendance tracking	Included	This feature will require little time to prototype and is a useful feature of the final system. I will include this feature because it's important for the attendance of the user to be automatically tracked to save time during the lessons.
4		Automatic determination of the reason for a user missing a session	Omitted	This feature is not essential to the system. It would also complicate the testing process. This is because every time a user logged in, the dialog asking whether the last session was running or not would open. For these reasons I will not be including this feature.
5		Levels of access	Omitted	Although, in the final system, there will be different levels of access, adding different levels of access to the prototype would just complicate the testing process as care would have to be taken to ensure a user that had access to the admin functions was logged in if the admin functions needed testing. It would also take additional time to program the functionality to restrict access to different areas of the system based on which user was logged in.
6	View Questions	View	Included	This is an essential feature of the system as the user needs to be able to view the questions that are currently in the system in order to choose one to attempt. Therefore, I will be including it.
7		Sort, filter	Included	It's important that the users are able to find questions in the system that they want to attempt. This feature needs to be well implemented in the final system so that it is as easy as possible for the users to find the questions in the system. By including it in the prototype, the sorts and searches will be able to be fully tested and evaluated and so improvements that can be made for the final system would be able to be found.
8		Delete	Included	This feature is important to make the testing process easier. If the question creation system goes wrong and creates a malformed question it would be much easier to remove it using a built in feature rather than cleaning the databases of

				it by hand. Therefore, including this feature will make the prototyping process easier. It's also a useful feature of the final system.
9		Attempt	Included	Attempting questions is a major feature of the final system therefore it is important that this feature is fully prototyped and tested.
10	View Forms	View	Included	This is an essential feature of the system as the user needs to be able to view the forms that are currently in the system in order to choose one to attempt. Therefore, I will be including it.
11		Sort, filter	Included	It's important that the users are able to find forms in the system that they want to attempt. This feature needs to be well implemented in the final system so that it is as easy as possible for the users to find the forms in the system. By including it in the prototype, the sorts and searches will be able to be fully tested and evaluated and so improvements that can be made for the final system would be able to be found.
12		Delete	Included	This feature is important to make the testing process easier. If the form creation system goes wrong and creates a malformed form it would be much easier to remove it using a built in feature rather than cleaning the databases of it by hand. Therefore, including this feature will make the prototyping process easier. It's also a useful feature of the final system.
13		Create form from user weaknesses	Included	This is a difficult feature to implement because it will be difficult to find which data about a user's attempt of each question should be used to determine which questions that they struggle with the most (It's the questions that they struggle with the most that would be added to the form). By prototyping this feature, I will be able to gather feedback (from the end user and third parties) to determine the how effective the system is at identifying these questions. This feedback would be used to refine the system's ability to determine the questions that the user struggles with.
14		Attempt	Included	Attempting forms is a major feature of the system therefore I will include it. See the attempt form breakdown further down.
15	Create questions	Add components	Included	I will include the feature to add all types of components that are detailed in the design (with the exception of calendar entry which is detailed below). It's important to prototype adding all of the types of components as it's likely that there will be some issues in adding certain components to questions. Prototyping this feature will allow me to work out these issues before the final system and means that when I program the final system I will be able to focus on making the feature as good as possible rather than just getting it to work.
16		Calendar entry	Omitted	I will omit this feature because I anticipate that it will take a large amount of time to implement. This is because creating a calendar would be very complex

				and would not be of much improvement to the system. The calendar entry is one of the least important types of components that can be added to questions. Despite this, however, I will implement a simple calendar entry that uses dropdowns to select the dates as a proof of concept.
17		Drag and drop	Omitted	Implementing a drag and drop system to add components to questions could potentially be a more user friendly way of adding components to questions compared just pressing buttons. However, I will not include it because it would take a large amount of time to implement as it is a complex feature. It would also slow down the process of creating questions as rather than just pressing buttons the components would have to be accurately dragged onto the question. This would complicate the prototyping and testing process and so I will not be including it.
18		Add basic information	Included	It's important that basic information is stored about each question. This is essential to searching and sorting the questions, and also in finding particular questions in the system. It's important that the user is able to easily find the questions that they want in the system so that they can attempt them it's also important that the user can easily find questions when creating forms. This makes the basic information stored about each question essential for good operation of the system therefore I will be including it.
19	Create forms	Add questions	Included	Adding questions to the form is how forms are created therefore this feature is essential as without it forms would not be able to be created. Allowing the admin to create new forms a major feature of the system therefore it's important that it is fully tested and prototyped so that it is as refined as possible in the final system.
20		Add basic information	Included	It's important that the admin is able to enter a description of the form and its difficulty into the system after creating a form. This is so that the user can easily find the forms that they want to attempt in the system because this data is needed in order to perform the sorts and searches. This feature will not take a large amount of time to develop therefore I will be including it.
21	Edit forms	Rearrange questions	Included	If an error is made creating a form in the system being able to edit forms allows the error to be corrected easily without having to recreate the entire form. This will make testing form creation easier, and therefore I will be including this feature.
22		Remove questions	Included	Removing questions from a form is an important feature of editing the forms of the system and means that errors can be corrected if a question is mistakenly added to the form. Implementing this feature will save time during the testing

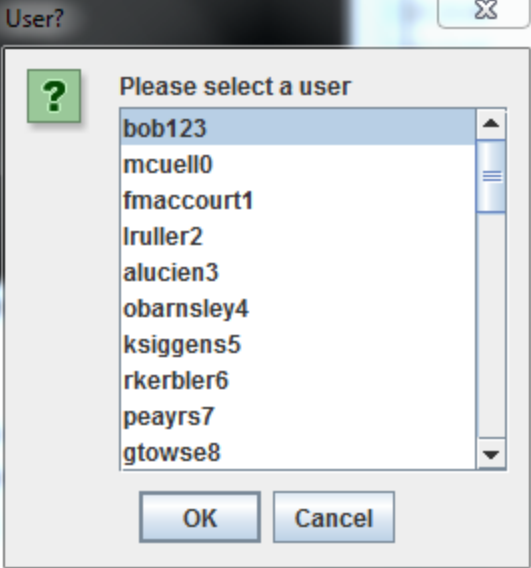
				process as it means that if a question is mistakenly added to a form it can be easily removed without having to recreate the entire form.
23	Users	Add	Included	It's important that the admin is able to add new users to the system. Prototyping this feature means that it will be able to be fully refined, before implementing into the final system, as feedback can be gathered.
24		Edit	Included	Editing the stored details of a user is an important feature of the system that needs to be fully tested even though it would be an infrequent action.
25		View	Included	There is a large amount of data that is stored about the users. Including the facility to view the users in the prototype means that feedback can be gathered and the best way to display the information can be determined.
26		Search	Included	Because there will be a large amount of users in the system, it's important that the users are able to be searched. This feature is relatively simple to implement because it's only a linear search therefore it will not take a large amount of development time.
27		Delete	Included	This feature is important to make the testing process easier. If the user creation system goes wrong and creates a malformed form it would be much easier to remove it using a built in feature rather than cleaning the databases of it by hand. Therefore, including this feature will make the prototyping process easier. It's also a useful feature of the final system.
28		Take register	Included	It's important that the take register feature is included so that it can be fully refined and streamlined. This is important because Steph will take a register at the start of every session. The faster that she can complete the register, the more time that is able to be spent teaching the users.
29		Encryption	Omitted	Although the final system will store sensitive information about the users, encryption is not important for the prototype. This is because the prototype will only use fake 'dummy' data and so there is no risk of sensitive information being leaked.
30	Statistics	Produce report	Included	Progress tracking is a major feature of the system and a large part of this feature is being able to produce reports on each user's progress therefore it is important that it is included in the prototype. Producing printable reports is also a unique area of the system as no other areas of the system print anything so it's important that it's prototyped to ensure that any issues are worked out before it is implemented in the final system.
31		Produce multiple reports	Omitted	This feature is only a small addition to the produce report feature and doesn't play a major role in the system. Therefore I will not be including it.

32		View stats for individual question	Included	Implementing this feature means that the automatic progress tracking features can be fully tested because the data that they produce can be easily viewed. For this reason it's important that this feature is included.
33		Graphing	Omitted	Although the final system will make use of graphs to display the progress, I will not be including this feature in the final system. This is because they would be complex to program and do not affect other areas of the system.
34	Import	Import questions	Included	This is an important feature of the system because the admin needs to be able transfer questions and forms to different instances of the system otherwise the club would not be able to run because it's essential that the users have the same forms to practice. In addition, it would take a large amount of time to recreate each form and question by hand on different instances of the system. Therefore it's incredibly important that all these features are included in the prototype
35		Import forms	Included	See above
36	Export	Export questions	Included	See above
37		Export forms	Included	See above
38	Component validation	Email address	Included	Email address validation will be a fairly simple feature to implement and it's important that the validation routine works well. Therefore, I will be including it so that it can be refined before it is added to the final system.
39		Date	Included	Date validation will be a fairly simple feature to implement and it's important that the validation routine works well. Therefore, I will be including it so that it can be refined before it is added to the final system.
40		Phone number	Included	Phone number validation will be a fairly simple feature to implement and it's important that the validation routine works well. Therefore, I will be including it so that it can be refined before it is added to the final system.
41		Address lookup via google maps	Omitted	Although this will be an important feature of the final system, because it will make address entry as accurate as possible, I will not be including it. This is because it would be very complex to program and its omission doesn't affect other areas of the system. It would also complicate the testing process as there are a large number of ways that it could fail (e.g. no internet connection, api request limit exceeded etc.) and it also limits development only to places that have a working internet connection.
42		Presence	Included	It's important that all of the fields of a question have presence checks. This is because presence checks are a major part of online forms. Presence checks will also be quick to implement and not take much development time.
43		Double entry verification	Included	Without double entry verification, the password entry questions would not be very accurate. This feature will be quick to implement and require little development time therefore I will include it.

44		Document upload fields	Included	Steph mentioned in her interview that this would be an important feature of the system and that currently she doesn't have much of a way of letting the users practice uploading documents to forms. Therefore, this feature is an important feature of the system. Including it in the prototype means that it can be refined before the final system and so it can be ensured that it will work well in the final system.
45	Attempt form	Save progress	Included	This will be a difficult feature to implement but it's important that it works for the final system. This is because users often don't have time to complete an entire form in one session therefore it's essential that they are able to save their progress between sessions.
46		Validate answers	Included	This is the most important feature of the system because the system's purpose is to teach the users how to fill forms in correctly so that their answers pass the validation checks. Without this feature the system would not be able to do this therefore the feature must be included.
47		Track statistics	Included	This is a major feature of the system. It will also be complex to implement as there are many different statistics that can be gathered. Implementing it in the prototype will allow me to be able to refine the feature and gather feedback to determine which statistics are most important and should be tracked and so will make the feature in the final system better.
48	Interface	Context sensitive help for every screen	Omitted	While this is an important feature in the final system, no actual users will be using to the prototype - only competent third parties. This means that, a help feature for every screen is unnecessary. A small help dialog will implemented in a few key screens, however, so that the location of the help button and the help available can be refined.
49		Main menu screen	Omitted	This screen is not essential to the system as it mainly contains buttons to navigate to the other areas of the system. These buttons are not necessary as the tabs at the top of the program can just be clicked instead. Therefore, I will not be including this in the prototype.

4.2 Screens and Outputs

4.21 Screens (no Data)

Login screen	Features
	● Allows the selected user to login.

View Questions Screen
Features

Form Filler
[Close] [Help]

View Questions
View Forms
Create questions
Create forms
Users
Statistics
Import and Export

Difficulty	Type
1	Name
3	Gender
4	Country of Birth
3	Birthday
6	Email
4	Password
8	File Upload

Difficulty Sort
1 2 3 4 5 6 7 8 9 10

Type Sort

Attempt Question

Delete Question

Filter by difficulty

☐ Email
 ☐ Name
 ☐ Phone
 ☐ Gender

☐ File Uplo...
 ☐ Passwo...
 ☐ Calendar
 ☐ Checkb...

☐ Drop-do...
 ☐ Radio B...
 ☐ Birthday
 ☐ Address

☐ James
 ☐ date of ...
 ☐ Country ...

Help screen

Message [Close]

i This is the view questions screen. From here you can select questions and attempt them by selecting them in the table and pressing the attempt button. You can filter and sort questions by using the buttons on the right.

OK

- Displays the questions to the user and displays their:
 - Difficulty
 - Type
- Allows the following sorts:
 - Difficulty sort
 - Type sort
- Allows the use of a slider to select the difficulty in order to filter the questions by difficulty
- Allows the user to select a type of question using radio buttons and then to filter by this type.
- Allows the admin to delete a question from the system by selecting it in the table and pressing the delete button.
- Allows the user to attempt a question by selecting the question in the table and then pressing the attempt question button.
- The user can also get help on this screen by pressing the help button at the top of the screen.

11

View Forms Screen
Features

Form Filler
_ □ ×

View Questions View Forms Create questions Create forms Users Statistics Import and Export

Help

Questions	Description	Main Skill...	Difficulty	Percent C...
Q815587...	large am...	Name Ge...	6	100%
Q815587...	Name an...	Name Fil...	5	100%
Q562800...	Gender a...	Gender C...	3	100%
Q815587...	Name pa...	Name Pa...	2	100%

Difficulty Sort

Attempt Form

Delete Form

Attempt form based on weaknesses

Filter by difficulty

☐ Email ☐ Name ☐ Phone ☐ Gender

☐ File Uplo... ☐ Passwo... ☐ Calendar ☐ Checkb...

☐ Drop-do... ☐ Radio B... ☐ Birthday ☐ Address

☐ James ☐ date of ... ☐ Country ...

- Displays the forms to the user and displays:
 - The list of questions in the form
 - The types of questions in the form
 - The description of the form
 - The percentage of that form that the user has completed
 - The difficulty of the form
- Allows the user to sort the forms by difficulty
- Allows the use of a slider to select the difficulty in order to filter the forms by difficulty.
- Allows the user to select multiple types of question using checkboxes and then to filter the table to only show forms that contain questions of all these types.
- Allows the admin to delete a form from the system by selecting it in the table and pressing the delete button.
- Allows the user to attempt a form by selecting the form in the table and then pressing the attempt form button.
- Allows the user to have a form automatically created and opened for them to attempt which is created from the questions that they struggle with the most.
- The user can also get help on this screen by pressing the help button at the top of the screen.

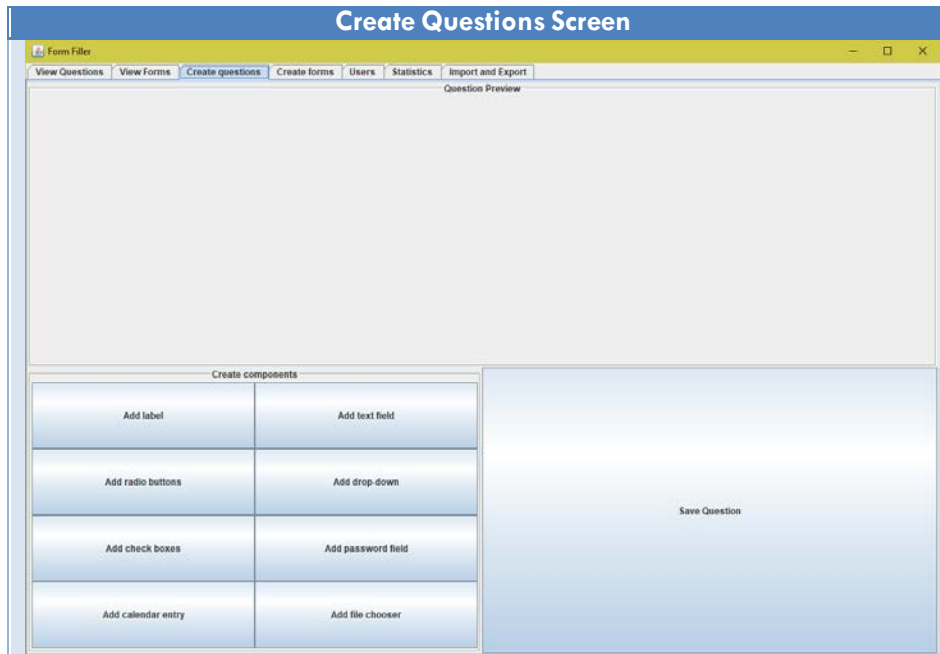
Help screen

Message _ ×

i This is the form display panel from here you can attempt a form by selecting one from the table and pressing attempt. By pressing attempt form based on weaknesses you can attempt a form based on your weak areas. You can filter and sort forms using the buttons on the right.

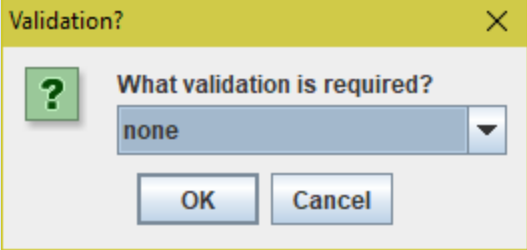
OK

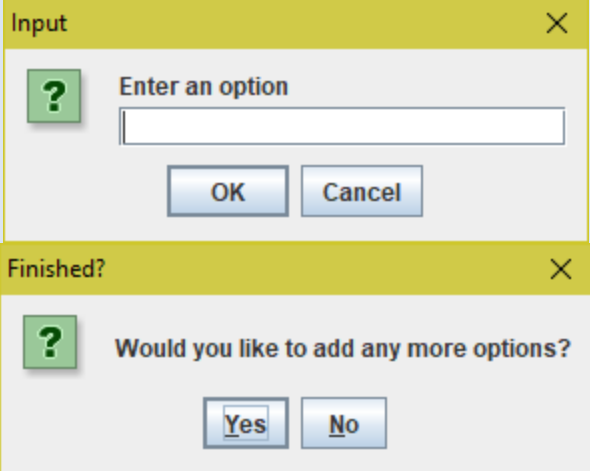
Create Questions Screen

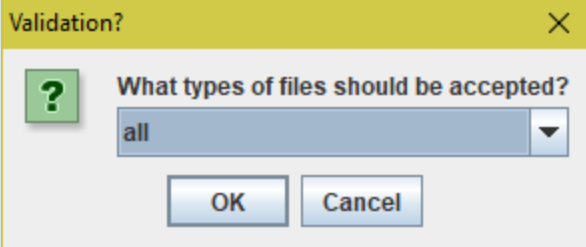


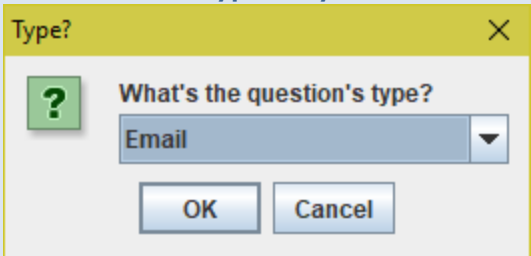
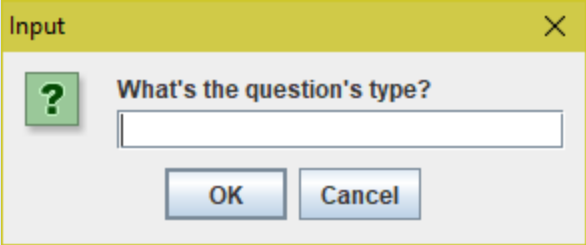
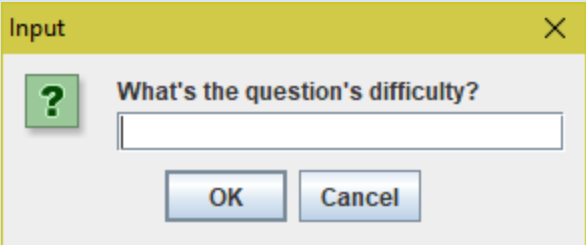
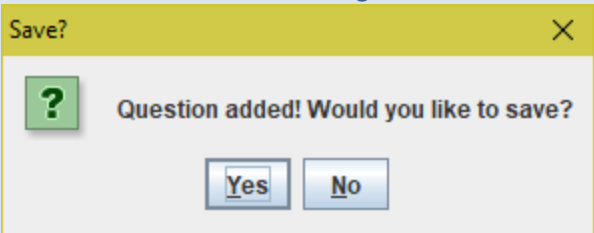
Features

- This screen allows the admin to create new questions by adding components to a blank panel
- There is a preview at the top of the screen which shows the admin how the question will look. This means that they are able to easily check how the question looks and are able to spot errors that they have made in creating the question (e.g. adding components in the wrong order).
- There are buttons for each type of component that the admin can add to a question.
- There is a save question button that allows the admin to save the question and add it to the system.

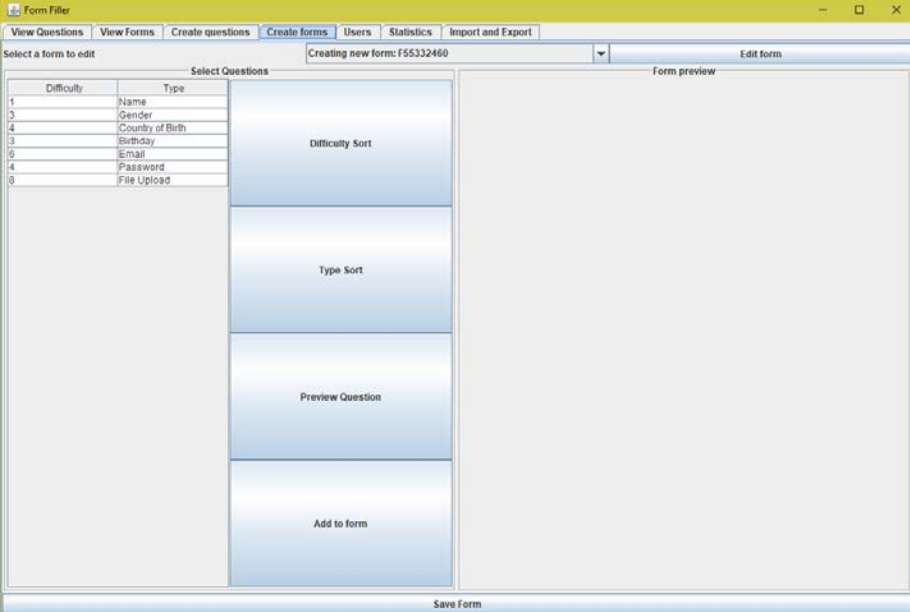
Add text field dialog	Features
 The dialog box titled "Validation?" has a yellow header bar with a close button (X). It contains a green question mark icon, the text "What validation is required?", a dropdown menu with "none" selected, and "OK" and "Cancel" buttons at the bottom.	● This dialog allows the admin to specify which type of validation rule should be applied to the text field that they are about to add to the question. They can select “none”, “phone number”, or “email address”

Add options dialog	Features
 The "Add options dialog" consists of two stacked dialog boxes. The top one, titled "Input", has a yellow header bar, a green question mark icon, the text "Enter an option", a text input field, and "OK" and "Cancel" buttons. The bottom one, titled "Finished?", also has a yellow header bar, a green question mark icon, the text "Would you like to add any more options?", and "Yes" and "No" buttons.	● These dialogs allow the user to specify the options that they want the user to be able to select using the radio buttons, checkboxes, or drop-down that they’ve just added to the question. ● The first dialog allows the admin to enter an option. ● The second dialog allows the admin to specify whether they have finished entering options or not.

Add file chooser dialog		Features
		● This allows the user to specify what types of files the file chooser that they are about to add to the system should accept.

Enter final question details dialog	Features
Type entry   Difficulty entry  Save dialog 	● These dialogs allow the admin to enter the required details about the question that they have just created. Type Entry: ● The first dialog allows them to select the type from a drop down of all of the types currently in the system. ● The last option of the dialog is “new type” and if they select this type the second dialog will open. ● The second dialog allows the admin to enter a brand new type into the system. Difficulty Entry: ● This allows the admin to enter the difficulty of the question into the system. Save dialog: ● This allows the admin to specify whether they want to permanently save the question to the database.

Create Forms Screen



Form with questions added

Features

- The admin can use this screen to create new forms (or edit existing ones) and add them to the system.
- The drop down at the top allows them to choose a form that is already in the system to edit.
- The table on the left allows the admin to view all questions in the system along with their type and difficulty.
- There are various sorts that the admin can perform on the list of questions using the buttons to the right of the table.
- The admin can also select a question in the table and press the preview question button to have the question opened in a new window so that they can see what it looks like.
- Once the admin has selected a question and pressed the add to form button, the question will be added to the form and displayed in the form preview on the right.
- There is a preview of the form on the right of window so that the admin can see what the final form will look like.
- When questions are added to the form (as shown in the second screenshot) buttons appear to the right of the question which allow the admin to rearrange the questions in the form or to remove a question from the form.
- The “Save form” button at the bottom of the screen opens dialogs which allow the admin to enter final details about the form before adding it to the system.

Form Filler

View QuestionsView FormsCreate questionsCreate formsUsersStatisticsImport and Export

Select a form to editCreating new form: F5532460Edit form

Difficulty	Type
1	Name
3	Gender
4	Country of Birth
3	Birthday
5	Email
4	Password
8	File Upload

Difficulty Sort

Type Sort

Preview Question

Add to form

First name

Date of birth

Email address

Password

delete

^

v

Day

▼

Month

▼

Year

▼

delete

^

v

delete

^

v

delete

^

v

Save Form

A question being previewed

—

□

×

Date of birth

Day

▼

M...

▼

Y...

▼

18

Add final form details dialog	Features
Enter description dialog Input ? Please provide a description of the form. OK Cancel Enter difficulty dialog Input ? What's the form's difficulty? OK Cancel Save form dialog Save? ? Form added! Would you like to save? Yes No	Description Entry: ● This allows the admin to enter a description of the form. Difficulty Entry: ● This allows the admin to enter the difficulty of the form into the system. Save dialog: ● This allows the admin to specify whether they want to permanently save the form to the database.

Users screen

Form Filter

View Questions View Forms Create questions Create forms **Users** Statistics Import and Export

Username	Password	First Name	Last Name	Date of Birth	Phone Number	Sessions attended
bet123	bob	bob	bobina	01/04/03	12345678	
mcuett0	u22q40j	Mag	Cuett	16/05/1992	5915148659	
fmaccourt1	p0otwgg9	Forster	MacCourt	29/04/1987	2478200469	2018-07-22
trulier2	928LQZAJA	Leela	Rulier	18/03/1997	2386249415	
aluden3	9ec3l8FE	Ash	Luden	26/07/1991	5851009544	2018-07-22
obamsity4	M0M610	Oratee	Bamsley	16/08/1970	9243490532	

First name to search for

Search

Add new user

Edit user

Delete user

Take register

Features

- This screen allows the admin to view details about the users in the system. They are also able to add, edit, and delete users.
- The table at the top of the screen displays all of the information (other than the statistics shown in the statistics screen) that is stored about the users in the system.
- The admin can search for a particular user by first name by using the text field.
- The admin can add a user by pressing the add new user button.
- The admin can edit, or delete a user by selecting the user in the table and pressing the corresponding button.
- The admin use the take register button to open a register that they can take.

Add/edit user dialogs	Features
Enter username Input  Please enter a username OK Cancel Enter password Input  Please enter a password OK Cancel Enter first name Input  Please enter your first name OK Cancel Enter last name Input  Please enter your last name OK Cancel Enter date of birth	● These dialogs allow the admin to enter information about the user into the system. ● If the admin is editing a user the dialogs will already be populated with the information stored about the user so that they can edit it.

Input ✕

 Please enter your date of birth as dd/mm/yyyy

Enter phone number

Input ✕

 Please enter your phone number

User added notification

Message ✕

 User added!

Register dialog

Register

☐ bob bobbins

☐ Mag Cuell

☐ Forster MacCourt

☐ Leela Ruller

☐ Ash Lucien

☐ Oralee Barnsley

☐ Karlyn Siggins

☐ Rockie Kerbler

☐ Parke Eayrs

☐ Gal Towse

☐ Ruth Figger

☐ Latia Slaight

☐ Daniella Goard

☐ Erin Dutteridge

☐ Hyacinthia Byllam

☐ Magda Lauder

☐ Emelina Abriani

☐ Graehme Daintrey

☐ Mellisent Main

☐ Marylee Hugland

☐ Roddy Jiggins

☐ Inga Rizzillo

☐ Loralyn Prangle

☐ Lynn Doneld

☐ Iormina Jobs

☐ Tannie Plenty

☐ Sanford Cuffe

☐ Emalee Bittlestone

☐ Tiler Moorman

☐ Kelsi Prynne

☐ Anica Sydney

☐ Cody Trunkfield

☐ Nerta Vamplew

☐ Shirlene Chazelle

☐ Hannie Adamczyk

☐ Drusilla De Carlo

☐ Nikoletta Biaggiotti

☐ Garner Jaycox

☐ Abbe Gobolos

☐ Eliza Fenty

Save register

Features

- This is a list of all the users in the system with checkboxes next to their name.
- The admin can tick the boxes and once they've pressed save, those users with ticked boxes will be marked as present at the session on the current date.

Statistics screen

Form Filler

View Questions

View Forms

Create questions

Create forms

Users

Statistics

Import and Export

Help

Produce report

Q56280064

View question

Selected question:

Times failed validation:

5 most recent number of attempts taken to correct an error:

5 most recent time taken to complete the question:

Average time taken to complete:

Features

- This screen allows the user to produce a report which they can print off.
- The user can select the questions that they've attempted using the drop down.
- Once they have selected a question and pressed the view question button, the labels at the bottom of the screen display the statistics stored about the user for that question.
- The user can also get help on this screen.

Form Filler

View Questions

View Forms

Create questions

Create forms

Users

Statistics

Import and Export

Help

Produce report

Q47136949

View question

Selected question: Q47136949

Times failed validation: 3

5 most recent number of attempts taken to correct an error: 1,1,1,

5 most recent time taken to complete the question: 1:39,0:52,0:01,

Average time taken to complete: 0:50

Help dialog

Message

i

This is the statistics panel. You can view information about each question that you've filled in.

To do this select the question from the drop down and press view question.

You can also press produce report to produce a printable report detailing your progress with each question type.

OK

Report preview dialog

bob123's report

bob123's report

Question type	Number of times failed validation	Average number of attempts needed to correct	Average time to complete
Email	0	0	5
Name	0	0	8
Phone			
Gender	2	2	15
File Upload	2	2	9
Password	2	2	1
Calendar			
Checkboxes			
Drop-down			
Radio Buttons			
Birthday	1	1	1
Address			
James			
date of birth			
Country of Birth	2	2	6

Print

Features

- This dialog shows the generated report to the user so that they can preview it.
- The print button opens the system print dialog so that the user can print the report.

Import and Export Screen
Features

Form Filler

View Questions View Forms Create questions Create forms Users Statistics **Import and Export**

Difficulty	Type
1	Name
3	Gender
4	Country of Birth
3	Birthday
6	Email
4	Password
8	File Upload

Questio...	Descrip...	Main Ski...	Difficulty
Q81558...	large a...	Name...	6
Q81558...	Name a...	Name.F...	5
Q56280...	Gender...	Gender...	3
Q81558...	Name p...	Name.P...	2

Difficulty Sort

Type Sort

Preview Question

Export

Import question

Difficulty Sort

Preview Form

Export

Import form

A question being previewed

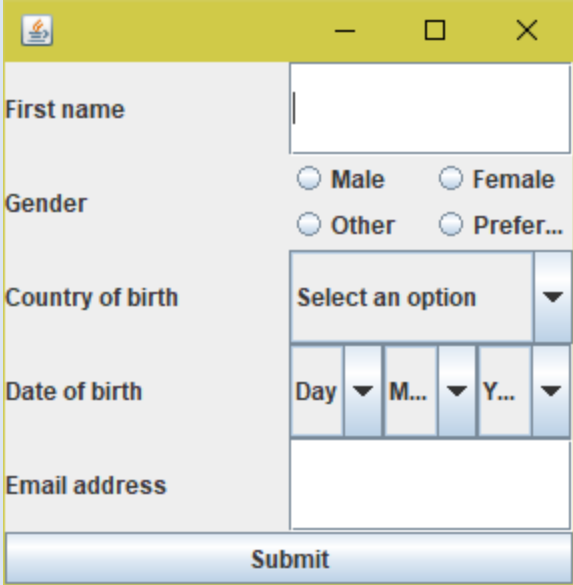
Date of birth

Day ▼
M... ▼
Y... ▼

A form being previewed

- This screen allows the user to import and export forms/questions to/from the system.
- The table in the top left allows the user to view all questions in the system along with their type and difficulty.
- There are various sorts that the user can perform on the list of questions using the buttons to the right of the table.
- The user can also select a question in the table and press the preview question button to have the question opened in a new window so that they can see what it looks like.
- Once the user has selected a question and pressed the export button a file chooser will then be opened allowing the user to select where to export the question to and what file name to give it.
- The table in the bottom left allows the user to view all forms in the system, along with the list of questions in the form, a description, the list of main skills tested, and the difficulty of each form.
- The user can perform a difficulty sort on the forms by pressing the difficulty sort button.
- The user can also select a form in the table and press the preview form button to have the form opened in a new window so that they can see what it looks like.
- Once the user has selected a form and pressed the export button a file chooser will then be opened allowing the user to select where to export the form to and what file name to give it.
- When the import question or import form buttons are pressed a file chooser opens allowing the user to select a question/form to import into the system.

 — □ ×	
First name	
Gender	☐ Male ☐ Female ☐ Other ☐ Prefer...
Country of birth	Select an option ▼
Date of birth	Day ▼ M... ▼ Y... ▼
Email address	

Form attempt window	Features
 The screenshot shows a window titled 'Form attempt window' with a yellow title bar. Inside, there's a form with the following fields: 'First name' (text input), 'Gender' (radio buttons for Male, Female, Other, Prefer...), 'Country of birth' (dropdown menu with 'Select an option' text), 'Date of birth' (four dropdown menus for Day, M..., Y..., and an unlabeled one), and 'Email address' (text input). A 'Submit' button is at the bottom. Red borders highlight the 'Country of birth' and 'Date of birth' fields, indicating failed validation checks.	● When the user selects a form to attempt and presses the attempt form button, this window opens. ● The window displays the questions in the form to the user and allows them to attempt them. ● There is a submit button at the bottom of the window that the user can press to submit the form. ● If a question fails its validation check, a red border is used to highlight this to the user. ● This window is also used when the user attempts an individual question. Rather than an entire form being displayed, just one question is shown in the window. ● If the form passes all validation checks a form complete dialog is shown to the user.

An example with some questions that have failed their validation checks

A screenshot of a web form with a yellow title bar. The form contains the following fields and controls:

- First name:** A text input field containing the text "Bob".
- Gender:** A group of four radio buttons labeled "Male", "Female", "Other", and "Prefer...". The "Male" radio button is selected.
- Country of birth:** A dropdown menu showing "England".
- Date of birth:** A date picker with four dropdown menus for "Day", "M...", "Y...", and an unlabeled month/year dropdown.
- Email address:** A text input field containing "bob....".
- Submit:** A blue button at the bottom of the form.

The "Date of birth" and "Email address" fields are highlighted with a red border.

A single question being attempted

A screenshot of a web form with a yellow title bar, showing only the "Country of birth" field and the "Submit" button. The "Country of birth" dropdown menu is open, showing "England".

Form complete dialog

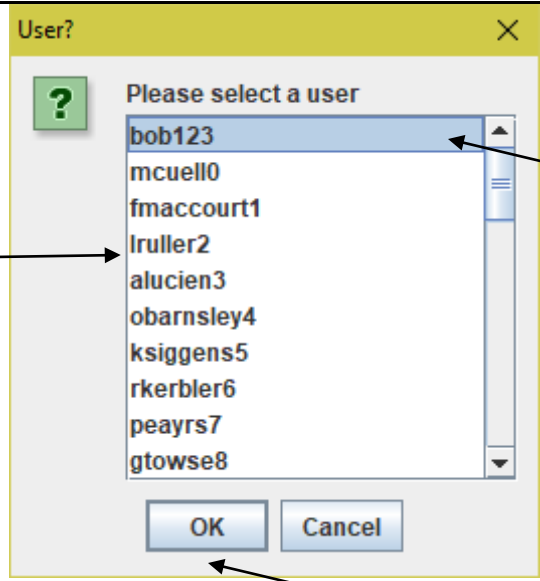
A screenshot of a message dialog box with a yellow title bar. The dialog box contains the following elements:

- Message:** A text label.
- Form complete!:** A message icon (a blue circle with a white 'i') followed by the text "Form complete!".
- OK:** A button at the bottom of the dialog box.

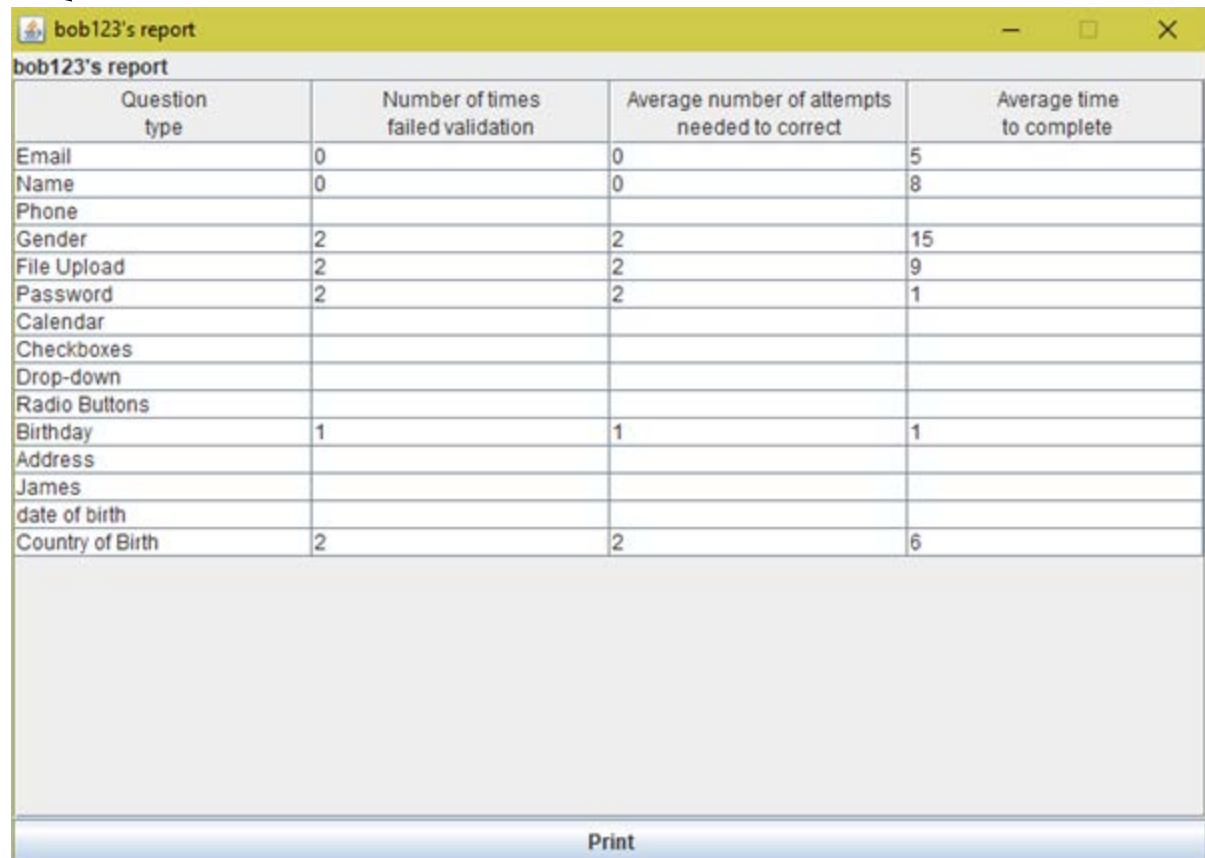
4.3 Functioning System (with Data)

Login

# (from 4.1)	2
Area of System	Login - Login facilities
Output & Storage Requirements	-
Data Used	Username, password

Evidence	User accounts in the system   Bob is selected  Press ok to login 
------------------------	--

The report generated has Bob's username showing that Bob is the account that is logged in



The screenshot shows a window titled "bob123's report" with a table containing the following data:

Question type	Number of times failed validation	Average number of attempts needed to correct	Average time to complete
Email	0	0	5
Name	0	0	8
Phone			
Gender	2	2	15
File Upload	2	2	9
Password	2	2	1
Calendar			
Checkboxes			
Drop-down			
Radio Buttons			
Birthday	1	1	1
Address			
James			
date of birth			
Country of Birth	2	2	6

At the bottom of the window is a "Print" button.

Objectives met	48
# (from 4.1)	3
Area of System	Automatic attendance tracking
Output & Storage Requirements	Appends the current date to the user's entry in userDB.txt
Data Used	Current date

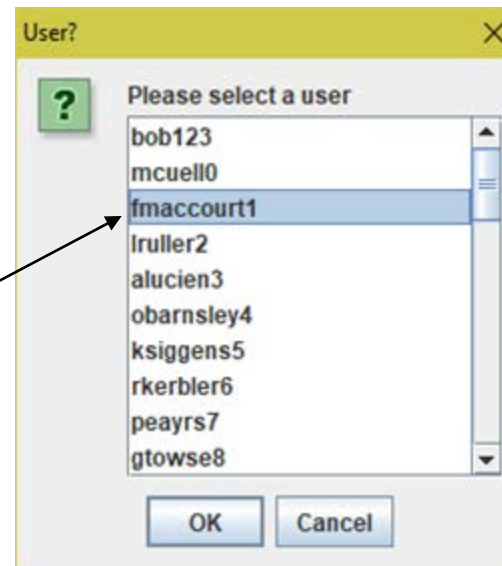
Evidence

fmaccourt1's row in the database

The current date (09/10/18) is not present

U54125754,fmaccourt1,p0oIwgp6,Forster,MacCourt,29/04/1987,2478200469,2018-07-22||

fmaccourt1 is logged in

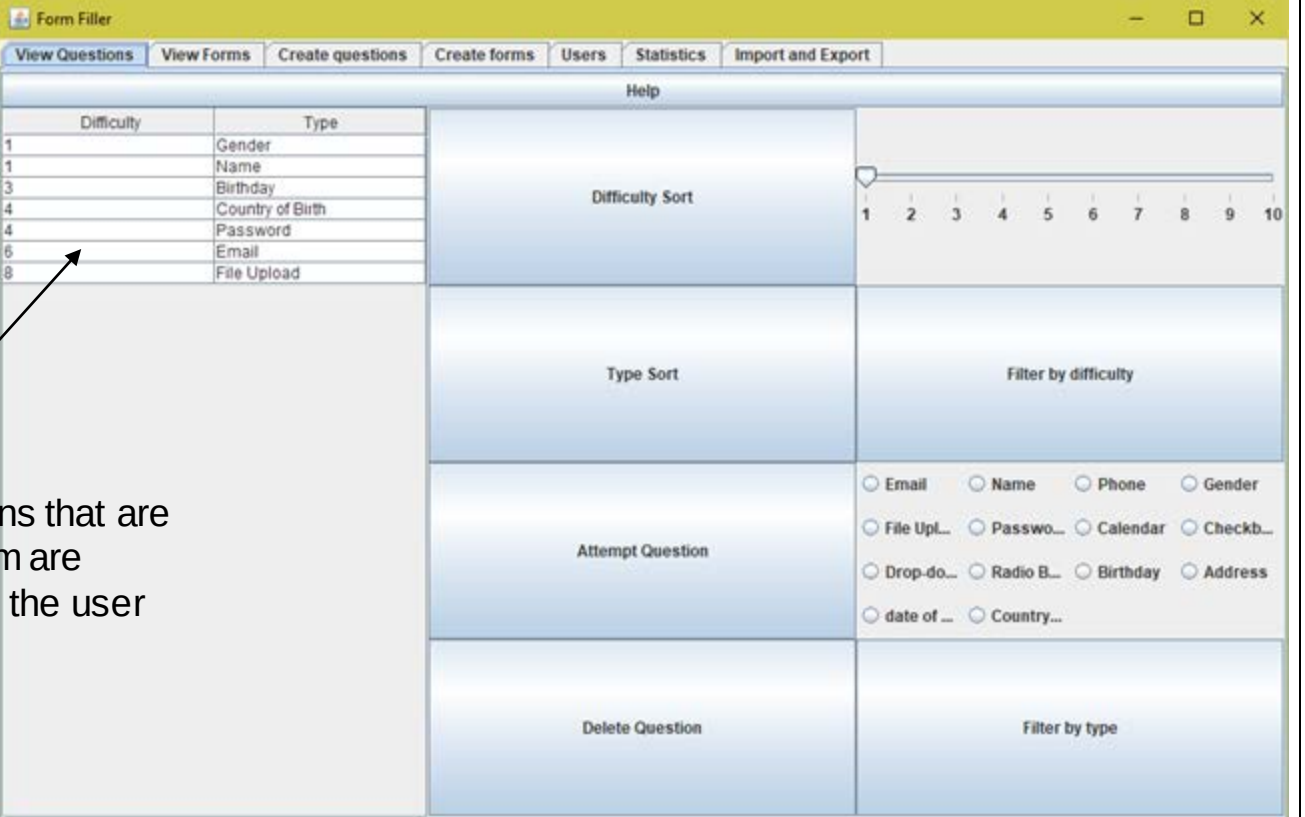


U54125754,fmaccourt1,p0oIwgp6,Forster,MacCourt,29/04/1987,2478200469,2018-07-22.2018-10-09||

The current date (09/10/18) is now present

Objectives met	45
-----------------------	----

View Questions

# (from 4.1)	6
Area of System	View Questions - view
Output & Storage Requirements	-
Data Used	QuestionList, difficulty (of each question), type (of each question)
Evidence	 The questions that are in the system are displayed to the user

Objectives met	53 (displays the questions to the user), 54 (a question can be selected by selecting its row in the table)
# (from 4.1)	7
Area of System	View questions - sort, filter
Output & Storage Requirements	-
Data Used	QuestionList, difficulty (of each question), type (of each question)

Evidence	Difficulty sort

The questions are not in difficulty order

Press difficulty sort button

Difficulty	Type
3	Birthday
4	Country of Birth
6	Email
8	File Upload
1	Gender
1	Name
4	Password

Difficulty	Type
3	Birthday
4	Country of Birth
6	Email
8	File Upload
1	Gender
1	Name
4	Password

The questions are now in difficulty order

The screenshot shows the 'Form Filler' application window. The 'View Questions' tab is active, displaying a list of questions sorted by difficulty. The list has two columns: 'Difficulty' and 'Type'. The questions are as follows:

Difficulty	Type
1	Gender
1	Name
3	Birthday
4	Country of Birth
4	Password
6	Email
8	File Upload

The application interface includes several other elements:

- Top Menu:** View Questions, View Forms, Create questions, Create forms, Users, Statistics, Import and Export.
- Left Panel:** Contains buttons for 'Difficulty Sort', 'Type Sort', 'Attempt Question', and 'Delete Question'.
- Right Panel:** Includes a 'Filter by difficulty' slider (1-10), a 'Filter by type' section with radio buttons for various question types (Email, Name, Phone, Gender, File UpL..., Passwo..., Calendar, Checkb..., Drop-do..., Radio B..., Birthday, Address, date of ..., Country...), and a 'Filter by type' button.

Type sort

Questions are not in order of type

Press type sort button

The screenshot shows the 'Form Filler' application window. It has a menu bar with 'View Questions', 'View Forms', 'Create questions', 'Create forms', 'Users', 'Statistics', and 'Import and Export'. Below the menu bar is a 'Help' button. The main area is divided into two columns. The left column contains a table of questions, and the right column contains sorting and filtering options.

Difficulty	Type
1	Gender
1	Name
3	Birthday
4	Country of Birth
4	Password
6	Email
8	File Upload

Below the table, there is a 'Type Sort' button. To the right of the table, there is a 'Difficulty Sort' button and a slider control ranging from 1 to 10. Below the 'Type Sort' button, there is a 'Filter by difficulty' section with radio buttons for 'Email', 'Name', 'Phone', 'Gender', 'File UpL...', 'Passwo...', 'Calendar', 'Checkb...', 'Drop-do...', 'Radio B...', 'Birthday', 'Address', 'date of ...', and 'Country...'. Below the 'Filter by difficulty' section, there is a 'Filter by type' section with a 'Delete Question' button.

Questions are now
in order of type

The screenshot shows the 'Form Filler' application window. The 'View Questions' tab is active, displaying a list of questions sorted by type. The list is as follows:

Difficulty	Type
3	Birthday
4	Country of Birth
6	Email
8	File Upload
1	Gender
1	Name
4	Password

The interface also includes a sidebar with the following options: Difficulty Sort, Type Sort, Attempt Question, and Delete Question. On the right, there is a 'Filter by difficulty' slider (1-10) and a 'Filter by type' section with radio buttons for various question types: Email, Name, Phone, Gender, File UpL..., Passwo..., Calendar, Checkb..., Drop-do..., Radio B..., Birthday, Address, date of ..., and Country...

Difficulty	Type
3	Birthday
4	Country of Birth
6	Email
8	File Upload
1	Gender
1	Name
4	Password

Difficulty filter

Questions of all difficulties are displayed

Select a difficulty to filter by using the slider

Press filter

The screenshot shows the 'Form Filler' application window. The main area displays a list of questions with columns for 'Difficulty' and 'Type'. The 'Difficulty' column contains values 3, 4, 6, 8, 1, 1, 4. The 'Type' column contains values Birthday, Country of Birth, Email, File Upload, Gender, Name, Password. Below this list is a smaller table with the same data. On the right side, there is a sidebar with various filtering options. At the top of the sidebar is a 'Difficulty Sort' section with a slider ranging from 1 to 10, currently set at 4. Below this is a 'Type Sort' section. Further down is an 'Attempt Question' section. At the bottom of the sidebar is a 'Delete Question' section. On the right side of the sidebar, there is a 'Filter by difficulty' section with a 'Press filter' button. Below this is a 'Filter by type' section with radio buttons for various question types: Email, Name, Phone, Gender, File Up..., Passwo..., Calendar, Checkb..., Drop-do..., Radio B..., Birthday, Address, date of ..., and Country....

Difficulty	Type
3	Birthday
4	Country of Birth
6	Email
8	File Upload
1	Gender
1	Name
4	Password

Only questions of difficulty 4 are displayed

The screenshot shows the 'Form Filler' application window. The 'View Questions' tab is active. The main list displays two questions, both with a difficulty of 4. The right sidebar contains various filters and sorting options. The 'Difficulty Sort' slider is set to 4. The 'Filter by difficulty' button is highlighted. The 'Filter by type' button is also visible. The 'Attempt Question' and 'Delete Question' buttons are at the bottom of the sidebar.

Difficulty	Type
4	Country of Birth
4	Password

Difficulty Sort: 1 2 3 4 5 6 7 8 9 10

Filter by difficulty

Filter by type

Attempt Question

Delete Question

Type filter

Questions of all types are displayed

The screenshot shows the 'Form Filler' application window. At the top, there is a menu bar with options: View Questions, View Forms, Create questions, Create forms, Users, Statistics, and Import and Export. Below the menu bar is a 'Help' button. The main area is divided into several sections. On the left, there is a table with two columns: 'Difficulty' and 'Type'. The table contains the following data:

Difficulty	Type
1	Gender
1	Name
3	Birthday
4	Country of Birth
4	Password
6	Email
8	File Upload

Below this table is another identical table. To the right of the tables, there are several sections: 'Difficulty Sort' with a slider from 1 to 10; 'Type Sort'; 'Attempt Question'; 'Delete Question'; 'Filter by difficulty' with a radio button selection for Gender (selected), Name, Phone, Email, File Up..., Passwo..., Calendar, Checkb, Drop-do..., Radio B..., Birthday, Address, date of ..., and Country...; and 'Filter by type'. An arrow points to the 'Gender' radio button in the 'Filter by difficulty' section.

Select a type to filter by using the radio buttons

Press filter

Only questions of type
"gender" are displayed

Objectives met

1 2 (sort questions by difficulty), 1 3 (sort questions by type), 1 4 (filter questions by difficulty), 1 5 (filter questions by type)

(from 4.1)

8

Area of System	View questions - Delete
Output & Storage Requirements	Removes the question from the question database.
Data Used	Question list, id (of the question to be deleted)

Evidence

Select question to delete

The screenshot shows the 'Form Filler' application window. The 'View Forms' tab is selected, and the 'View Questions' sub-tab is active. A list of questions is displayed on the left, with the third question (Difficulty 3, Type Gender) selected. The right side of the window contains several interactive elements: a 'Difficulty Sort' slider set to 4, a 'Type Sort' button, an 'Attempt Question' button, a 'Delete Question' button, and two filter sections: 'Filter by difficulty' and 'Filter by type'. The 'Filter by type' section has radio buttons for various question types, with 'Gender' selected.

Difficulty	Type
1	Gender
1	Name
3	Birthday
3	Gender
4	Country of Birth
4	Password
6	Email
8	File Upload

Difficulty Sort: 1 2 3 4 5 6 7 8 9 10

Type Sort

Attempt Question

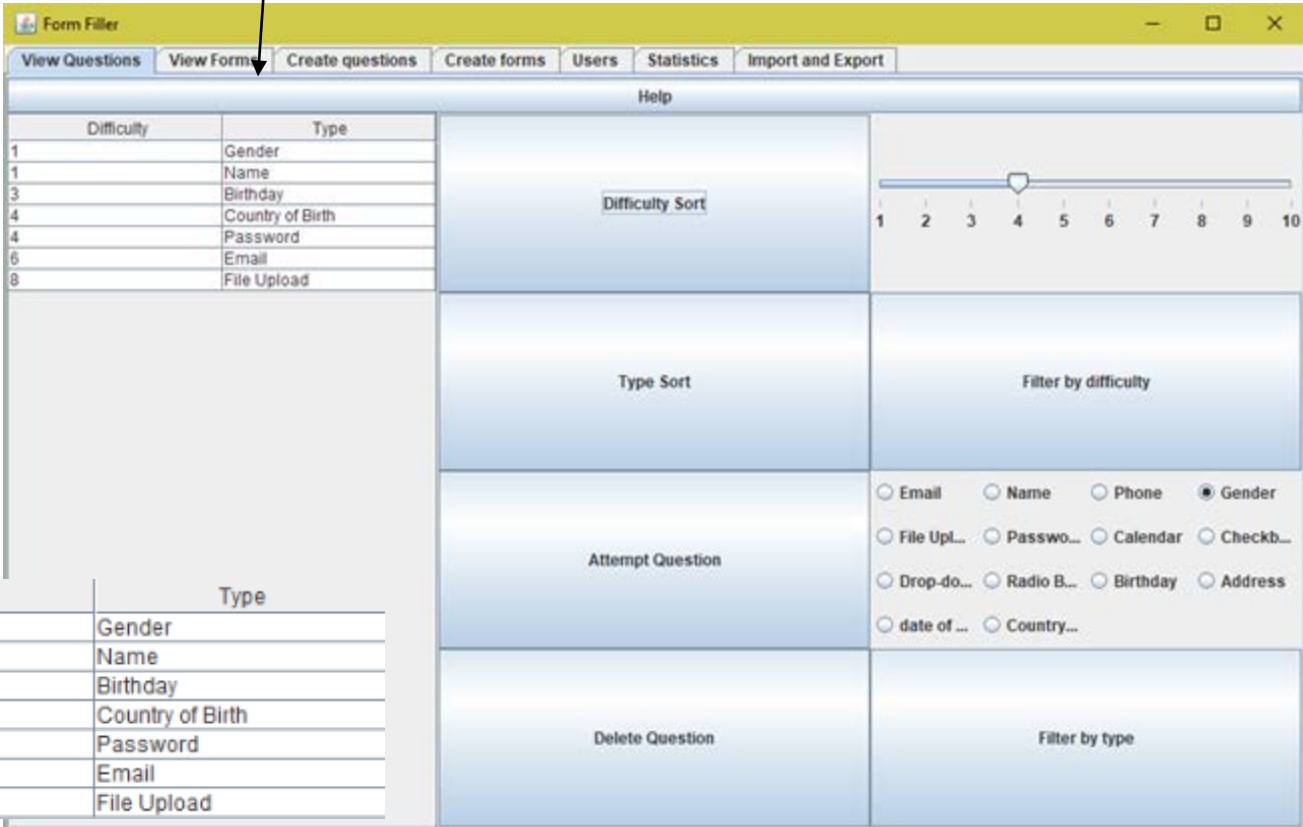
Delete Question

Filter by difficulty

Filter by type

☐ Email ☐ Name ☐ Phone ☒ Gender
☐ File UpL... ☐ Passwo... ☐ Calendar ☐ Checkb...
☐ Drop-do... ☐ Radio B... ☐ Birthday ☐ Address
☐ date of ... ☐ Country...

Press delete

	Question is no longer present  The screenshot shows the 'Form Filler' application window. The 'View Questions' tab is selected. On the left, a table lists questions with columns 'Difficulty' and 'Type'. The first question (Difficulty 1, Type Gender) is highlighted. An arrow points to the 'View Questions' tab. The main area contains buttons for 'Difficulty Sort', 'Type Sort', 'Attempt Question', and 'Delete Question'. A difficulty slider is visible at the top right, and a filter by difficulty section is on the right. A filter by type section is at the bottom right.
Objectives met	11 (Ability to delete questions)
# (from 4.1)	9
Area of System	View Questions - Attempt
Output & Storage	-

Requirements	
Data Used	id (of the question to be attempted), component list (list of components in the question)

Evidence

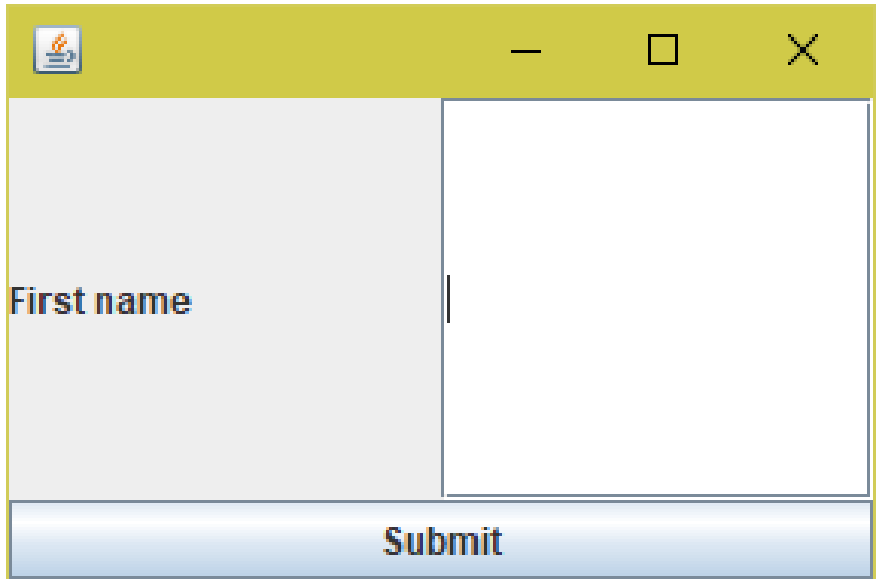
Select question to be attempted

The screenshot shows the 'Form Filler' application window with the 'View Questions' tab selected. The interface is divided into several sections:

- Question List Table:** A table with columns 'Difficulty' and 'Type'. The 'Name' question (Difficulty 1) is highlighted.
- Sorting and Filtering Options:** Includes 'Difficulty Sort', 'Type Sort', 'Filter by difficulty' (with a slider from 1 to 10), and 'Filter by type'.
- Attempt Question Button:** A button labeled 'Attempt Question' is visible, with an arrow pointing to it from the text 'Press attempt question'.
- Radio Buttons:** A set of radio buttons for filtering by type, including 'Email', 'Name', 'Phone', 'Gender' (selected), 'File Up...', 'Passwo...', 'Calendar', 'Checkb...', 'Drop-do...', 'Radio B...', 'Birthday', 'Address', 'date of ...', and 'Country...'.

Difficulty	Type
1	Gender
1	Name
3	Birthday
4	Country of Birth
4	Password
6	Email
8	File Upload

Press attempt
question

	Question is opened to be attempted 
Objectives met	23 (Attempt a question), 54 (Allow a user to select a question to be attempted).

View Forms

# (from 4.1)	10
Area of System	View Forms - view
Output & Storage Requirements	-

Data Used	FormList, Question Ids (of the questions in each form), Description (of each form), Main skills tested (of each form), difficulty (of each form), percent complete (of each form)
------------------	---

Evidence

The forms in the system are displayed to the user

The screenshot shows the 'Form Filler' application window. The 'View Forms' tab is active, displaying a table of questions and a sidebar with various form management options.

Questions	Descripti...	Main Skill...	Difficulty	Percent ...
Q815587...	large am...	Name.G...	6	100%
Q815587...	Name an...	Name.Fil...	5	100%
Q562800...	Gender a...	Gender....	3	100%
Q815587...	Name pa...	Name.Pa...	2	100%

The sidebar contains the following options:

- Difficulty Sort (with a slider from 1 to 10)
- Attempt Form
- Delete Form
- Attempt form based on weaknesses
- Filter by difficulty
- Filter by type (with checkboxes for Email, Name, Phone, Gender, File Up..., Passwo..., Calendar, Checkb..., Drop-do..., Radio B..., Birthday, Address, date of ..., Country...)

Objectives met

50 (There should be a panel to display the forms to the user)

# <i>(from 4.1)</i>	11
Area of System	View Forms - Sort, Filter
Output & Storage Requirements	-
Data Used	FormList, difficulty (of each form), main skills tested (of each form)
Evidence	Difficulty sort

Forms are not in
difficulty order

Press difficulty sort

Questions	Descripti...	Main Skill...	Difficulty	Percent ...
Q815587...	large am...	Name.G...	6	100%
Q815587...	Name an...	Name.Fil...	5	100%
Q562800...	Gender a...	Gender....	3	100%
Q815587...	Name pa...	Name.Pa...	2	100%

Questions	Description	Main Skill...	Difficulty	Percent ...
Q815587...	large am...	Name.G...	6	100%
Q815587...	Name an...	Name.Fil...	5	100%
Q562800...	Gender a...	Gender....	3	100%
Q815587...	Name pa...	Name.Pa...	2	100%

Forms are now in difficulty order

The screenshot shows the 'Form Filler' application window. The 'View Forms' tab is selected, and an arrow points to the 'View Forms' tab. The main area displays a table of questions sorted by difficulty. The table has columns: Questions, Description, Main Skill, Difficulty, and Percent. The questions are listed in ascending order of difficulty (2, 3, 5, 6).

Questions	Description	Main Skill	Difficulty	Percent
Q815587...	Name pa...	Name.Pa...	2	100%
Q562800...	Gender a...	Gender....	3	100%
Q815587...	Name an...	Name.Fil...	5	100%
Q815587...	large am...	Name.G...	6	100%

Below the table, there are several buttons and filters:

- Difficulty Sort**: A button with a slider ranging from 1 to 10.
- Attempt Form**: A button.
- Delete Form**: A button.
- Attempt form based on weaknesses**: A button.
- Filter by difficulty**: A button.
- Filter by type**: A button.
- Filters**: A list of checkboxes for various fields: Email, Name, Phone, Gender, File UpL..., Passwo..., Calendar, Checkb..., Drop-do..., Radio B..., Birthday, Address, date of ..., and Country...

Difficulty filter

Forms of all difficulties are displayed

Press filter Select a difficulty to filter by using the slider.

The screenshot shows the 'Form Filler' application window. The 'View Forms' tab is selected. The main area displays a table of questions with columns: Questions, Description, Main Skill, Difficulty, and Percent. The table contains four rows of data. The sidebar on the right contains several sections: 'Difficulty Sort' with a slider set to 5, 'Attempt Form', 'Delete Form', 'Attempt form based on weaknesses', 'Filter by difficulty', and 'Filter by type'. The 'Filter by difficulty' section has a slider and a 'Filter' button. The 'Filter by type' section has a list of checkboxes for various question types.

Questions	Description	Main Skill	Difficulty	Percent
Q815587...	Name pa...	Name Pa...	2	100%
Q562800...	Gender a...	Gender...	3	100%
Q815587...	Name an...	Name Fil...	5	100%
Q815587...	large am...	Name G...	6	100%

Only forms of difficulty 5 are displayed

The screenshot shows the 'Form Filler' application window. The 'View Forms' tab is selected. On the left, a table lists forms with columns: Questions, Descripti..., Main Skill..., Difficulty, and Percent ... The table contains one entry: Q815587..., Name an..., Name.Fil..., 5, 100%. The main area of the window is divided into several sections: 'Difficulty Sort' with a slider set to 5, 'Attempt Form', 'Delete Form', and 'Filter by difficulty'. The 'Filter by difficulty' section has a slider from 1 to 10 with a marker at 5. Below this, there are checkboxes for various form types: Email, Name, Phone, Gender, File Upl..., Passwo..., Calendar, Checkb..., Drop-do..., Radio B..., Birthday, Address, date of ..., and Country... The 'Filter by type' section is at the bottom right. A table at the bottom of the window shows the same data as the top table.

Questions	Descripti...	Main Skill...	Difficulty	Percent ...
Q815587...	Name an...	Name.Fil...	5	100%

Questions	Description	Main Skill...	Difficulty	Percent ...
Q815587...	Name an...	Name.Fil...	5	100%

Type filter

Forms containing all types of questions are displayed

Select type(s) to filter by

The screenshot shows the 'Form Filler' application window. The 'View Forms' tab is selected. On the left, there are two identical tables of question data. The main area contains several buttons: 'Difficulty Sort', 'Attempt Form', 'Delete Form', and 'Attempt form based on weaknesses'. On the right, there is a 'Filter by difficulty' section with a slider set to 5, and a 'Filter by type' section with checkboxes for various question types. The 'Filter by type' section includes checkboxes for Email, Name (checked), Phone, Gender (checked), File Upload, Password, Calendar, Checkboxes, Drop-down, Radio Buttons, Birthday, Address, Date of Birth, and Country.

Questions	Descripti...	Main Skills Tested	Diffi...	Per...
Q815587...	Name pa...	Name.Password.E...	2	100...
Q562800...	Gender a...	Gender.Country of B...	3	100...
Q815587...	Name an...	Name.File Upload	5	100...
Q815587...	large am...	Name.Gender.Coun...	6	100...

Questions	Descripti...	Main Skills Tested	Diffi...	Per...
Q815587...	Name pa...	Name.Password.E...	2	100...
Q562800...	Gender a...	Gender.Country of B...	3	100...
Q815587...	Name an...	Name.File Upload	5	100...
Q815587...	large am...	Name.Gender.Coun...	6	100...

Press filter

Only forms containing questions of type name and gender are displayed

Questions	Descripti...	Main Skills Tested	Diffi...	Per...
Q815587...	large am...	Name.Gender.Coun...	6	100...

Objectives met

27 (Ability to sort forms by difficulty), 28 (Ability to filter forms by difficulty), 29 (Ability to filter forms by types of questions tested)

# <i>(from 4.1)</i>	12
Area of System	View Forms - Delete
Output & Storage Requirements	-
Data Used	FormList, id (of the form to be deleted)

Evidence

Form selected to be deleted

Press delete form

Form Filler

View Questions View Forms Create questions Create forms Users Statistics Import and Export

Help

Questions	Descripti...	Main Skills Tested	Diffi...	Per...
Q815587...	Name pa...	Name.Password.E...	2	100...
Q815587...	Asks the ...	Name	2	0%
Q562800...	Gender a...	Gender.Country of B...	3	100...
Q815587...	Name an...	Name.File Upload	5	100...
Q815587...	large am...	Name.Gender Coun...	6	100...

Difficulty Sort

Attempt Form

Delete Form

Filter by difficulty

Filter by type

pt form based on weaknesses

Questions	Descripti...	Main Skills Tested	Diffi...	Per...
Q815587...	Name pa...	Name.Password.E...	2	100...
Q815587...	Asks the ...	Name	2	0%
Q562800...	Gender a...	Gender.Country of B...	3	100...
Q815587...	Name an...	Name.File Upload	5	100...
Q815587...	large am...	Name.Gender Coun...	6	100...

Form no longer present in the table (i.e. it has been deleted from the system)

Questions	Descripti...	Main Skills Tested	Diffi...	Per...
Q815587...	Name pa...	Name.Password.E...	2	100...
Q562800...	Gender a...	Gender.Country of B...	3	100...
Q815587...	Name an...	Name.File Upload	5	100...
Q815587...	large am...	Name.Gender.Coun...	6	100...

Objectives met

31 (Ability to delete forms)

(from 4.1)

13

Area of System	View forms - Create form from user weaknesses
Output & Storage Requirements	-
Data Used	QuestionStatList, Average number of attempts to correct an error (for each question stat)

Evidence

Press attempt form based on user weaknesses

Form Filler

View Questions View Forms Create questions Create forms Users Statistics Import and Export

Help

Questions	Descripti...	Main Skills Tested	Diffi...	Per...
Q815587...	Name pa...	Name.Password.E...	2	100...
Q562800...	Gender a...	Gender.Country of B...	3	100...
Q815587...	Name an...	Name.File Upload	5	100...
Q815587...	large am...	Name.Gender.Coun...	6	100...

Difficulty Sort

1 2 3 4 5 6 7 8 9 10

Attempt Form

Filter by difficulty

Delete Form

Attempt form based on weaknesses

Filter by type

☐ Email
 ☒ Name
 ☐ Phone
 ☒ Gender

☐ File Up...
 ☐ Passwo...
 ☐ Calendar
 ☐ Checkb...

☐ Drop-do...
 ☐ Radio B...
 ☐ Birthday
 ☐ Address

☐ date of ...
 ☐ Country...

A form containing the questions that they struggle with the most opens

Country of birth

Select an option

Password

Photo id

Upload File

Submit

Objectives met

26 (Ability to create forms from user weaknesses)

# <i>(from 4.1)</i>	14
Area of System	View Forms - Attempt
Output & Storage Requirements	-
Data Used	id (of the form to be attempted), question list (list of questions in the form), component list (list of components in each question)

Evidence

Select a form

Press attempt form

Form Filler

View Questions View Forms Create questions Create forms Users Statistics Import and Export

Help

Questions	Descripti...	Main Skills Tested	Diffi...	Per...
Q815587...	Name pa...	Name.Password.E...	2	100...
Q562800...	Gender a...	Gender.Country of B...	3	100...
Q815587...	Name an...	Name.File Upload	5	100...
Q815587...	large am...	Name.Gender.Coun...	6	100...

Difficulty Sort

1 2 3 4 5 6 7 8 9 10

Attempt Form

Filter by difficulty

Delete Form

☐ Email
 ☒ Name
 ☐ Phone
 ☒ Gender

☐ File Up...
 ☐ Passwo...
 ☐ Calendar
 ☐ Checkb...

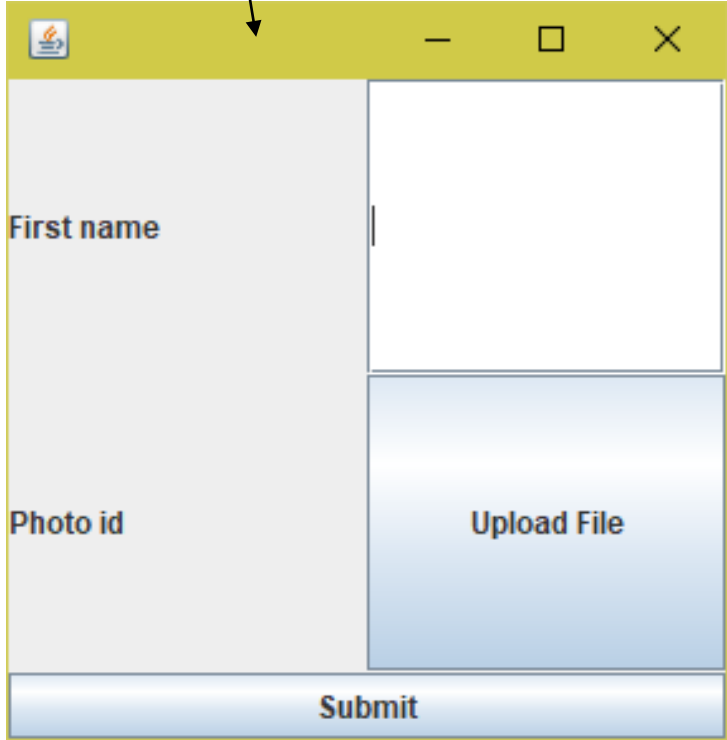
☐ Drop-do...
 ☐ Radio B...
 ☐ Birthday
 ☐ Address

☐ date of ...
 ☐ Country...

Questions	Descripti...	Main Skills Tested	Diffi...	Per...
Q815587...	Name pa...	Name.Password.E...	2	100...
Q562800...	Gender a...	Gender.Country of B...	3	100...
Q815587...	Name an...	Name.File Upload	5	100...
Q815587...	large am...	Name.Gender.Coun...	6	100...

Attempt form based on weaknesses

Filter by type

	The form is opened for the user to attempt 
Objectives met	34 (Ability to attempt a form), 51 (Allow user to select a form to be attempted)

Create Questions

# (from 4.1)	15
--------------	----

Area of System	Create Questions - Add components
Output & Storage Requirements	-
Data Used	component list (list of components in the question), type of validation required (where appropriate), options that the user can select (where appropriate)
Evidence	Add label

Press add label

Form Filler

View Questions View Forms Create questions Create forms Users Statistics Import and Export

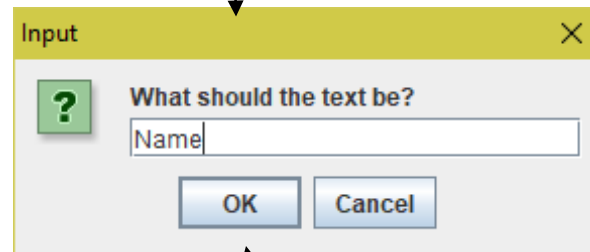
Question Preview

Create components

Add label	Add text field
Add radio buttons	Add drop-down
Add check boxes	Add password field
Add calendar entry	Add file chooser

Save Question

Dialog opens to enter
label text



Press ok

Label added to question

The screenshot displays the 'Form Filler' application window. The title bar is yellow and contains the text 'Form Filler' and standard window controls. Below the title bar is a menu bar with the following tabs: 'View Questions', 'View Forms', 'Create questions', 'Create forms', 'Users', 'Statistics', and 'Import and Export'. The main area is divided into two sections. The top section, titled 'Question Preview', shows a large, empty light gray area. The bottom section, titled 'Create components', contains a grid of buttons for adding different form elements. An arrow points from the text 'Label added to question' to the 'Add label' button in the 'Create components' section.

Create components	
Add label	Add text field
Add radio buttons	Add drop-down
Add check boxes	Add password field
Add calendar entry	Add file chooser

Save Question

Add Text field

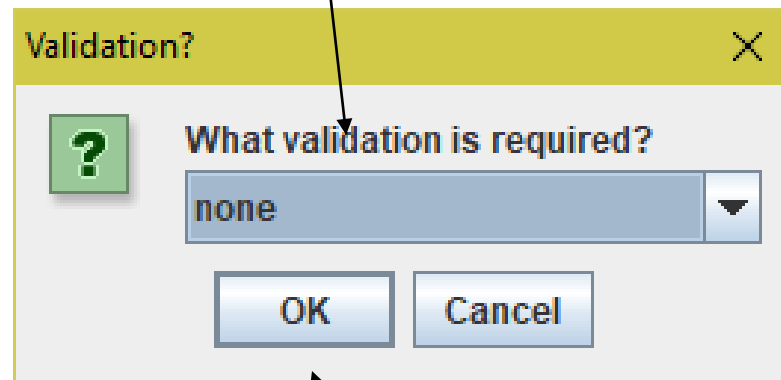
Press add text field

The screenshot shows the 'Form Filler' application window. The title bar is yellow and contains the text 'Form Filler' and standard window controls. Below the title bar is a tabbed interface with the following tabs: 'View Questions', 'View Forms', 'Create questions' (which is the active tab), 'Create forms', 'Users', 'Statistics', and 'Import and Export'. The main area of the window is labeled 'Question Preview' and is currently empty. At the bottom of the window, there is a 'Create components' section. This section contains a grid of buttons for adding different form elements: 'Add label', 'Add text field', 'Add radio buttons', 'Add drop-down', 'Add check boxes', 'Add password field', 'Add calendar entry', and 'Add file chooser'. An arrow points from the text 'Press add text field' to the 'Add text field' button. To the right of the 'Create components' section is a large blue button labeled 'Save Question'.

Create components	
Add label	Add text field
Add radio buttons	Add drop-down
Add check boxes	Add password field
Add calendar entry	Add file chooser

Save Question

Select the validation required
from the drop down in the dialog



Press ok

Text field is added to question

The screenshot shows the 'Form Filler' application window. The title bar is yellow and contains the text 'Form Filler' and standard window controls. Below the title bar is a tabbed interface with the following tabs: 'View Questions', 'View Forms', 'Create questions' (which is the active tab), 'Create forms', 'Users', 'Statistics', and 'Import and Export'. The main area is divided into two panes. The left pane is labeled 'Name' and is currently empty. The right pane is labeled 'Question Preview' and is also empty. Below the 'Name' pane is a section titled 'Create components' which contains a grid of buttons: 'Add label', 'Add text field', 'Add radio buttons', 'Add drop-down', 'Add check boxes', 'Add password field', 'Add calendar entry', and 'Add file chooser'. To the right of the 'Create components' section is a large blue button labeled 'Save Question'. An arrow points from the text 'Text field is added to question' to the 'Add text field' button.

Add Radio buttons

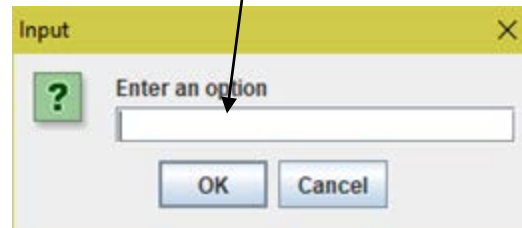
Press add radio buttons

The screenshot shows the 'Form Filler' application window. The title bar is yellow and contains the text 'Form Filler' and standard window controls. Below the title bar is a tabbed interface with the following tabs: 'View Questions', 'View Forms', 'Create questions' (which is the active tab), 'Create forms', 'Users', 'Statistics', and 'Import and Export'. The main area of the window is divided into two sections. The top section is labeled 'Question Preview' and is currently empty. The bottom section is labeled 'Create components' and contains a grid of buttons for adding different form elements. The buttons are arranged in two columns and four rows:

Create components	
Add label	Add text field
Add radio buttons	Add drop-down
Add check boxes	Add password field
Add calendar entry	Add file chooser

To the right of the 'Create components' grid is a large blue button labeled 'Save Question'. An arrow points from the text 'Press add radio buttons' to the 'Add radio buttons' button in the grid.

Enter an option into
dialog that appears



Press ok

After pressing ok this
dialog appears



Press yes to
continue entering
options

Press no to stop entering options and to
add the component to the question

Radio buttons added to the question

The screenshot shows the 'Form Filler' application window. The title bar is yellow and contains the text 'Form Filler' and standard window controls. Below the title bar is a menu bar with tabs: 'View Questions', 'View Forms', 'Create questions' (selected), 'Create forms', 'Users', 'Statistics', and 'Import and Export'. The main area is divided into two sections. The top section, titled 'Question Preview', shows a form with a 'Name' label and a text input field. Below it is a 'Gender' label followed by three radio buttons labeled 'Male', 'Female', and 'Other'. An arrow points from the text 'Radio buttons added to the question' to the 'Male' radio button. The bottom section, titled 'Create components', contains a grid of buttons: 'Add label', 'Add text field', 'Add radio buttons' (highlighted with a blue border), 'Add drop-down', 'Add check boxes', 'Add password field', 'Add calendar entry', and 'Add file chooser'. To the right of this grid is a large blue button labeled 'Save Question'.

Add drop down

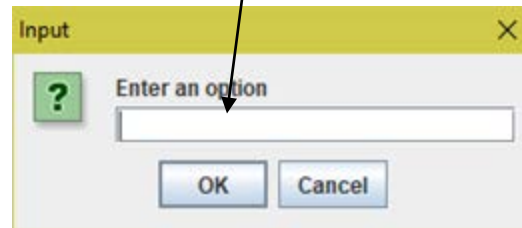
Press add drop down

The screenshot shows the 'Form Filler' application window. The title bar is yellow and contains the text 'Form Filler' and standard window controls. Below the title bar is a tabbed interface with the following tabs: 'View Questions', 'View Forms', 'Create questions' (which is the active tab), 'Create forms', 'Users', 'Statistics', and 'Import and Export'. The main area of the window is divided into two sections. The top section, labeled 'Question Preview', is a large, empty light gray rectangle. The bottom section, labeled 'Create components', contains a grid of buttons for adding different form elements. The buttons are arranged in two columns and four rows: 'Add label', 'Add text field', 'Add radio buttons', 'Add drop-down', 'Add check boxes', 'Add password field', 'Add calendar entry', and 'Add file chooser'. An arrow points from the text 'Press add drop down' to the 'Add drop-down' button. To the right of the 'Create components' grid is a large light blue rectangle labeled 'Save Question'.

Create components	
Add label	Add text field
Add radio buttons	Add drop-down
Add check boxes	Add password field
Add calendar entry	Add file chooser

Save Question

Enter an option into dialog
that appears



Press ok

After pressing ok this
dialog appears



Press yes to
continue entering
options

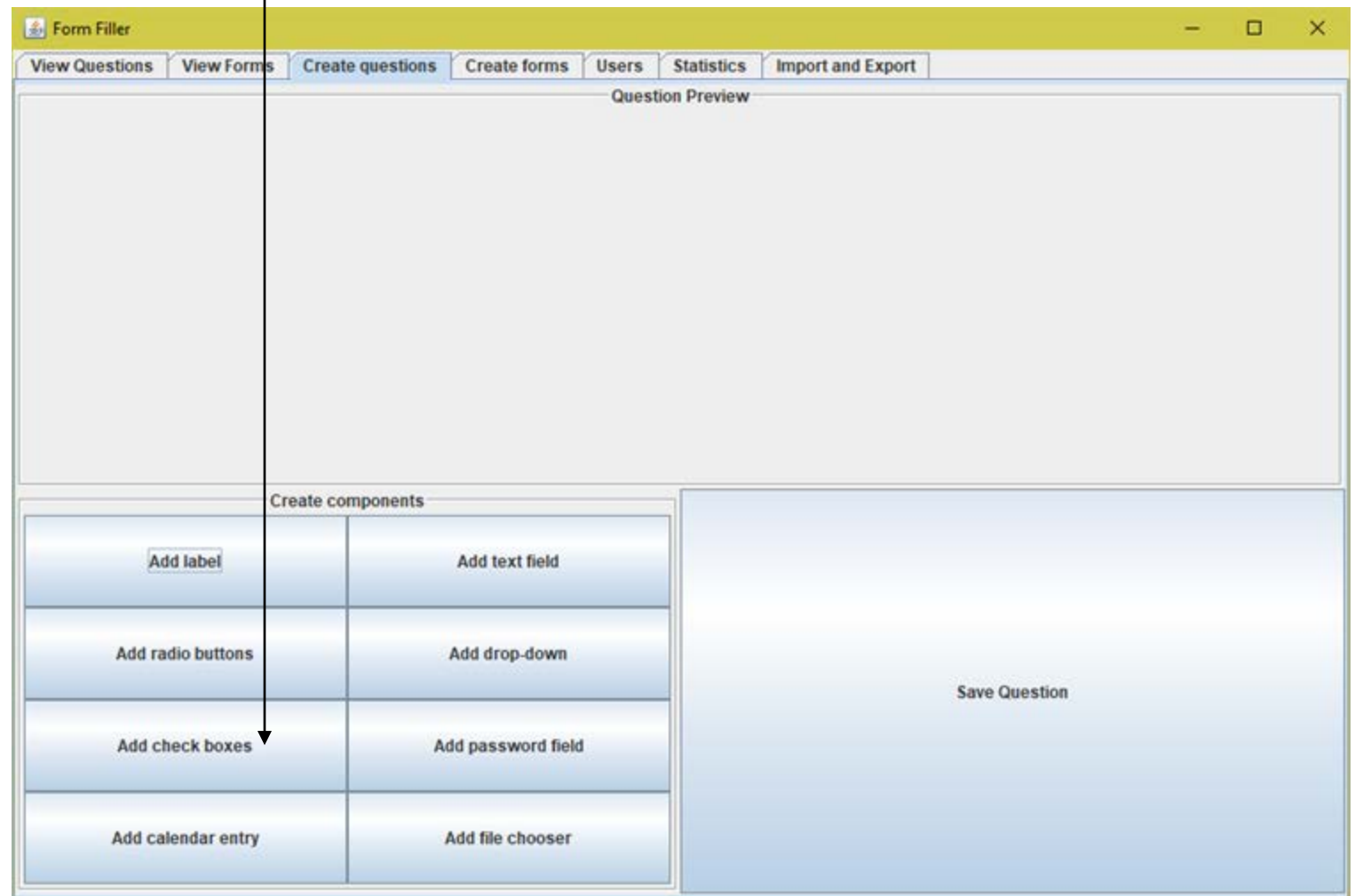
Press no to stop entering options and
to add the component to the question

Drop down added to question

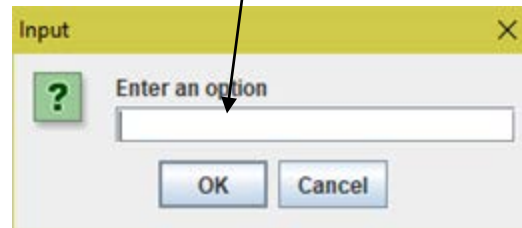
The screenshot displays the 'Form Filler' application window. The title bar is yellow and contains the text 'Form Filler' and standard window controls. Below the title bar is a tabbed interface with the following tabs: 'View Questions', 'View Forms', 'Create questions' (which is the active tab), 'Create forms', 'Users', 'Statistics', and 'Import and Export'. The main workspace is divided into three sections. The top-left section is a large light gray area labeled 'Choose an option'. The top-right section is a 'Question Preview' area, also light gray, showing a preview of the question with the text 'Select an option' and a small downward arrow on the right side. An arrow from the text 'Drop down added to question' points to this preview area. The bottom-left section is a 'Create components' palette with a grid of buttons: 'Add label', 'Add text field', 'Add radio buttons', 'Add drop-down' (which is highlighted with a dashed border), 'Add check boxes', 'Add password field', 'Add calendar entry', and 'Add file chooser'. The bottom-right section is a large light blue area with the text 'Save Question'.

Add checkboxes

Press add check boxes

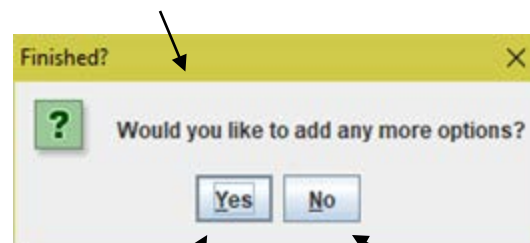


Enter an option into
dialog that appears



Press ok

After pressing ok this dialog
appears



Press yes to
continue entering
options

Press no to stop entering options and
to add the component to the question

Checkboxes have been added to the question

The screenshot shows the 'Form Filler' application window. The title bar is yellow and contains the text 'Form Filler' and standard window controls. Below the title bar is a menu bar with tabs: 'View Questions', 'View Forms', 'Create questions' (selected), 'Create forms', 'Users', 'Statistics', and 'Import and Export'. The main area is titled 'Question Preview' and displays a question: 'Which of these places have you lived in'. Below the question are three checkboxes: 'UK', 'In the EU', and 'Outside the EU'. An arrow points from the text 'Checkboxes have been added to the question' to the 'UK' checkbox. At the bottom of the window is a 'Create components' panel with a grid of buttons: 'Add label', 'Add text field', 'Add radio buttons', 'Add drop-down', 'Add check boxes' (highlighted with a blue border), 'Add password field', 'Add calendar entry', and 'Add file chooser'. To the right of this panel is a large blue button labeled 'Save Question'.

Add password field

Press add password field

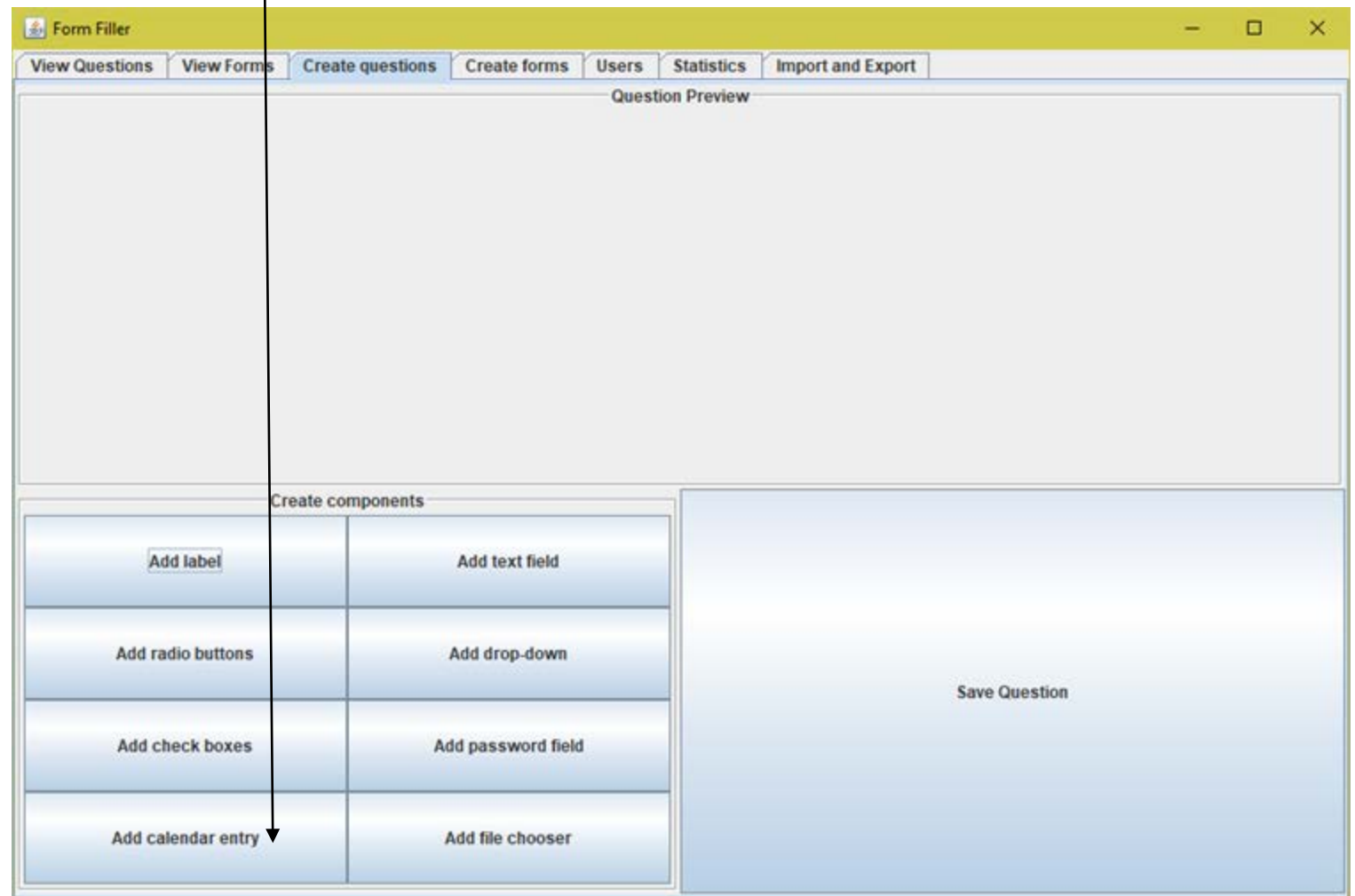
The screenshot shows the 'Form Filler' application window. The title bar is yellow and contains the text 'Form Filler' and standard window controls. Below the title bar is a menu bar with tabs: 'View Questions', 'View Forms', 'Create questions' (which is selected), 'Create forms', 'Users', 'Statistics', and 'Import and Export'. The main area of the window is divided into two sections. The top section is labeled 'Question Preview' and is currently empty. The bottom section is labeled 'Create components' and contains a grid of buttons for adding different field types. The buttons are: 'Add label', 'Add text field', 'Add radio buttons', 'Add drop-down', 'Add check boxes', 'Add password field' (which is pointed to by a black arrow from the text 'Press add password field' above), 'Add calendar entry', and 'Add file chooser'. To the right of the 'Create components' grid is a large blue button labeled 'Save Question'.

Password field added to question

The screenshot displays the 'Form Filler' application window. The title bar is yellow and contains the text 'Form Filler' and standard window controls. Below the title bar is a menu bar with the following options: 'View Questions', 'View Forms', 'Create questions', 'Create forms', 'Users', 'Statistics', and 'Import and Export'. The 'Create questions' menu item is currently selected. The main workspace is divided into three sections. On the left, a large grey rectangular area contains the text 'Password'. On the right, a 'Question Preview' section shows a white rectangular area. At the bottom, a 'Create components' section contains a grid of eight buttons: 'Add label', 'Add text field', 'Add radio buttons', 'Add drop-down', 'Add check boxes', 'Add password field', 'Add calendar entry', and 'Add file chooser'. The 'Add password field' button is highlighted with a blue border. To the right of this grid is a large blue rectangular area with the text 'Save Question'. An arrow points from the text 'Password field added to question' to the 'Add password field' button.

Add calendar entry

Press add calendar entry

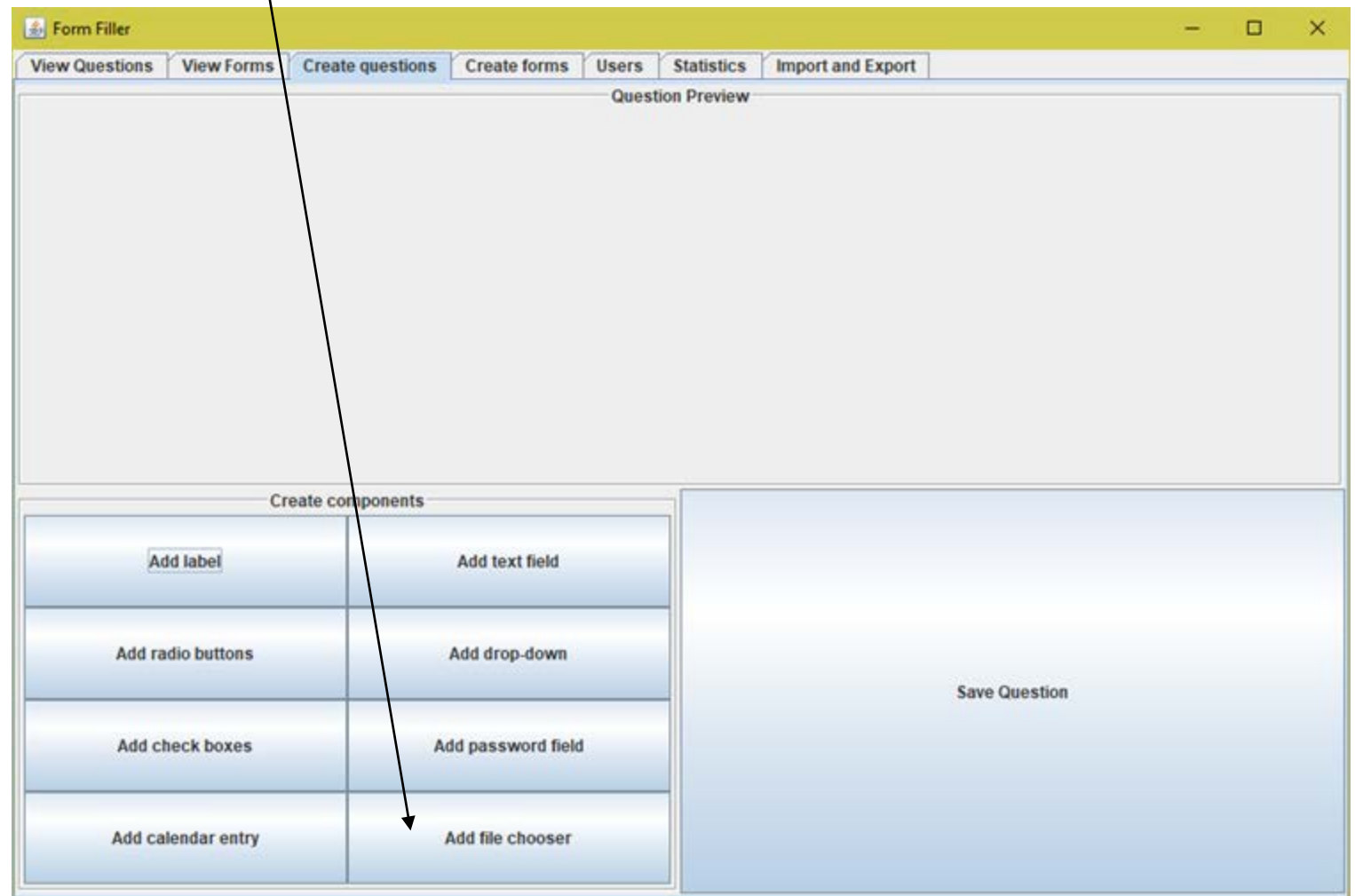


Calendar entry added to
question

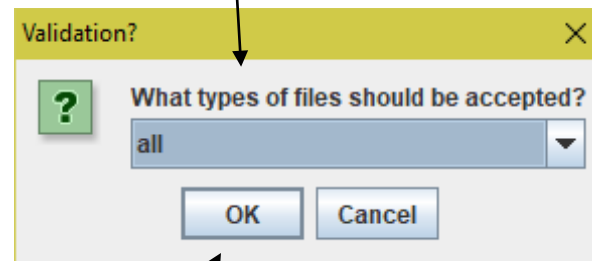
The screenshot displays the 'Form Filler' application window. The title bar reads 'Form Filler'. Below it is a menu bar with the following options: 'View Questions', 'View Forms', 'Create questions', 'Create forms', 'Users', 'Statistics', and 'Import and Export'. The 'Create questions' tab is currently selected. The main workspace is divided into two sections. The top section, titled 'Question Preview', shows a form layout for a 'Date of birth' question. It consists of a large text input field on the left and three dropdown menus on the right labeled 'Day', 'Month', and 'Year'. The bottom section, titled 'Create components', contains a grid of buttons for adding different form elements: 'Add label', 'Add text field', 'Add radio buttons', 'Add drop-down', 'Add check boxes', 'Add password field', 'Add calendar entry', and 'Add file chooser'. The 'Add calendar entry' button is highlighted with a black border. To the right of this grid is a large blue button labeled 'Save Question'.

Add file chooser

Press add file chooser



Select which types of files should be accepted by using the drop down in the dialog that appears



Press ok

File chooser added to the question

The screenshot displays the 'Form Filler' application window. The title bar reads 'Form Filler'. The menu bar includes 'View Questions', 'View Forms', 'Create questions', 'Create forms', 'Users', 'Statistics', and 'Import and Export'. The 'Create questions' tab is active. The main area is divided into two sections: 'Question Preview' on the right and 'Create components' on the left. The 'Question Preview' section shows a question titled 'Photo ID' with a large blue box labeled 'Upload File'. The 'Create components' section contains a grid of buttons: 'Add label', 'Add text field', 'Add radio buttons', 'Add drop-down', 'Add check boxes', 'Add password field', 'Add calendar entry', and 'Add file chooser'. An arrow points from the text 'File chooser added to the question' to the 'Add file chooser' button. The 'Save Question' button is located at the bottom right of the 'Question Preview' section.

Create components	
Add label	Add text field
Add radio buttons	Add drop-down
Add check boxes	Add password field
Add calendar entry	Add file chooser

Question Preview

Photo ID

Upload File

Save Question

Objectives met	1 (Ability to create questions), 2 (Ability to add labels), 3 (Ability to add text fields), 4 (Ability to specify the type of data that should be entered into the text field), 5 (Ability to add radio buttons), 6 (Ability to add drop-downs), 7 (Ability to add checkboxes), 8 (Ability to add password fields), 9 (Ability to add calendar entry), 10 (Ability to add a document upload field (file chooser))
# (from 4.1)	18
Area of System	Create Questions - Add basic information
Output & Storage Requirements	QuestionDB, QuestionPanelDB
Data Used	difficulty, type, component list (list of components in the question)

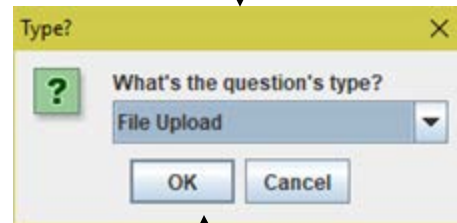
Evidence

Press save question

The screenshot displays the 'Form Filler' application window. The title bar reads 'Form Filler'. The menu bar includes 'View Questions', 'View Forms', 'Create questions' (which is the active tab), 'Create forms', 'Users', 'Statistics', and 'Import and Export'. The main workspace is divided into three sections: a large light gray area on the left labeled 'Photo ID', a top right area labeled 'Question Preview' containing an 'Upload File' button, and a bottom right area containing a 'Save Question' button. Below the 'Photo ID' section is a 'Create components' panel with a grid of buttons: 'Add label', 'Add text field', 'Add radio buttons', 'Add drop-down', 'Add check boxes', 'Add password field', 'Add calendar entry', and 'Add file chooser'. An arrow points from the text 'Press save question' to the 'Save Question' button.

Create components	
Add label	Add text field
Add radio buttons	Add drop-down
Add check boxes	Add password field
Add calendar entry	Add file chooser

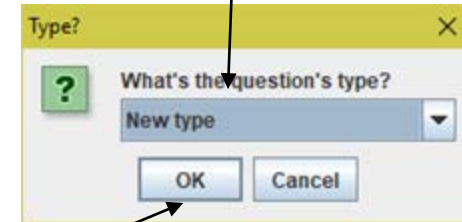
Select the
type of
question



Press ok

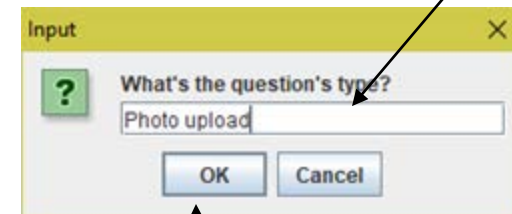
Or

Select new type



Press ok

Enter the question's type in



Press ok

Enter the
question's
difficulty

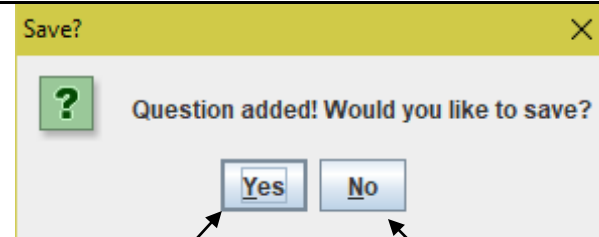
Input

What's the question's difficulty?

3

OK Cancel

Press ok



Press yes to permanently save the question to the database

Press no to only save it for this session

Question is now added to the system and can be seen in the view

The screenshot shows the 'Form Filler' application window. The 'View Questions' tab is active, displaying a list of questions with columns for 'Difficulty' and 'Type'. The list includes questions with difficulties ranging from 1 to 8 and types such as Gender, Name, Birthday, Photo upload, Country of Birth, Password, Email, and File Upload. To the right of the list are several interactive panels: 'Difficulty Sort' with a slider from 1 to 10, 'Type Sort', 'Filter by difficulty', 'Attempt Question', 'Delete Question', and 'Filter by type' which includes radio buttons for various question types like Email, Name, Phone, Gender, File Upload, Password, Calendar, Checkboxes, Dropdown, Radio Buttons, Birthday, Address, Date of Birth, and Country.

Difficulty	Type
1	Gender
1	Name
3	Birthday
3	Photo upload
4	Country of Birth
4	Password
6	Email
8	File Upload

Objectives met

1 (Ability to create questions)

Create Forms

# (from 4.1)	19
Area of System	Create Form - Add questions
Output & Storage Requirements	-
Data Used	id (of the question to add), question panel id (of the question to add), component list (of the question to add)

Evidence

Select a question in the table

Press add to form

The screenshot shows the 'Form Filler' application window. The 'Create questions' tab is active. The interface includes a 'Select a form to edit' dropdown (set to 'Creating new form: F35471340') and an 'Edit form' button. The main area is divided into three sections: 'Select Questions', 'Form preview', and a bottom 'Save Form' bar.

The 'Select Questions' section contains a table with the following data:

Difficulty	Type
1	Gender
1	Name
3	Birthday
4	Country of Birth
4	Password
6	Email
8	File Upload

Below this table is a smaller, identical table. An arrow points from the text 'Select a question in the table' to the 'Birthday' row in the top table. Another arrow points from the text 'Press add to form' to the 'Add to form' button in the bottom section of the 'Select Questions' area.

The bottom section of the 'Select Questions' area contains three buttons: 'Difficulty Sort', 'Type Sort', and 'Preview Question'. The 'Add to form' button is located at the bottom of this section.

Question is added to the form

The screenshot shows the 'Form Filler' application window. The 'Create forms' tab is active, displaying 'Creating new form: F35471340'. The interface is divided into two main sections: 'Select Questions' on the left and 'Form preview' on the right.

Select Questions:

Difficulty	Type
1	Gender
1	Name
3	Birthday
4	Country of Birth
4	Password
6	Email
8	File Upload

Below the table are four buttons: 'Difficulty Sort', 'Type Sort', 'Preview Question', and 'Add to form'.

Form preview:

The preview area shows a vertical stack of form elements. The top element is a 'Date of birth' field, which is currently expanded to show three dropdown menus: 'Day', 'Mo...', and 'Year'. To the right of these dropdowns is a 'delete' button. Below the 'Date of birth' field, there are two more empty slots, each with a small upward arrow (^) and a downward arrow (v) on its right side.

At the bottom of the window is a 'Save Form' button.

Objectives met	25 (Ability to create forms), 57 (The user should be able to create forms by selecting questions from a list)
# (from 4.1)	20
Area of System	Create forms - Add basic information
Output & Storage Requirements	formDB
Data Used	id (of the form), question panel id list, description, difficulty

Evidence

The screenshot shows the 'Form Filler' application window. The 'Create forms' tab is active, displaying a form titled 'Creating new form: F35471340'. The interface is divided into several sections:

- Select Questions:** A table with two columns: 'Difficulty' and 'Type'. The 'Birthday' row is selected.
- Form preview:** A preview of the form structure, showing a 'Date of birth' field with sub-fields for 'Day', 'Mo...', and 'Year', and a 'delete' button.
- Buttons:** 'Add to form' and 'Save Form'.

Difficulty	Type
1	Gender
1	Name
3	Birthday
4	Country of Birth
4	Password
6	Email
8	File Upload

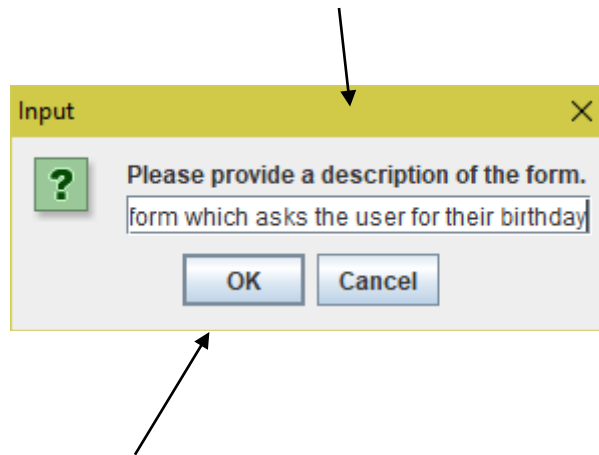
Form preview details:

- Field: Date of birth
- Sub-fields: Day, Mo..., Year
- Action: delete

Buttons: Add to form, Save Form

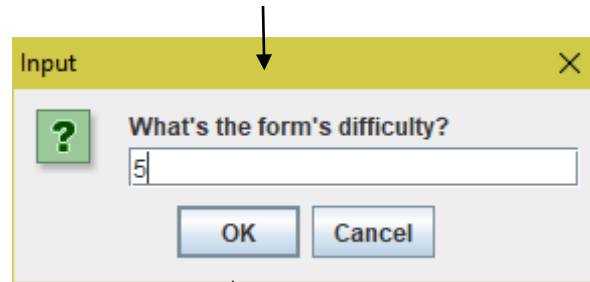
Press save
form

Enter a description of the form in the dialog box that appears



Press ok

Enter the form's difficulty in the next dialog box



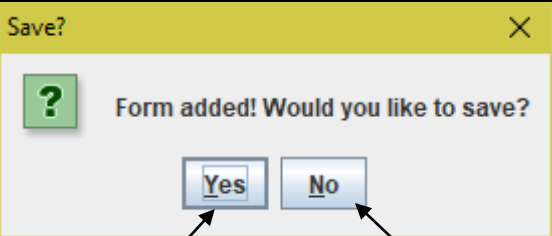
Input

What's the form's difficulty?

5

OK Cancel

Press ok

	 Press yes to permanently save the form to the database Press no to only save it for this session
--	---

Form can be seen in the view forms table

The screenshot shows the 'Form Filler' application with the 'View Forms' tab selected. The interface is divided into several sections:

- Top Menu:** View Questions, View Forms, Create questions, Create forms, Users, Statistics, Import and Export.
- Questions Table:** A table with columns: Questions, Descripti..., Main Skil..., Difficulty, Percent ...
- Difficulty Sort:** A slider control ranging from 1 to 10.
- Attempt Form:** A button to attempt a form.
- Delete Form:** A button to delete a form.
- Filter by difficulty:** A section with checkboxes for various fields: Email, Name, Phone, Gender, File UpL..., Passw..., Calendar, Checkb..., Drop-d..., Radio B..., Birthday, Address, date of ..., Country...
- Attempt form based on weaknesses:** A button to attempt a form based on weaknesses.
- Filter by type:** A section for filtering forms by type.

The 'Questions' table contains the following data:

Questions	Descripti...	Main Skil...	Difficulty	Percent ...
Q81558...	Name p...	Name.P...	2	100%
Q56280...	Gender ...	Gender....	3	100%
Q81558...	Name a...	Name.Fil...	5	0%
Q47136...	A form w...	Birthday	5	0%
Q81558...	large am...	Name.G...	6	100%

Objectives met

25 (Ability to create forms)

Edit forms

# (from 4.1)	21
Area of System	Edit Forms - Rearrange questions
Output & Storage Requirements	-
Data Used	id (of the question to move), position in form to move it to

Evidence

Select a form to edit

Press edit form

The screenshot shows the 'Form Filler' application window. The 'Create forms' tab is selected in the top menu. Below the menu, there is a dropdown menu labeled 'Select a form to edit' with the value 'F81178233'. To the right of this dropdown is an 'Edit form' button. Below the dropdown, there is a 'Select Questions' section with a table of questions and a 'Form preview' section on the right. At the bottom, there is a 'Save Form' button.

Difficulty	Type
1	Gender
1	Name
3	Birthday
4	Country of Birth
4	Password
6	Email
8	File Upload
3	Photo upload

Buttons in the interface include: View Questions, View Forms, Create questions, Create forms, Users, Statistics, Import and Export, Edit form, and Save Form.

Form is loaded to edit

Press an arrow next to a question to move it up or down

The screenshot shows the 'Form Filler' application window. The title bar is yellow and contains the text 'Form Filler' and standard window controls. Below the title bar is a menu bar with the following items: 'View Questions', 'View Forms', 'Create questions', 'Create forms', 'Users', 'Statistics', and 'Import and Export'. The 'Create forms' menu item is selected. Below the menu bar is a section titled 'Select a form to edit' with a dropdown menu showing 'F81178233' and an 'Edit form' button. The main workspace is divided into several sections. On the left, there is a 'Select Questions' section with a table of questions. The table has two columns: 'Difficulty' and 'Type'. The questions are: Gender (Difficulty 1, Type 1), Name (Difficulty 1, Type 1), Birthday (Difficulty 3, Type 3), Country of Birth (Difficulty 4, Type 4), Password (Difficulty 4, Type 4), Email (Difficulty 6, Type 6), File Upload (Difficulty 8, Type 8), and Photo upload (Difficulty 3, Type 3). Below the table are four buttons: 'Difficulty Sort', 'Type Sort', 'Preview Question', and 'Add to form'. In the center, there is a large grey area with the text 'First name', 'Password', and 'Email address'. On the right, there is a 'Form preview' section with a large white area and a vertical list of buttons: 'delete', 'V', 'A', 'delete', 'V', 'A', 'delete', 'V'. An arrow points from the text 'Form is loaded to edit' to the 'Edit form' button. Another arrow points from the text 'Press an arrow next to a question to move it up or down' to the 'V' button next to the 'Email address' question in the preview section.

Difficulty	Type
1	Gender
1	Name
3	Birthday
4	Country of Birth
4	Password
6	Email
8	File Upload
3	Photo upload

Buttons: Difficulty Sort, Type Sort, Preview Question, Add to form

Form preview buttons: delete, V, A, delete, V, A, delete, V

Form fields: First name, Password, Email address

Save Form

Question has moved 1
position upwards

The screenshot shows the 'Form Filler' application window. The title bar is yellow and contains the text 'Form Filler' and standard window controls. Below the title bar is a menu bar with the following options: 'View Questions', 'View Forms', 'Create questions', 'Create forms', 'Users', 'Statistics', and 'Import and Export'. The 'Create forms' menu is currently selected. Below the menu bar, there is a section titled 'Select a form to edit' with a dropdown menu showing 'F81178233' and an 'Edit form' button. The main area is divided into two columns. The left column is titled 'Select Questions' and contains a table of questions. The right column is titled 'Form preview' and shows a preview of the form with fields for 'First name', 'Email address', and 'Password'. Each field has a 'delete' button and a 'V' button. At the bottom of the window is a 'Save Form' button.

Difficulty	Type
1	Gender
1	Name
3	Birthday
4	Country of Birth
4	Password
6	Email
8	File Upload
3	Photo upload

Buttons in the 'Select Questions' column: Difficulty Sort, Type Sort, Preview Question, Add to form.

Form preview fields: First name, Email address, Password.

Buttons in the 'Form preview' column: delete, V (for each field).

Bottom button: Save Form

Press save form and then re-enter the description and difficulty to save

Form Filler

View Questions View Forms Create questions **Create forms** Users Statistics Import and Export

Select a form to edit: F81178233 Edit form

Select Questions

Difficulty	Type
1	Gender
1	Name
3	Birthday
4	Country of Birth
4	Password
6	Email
8	File Upload
3	Photo upload

Difficulty Sort

Type Sort

Preview Question

Add to form

Form preview

First name

Email address

Password

delete

delete

delete

Save Form

Objectives met

30 (Ability to edit forms)

# <i>(from 4.1)</i>	22
Area of System	Edit Forms - Remove Questions
Output & Storage Requirements	-
Data Used	id (of the question to delete)

Evidence

Select a form to edit

Press edit

The screenshot shows the 'Form Filler' application window. The 'Create forms' tab is active. A dropdown menu shows 'F81178233' selected. An arrow points to this dropdown with the label 'Select a form to edit'. Another arrow points to the 'Edit form' button with the label 'Press edit'. The main area is divided into 'Select Questions' and 'Form preview'. The 'Select Questions' section contains a table of questions and four buttons: 'Difficulty Sort', 'Type Sort', 'Preview Question', and 'Add to form'. The 'Form preview' section is empty. A 'Save Form' button is at the bottom.

Difficulty	Type
1	Gender
1	Name
3	Birthday
4	Country of Birth
4	Password
6	Email
8	File Upload
3	Photo upload

Buttons in 'Select Questions': Difficulty Sort, Type Sort, Preview Question, Add to form

Buttons in 'Form preview': (empty)

Bottom button: Save Form

Form is loaded to edit

Press a question's delete button to delete it

The screenshot shows the 'Form Filler' application window. The title bar is yellow and contains the text 'Form Filler' and standard window controls. Below the title bar is a menu bar with the following items: 'View Questions', 'View Forms', 'Create questions', 'Create forms', 'Users', 'Statistics', and 'Import and Export'. The 'Create forms' menu item is currently selected. Below the menu bar is a toolbar with a dropdown menu showing 'F81178233' and an 'Edit form' button. The main workspace is divided into several sections. On the left, there is a 'Select Questions' section with a table of questions. The table has two columns: 'Difficulty' and 'Type'. The questions are: Gender (Difficulty 1), Name (Difficulty 1), Birthday (Difficulty 3), Country of Birth (Difficulty 4), Password (Difficulty 4), Email (Difficulty 6), File Upload (Difficulty 8), and Photo upload (Difficulty 3). Below the table are four buttons: 'Difficulty Sort', 'Type Sort', 'Preview Question', and 'Add to form'. In the center, there is a large grey area representing the form being edited. It contains three text input fields labeled 'First name', 'Password', and 'Email address'. On the right, there is a 'Form preview' section showing a preview of the form. Below the preview, there are three 'delete' buttons, each corresponding to one of the form fields. At the bottom of the window, there is a 'Save Form' button.

Difficulty	Type
1	Gender
1	Name
3	Birthday
4	Country of Birth
4	Password
6	Email
8	File Upload
3	Photo upload

Buttons: Difficulty Sort, Type Sort, Preview Question, Add to form

Form fields: First name, Password, Email address

Form preview: (Preview of the form fields)

Buttons: delete, delete, delete

Save Form

Question has been removed from the form

The screenshot shows the 'Form Filler' application window. The title bar is yellow and contains the text 'Form Filler' and standard window controls. Below the title bar is a menu bar with the following items: 'View Questions', 'View Forms', 'Create questions', 'Create forms', 'Users', 'Statistics', and 'Import and Export'. The 'Create forms' menu item is highlighted. Below the menu bar is a toolbar with a dropdown menu showing 'F81178233' and an 'Edit form' button. The main area is divided into several sections. On the left, there is a table titled 'Select Questions' with two columns: 'Difficulty' and 'Type'. The table contains the following data:

Difficulty	Type
1	Gender
1	Name
3	Birthday
4	Country of Birth
4	Password
6	Email
8	File Upload
3	Photo upload

Below the table is a large grey rectangular area. To the right of the table is a vertical stack of four buttons: 'Difficulty Sort', 'Type Sort', 'Preview Question', and 'Add to form'. To the right of these buttons is a large grey rectangular area. To the right of this area is a 'Form preview' section. The preview section shows a form with two fields: 'First name' and 'Password'. To the right of the preview section is a vertical stack of four buttons: 'delete', 'V', 'A', and 'V'. At the bottom of the window is a 'Save Form' button.

Press save form and then re-enter the description and difficulty to save

The screenshot shows the 'Form Filler' application window. The 'Edit form' tab is active for form ID F81178233. The 'Select Questions' table lists various question types with their difficulties. The 'Form preview' section shows a form with fields for 'First name' and 'Password'. The 'Save Form' button is located at the bottom center of the window.

Difficulty	Type
1	Gender
1	Name
3	Birthday
4	Country of Birth
4	Password
6	Email
8	File Upload
3	Photo upload

Form preview fields:

- First name
- Password

Buttons: Difficulty Sort, Type Sort, Preview Question, Add to form, delete, V

Save Form

Objectives met

30 (Ability to edit forms)

Users

# (from 4.1)	23
Area of System	Users - Add
Output & Storage Requirements	UserDB
Data Used	id, username, password, first name, last name, date of birth, phone number, sessions present at (a list of the dates of the sessions that the user has attended)

Evidence

Press add new user

Username	Password	First Name	Last Name	Date of Birth	Phone Number	Sessions attended
bob123	bob	bob	bobbins	01/04/03	12345678	
mcuell0	juZ2q4Oyj	Mag	Cuell	16/05/1992	5915148609	
fmaccourt1	pOolwgp6	Forster	MacCourt	29/04/1987	2478200469	2018-07-22 2018-10-09
lruller2	dCfEQZA3jA	Leela	Ruller	19/03/1967	2386249416	
alucien3	9!ciOMPfF	Ash	Lucien	26/07/1991	5851009564	2018-07-22

First name to search for

Search

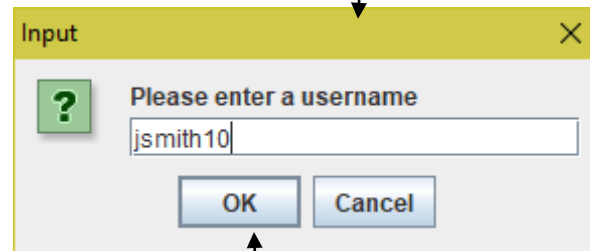
Add new user

Edit user

Delete user

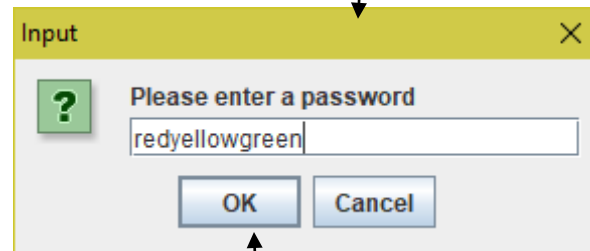
Take register

Enter a username in the
dialog that opens



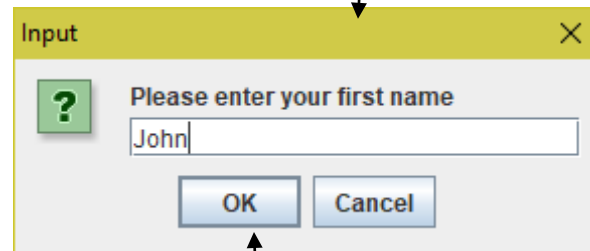
Press ok

Enter a password in the
dialog that opens



Press ok

Enter their first name in the
dialog that opens



Input

?

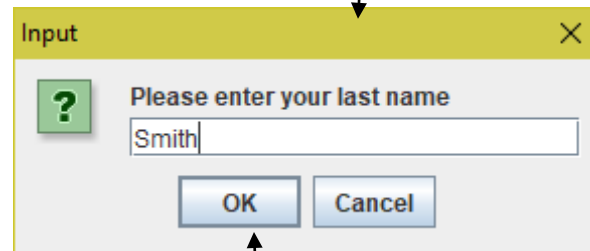
Please enter your first name

John

OK Cancel

Press ok

Enter their last name in the
dialog that opens



Input

?

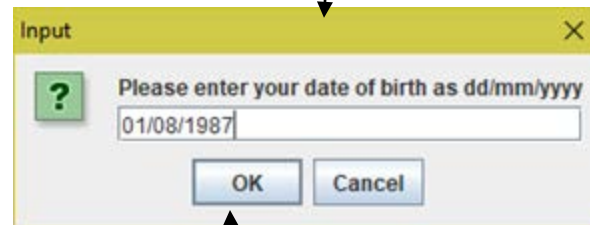
Please enter your last name

Smith

OK Cancel

Press ok

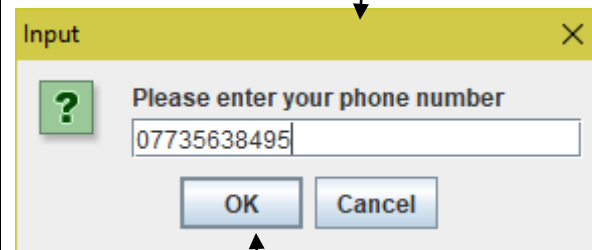
Enter their date of birth
in the dialog that opens



The image shows a screenshot of a software dialog box titled "Input" with a yellow header bar and a close button (X) in the top right corner. Inside the dialog, there is a green square icon with a white question mark. To the right of the icon, the text "Please enter your date of birth as dd/mm/yyyy" is displayed. Below this text is a text input field containing the date "01/08/1987". At the bottom of the dialog, there are two buttons: "OK" and "Cancel".

Press ok

Enter their phone number in
the dialog that opens



Input

?

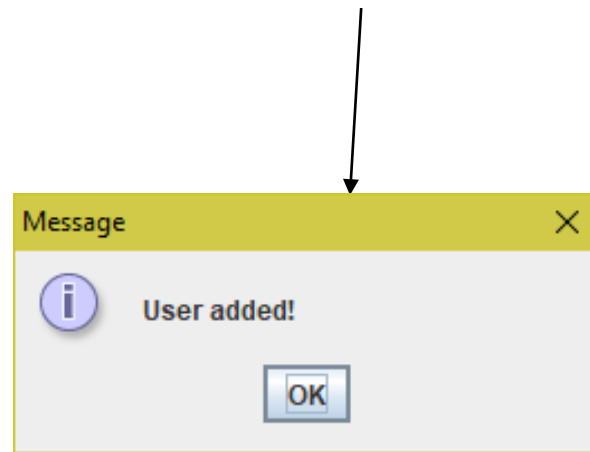
Please enter your phone number

07735638495

OK Cancel

Press ok

User added dialog appears



Press ok

The new user can be seen
in the user table

Form Filler

View Questions View Forms Create questions Create forms **Users** Statistics Import and Export

Username	Password	First Name	Last Name	Date of Birth	Phone Number	Sessions attended
uoccaroy	uocar azay	Orasma	De Caro	09/01/1950	811119922	
nbiaggiottiz	qSd40pYq3a	Nikolella	Biaggiotti	17/08/2005	4841217883	2018-07-22
gjaycox10	lNnctbRAF5	Garner	Jaycox	03/01/2007	8619603372	
agobolos11	LC0s18qssSAg	Abbe	Gobolos	24/07/1989	3724626831	
efenty12	uQ7HfYv6h5	Eliza	Fenty	13/04/1978	2134951916	
jsmith10	redyellowgreen	John	Smith	01/08/1987	07735638495	

First name to search for

Search

Add new user

Edit user

Delete user

jsmith10	redyellowgreen	John	Smith	01/08/1987	07735638495	
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Take register

Objectives met	38 (Ability to add users), 44 (First name, last name, Date of birth and phone number should be stored for each user)
# (from 4.1)	24
Area of System	Users - Edit
Output & Storage Requirements	UserDB
Data Used	id, username, password, first name, last name, date of birth, phone number, sessions present at (a list of the dates of the sessions that the user has attended)

Evidence

Select a user

Press edit

Username	Password	First Name	Last Name	Date of Birth	Phone Number	Sessions attended
uacanoj	uZar szarv	Orszida	De Cano	00/00/1900	4111144223	
nbiaggiottiz	qSdA0pYq3a	Nikolletta	Biaggiotti	17/08/2005	4841217883	2018-07-22
gjaycox10	lNnctbRAF5	Garner	Jaycox	03/01/2007	8619603372	
agobolos11	LC0sl8qssSAg	Abbe	Gobolos	24/07/1989	3724626831	
efenty12	uQ7HFYv6h5	Eliza	Fenty	13/04/1978	2134951916	
jsmith10	redyellowgreen	John	Smith	01/08/1987	07735638495	

First name to search for

Search

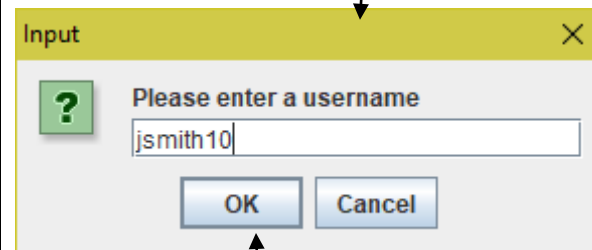
Add new user

Edit user

Delete user

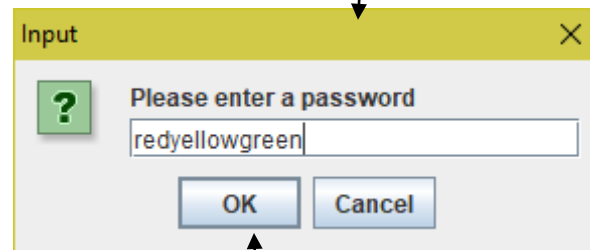
Take register

Edit their username in the
dialog that opens



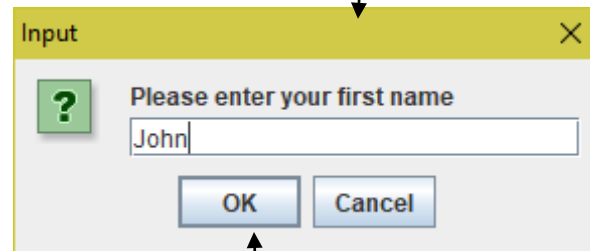
Press ok

Edit their password in the
dialog that opens



Press ok

Edit their first name in the
dialog that opens



Input

?

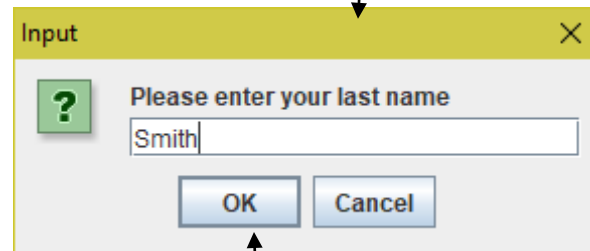
Please enter your first name

John

OK Cancel

Press ok

Edit their last name in the
dialog that opens



Input

?

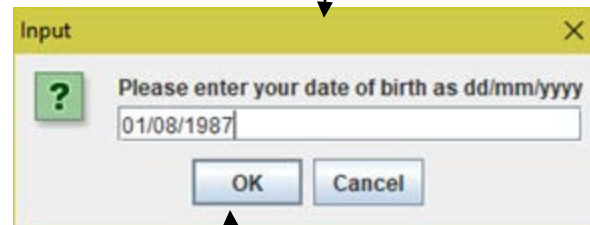
Please enter your last name

Smith

OK Cancel

Press ok

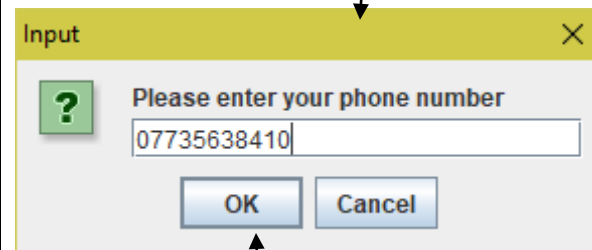
Edit their date of birth in
the dialog that opens



The image shows a screenshot of a software dialog box titled "Input" with a yellow header bar and a close button (X) in the top right corner. Inside the dialog, there is a green square icon with a white question mark on the left. To its right, the text "Please enter your date of birth as dd/mm/yyyy" is displayed. Below this text is a text input field containing the date "01/08/1987". At the bottom of the dialog, there are two buttons: "OK" and "Cancel".

Press ok

Edit their phone number in
the dialog that opens



Input

?

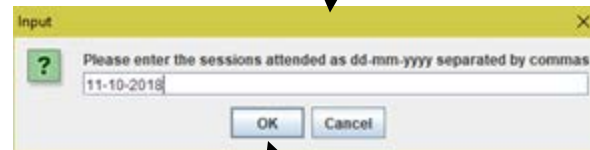
Please enter your phone number

07735638410

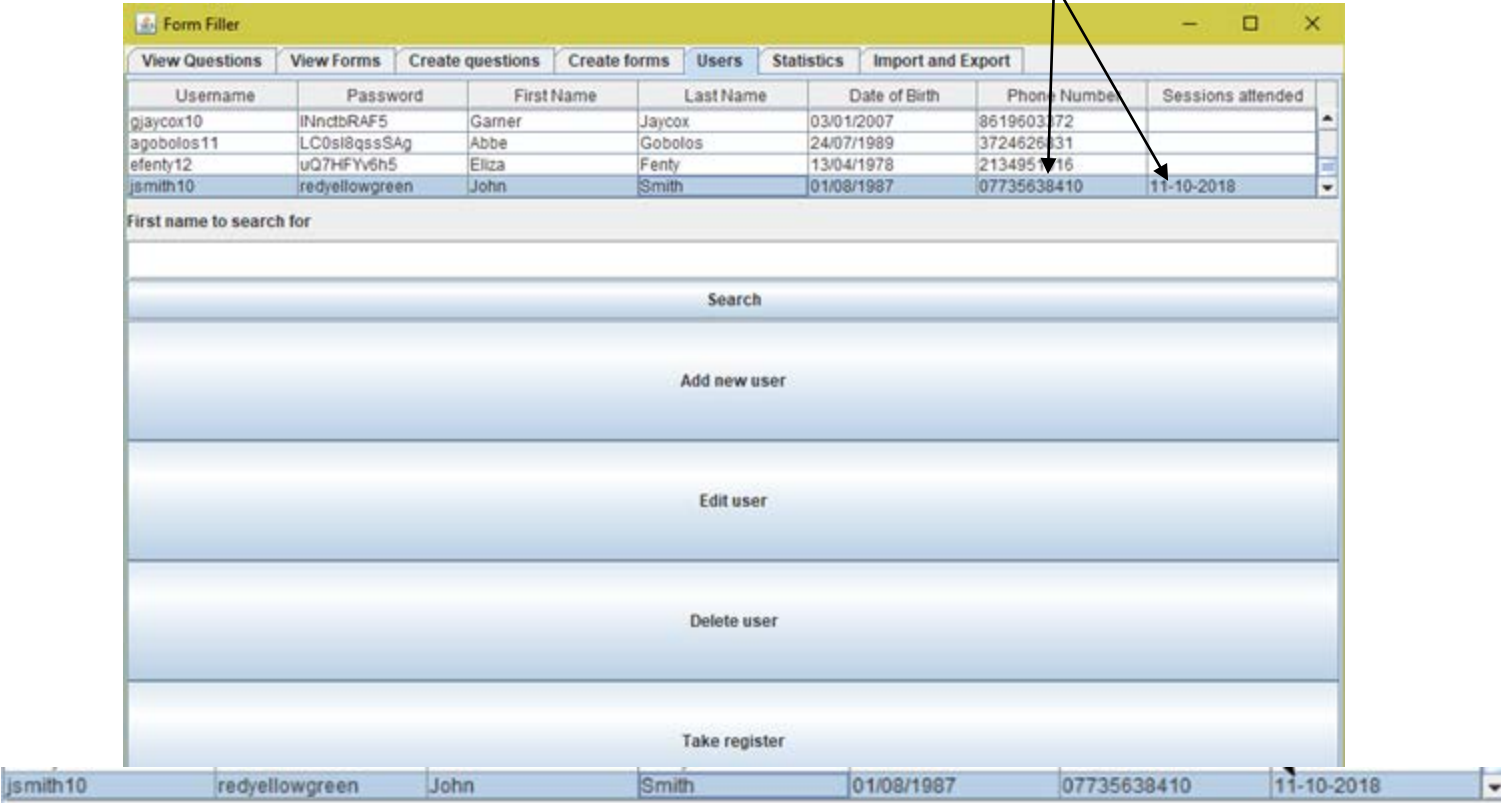
OK Cancel

Press ok

Edit the list of sessions that they have been present at in the dialog that appears



Press ok

	Changes have been made to the user  <table><thead><tr><th>Username</th><th>Password</th><th>First Name</th><th>Last Name</th><th>Date of Birth</th><th>Phone Number</th><th>Sessions attended</th></tr></thead><tbody><tr><td>gjaycox10</td><td>lNndtbrAF5</td><td>Garner</td><td>Jaycox</td><td>03/01/2007</td><td>8619603172</td><td></td></tr><tr><td>agobolos11</td><td>LC0sl8qssSAg</td><td>Abbe</td><td>Gobolos</td><td>24/07/1989</td><td>3724626131</td><td></td></tr><tr><td>efenty12</td><td>uQ7HfYv6h5</td><td>Eliza</td><td>Fenty</td><td>13/04/1978</td><td>213495116</td><td></td></tr><tr><td>jsmith10</td><td>redyellowgreen</td><td>John</td><td>Smith</td><td>01/08/1987</td><td>07735638410</td><td>11-10-2018</td></tr></tbody></table> First name to search for Search Add new user Edit user Delete user Take register <table><tr><td>jsmith10</td><td>redyellowgreen</td><td>John</td><td>Smith</td><td>01/08/1987</td><td>07735638410</td><td>11-10-2018</td></tr></table>	Username	Password	First Name	Last Name	Date of Birth	Phone Number	Sessions attended	gjaycox10	lNndtbrAF5	Garner	Jaycox	03/01/2007	8619603172		agobolos11	LC0sl8qssSAg	Abbe	Gobolos	24/07/1989	3724626131		efenty12	uQ7HfYv6h5	Eliza	Fenty	13/04/1978	213495116		jsmith10	redyellowgreen	John	Smith	01/08/1987	07735638410	11-10-2018	jsmith10	redyellowgreen	John	Smith	01/08/1987	07735638410	11-10-2018
Username	Password	First Name	Last Name	Date of Birth	Phone Number	Sessions attended																																					
gjaycox10	lNndtbrAF5	Garner	Jaycox	03/01/2007	8619603172																																						
agobolos11	LC0sl8qssSAg	Abbe	Gobolos	24/07/1989	3724626131																																						
efenty12	uQ7HfYv6h5	Eliza	Fenty	13/04/1978	213495116																																						
jsmith10	redyellowgreen	John	Smith	01/08/1987	07735638410	11-10-2018																																					
jsmith10	redyellowgreen	John	Smith	01/08/1987	07735638410	11-10-2018																																					
Objectives met	39 (Ability to edit users), 44 (First name, last name, Date of birth and phone number should be stored for each user), 62 (The admin user should be able to edit information about each user)																																										
# (from 4.1)	25																																										

Area of System	Users - View
Output & Storage Requirements	-
Data Used	UserList, for each user: [id, username, password, first name, last name, date of birth, phone number, sessions present at (a list of the dates of the sessions that the user has attended)]

Evidence

All of the users in the system can be seen in this table along with the data stored about them

The table can be scrolled to view more users

Username	Password	First Name	Last Name	Date of Birth	Phone Number	Sessions attended
bob123	bob	bob	bobbins	01/04/03	12345678	
mcuelli0	jvZ2q4Oyj	Mag	Cuell	16/05/1992	5915148609	
fmaccourt1	pOolwgp6	Forster	MacCourt	29/04/1987	2478200469	2018-07-22.2018-10...
lruller2	dCFEQZA3jA	Leela	Ruller	19/03/1967	2386249416	

First name to search for

Search

Add new user

Edit user

Delete user

Take register

Objectives met

60 (The admin user should be able to view information about each user)

# (from 4.1)	26
Area of System	User - Search
Output & Storage Requirements	-
Data Used	first name to search for, first name (of each user in the system)

Evidence

Enter a first name in the search box

Press search

Form Filler

View Questions View Forms Create questions Create forms Users Statistics Import and Export

Username	Password	First Name	Last Name	Date of Birth	Phone Number	Sessions attended
bob123	bob	bob	bobbins	01/04/93	12345678	
mcuelli0	juZ2q4Oyj	Mag	Cuelli	16/05/1992	5915148609	
fmaccourt1	pOolwgp6	Forster	MacCourt	29/04/1987	2478200469	2018-07-22 2018-10...
lruller2	dCfEQZA3jA	Leela	Ruller	19/03/1967	2386249416	

First name to search for

john

Search

Add new user

Edit user

Delete user

Take register

Only users with that first name are displayed

Username	Password	First Name	Last Name	Date of Birth	Phone Number	Sessions attended
jsmith10	redyellowgreen	John	Smith	01/08/1987	07735638410	11-10-2018

First name to search for

John

Search

Add new user

Edit user

Delete user

Username	Password	First Name	Last Name	Date of Birth	Phone Number	Sessions attended
jsmith10	redyellowgreen	John	Smith	01/08/1987	07735638410	11-10-2018

Take register

Objectives met	41 (Ability to search through users)
# (from 4.1)	27
Area of System	Users - Delete
Output & Storage	UserDB

Requirements	
Data Used	id (of user to delete)

Evidence

Select user in table to delete

Press delete user

Form Filler

View Questions View Forms Create questions Create forms Users Statistics Import and Export

Username	Password	First Name	Last Name	Date of Birth	Phone Number	Sessions attended
gjaycox10	lNnctbRAF5	Garner	Jaycox	03/01/2007	8619603372	
agobolos11	LC0sl8qssSAg	Abbe	Gobolos	24/07/1989	3724626831	
efenty12	uQ7HFYv6h5	Eliza	Fenty	13/04/1978	2124951916	
jsmith10	redyellowgreen	John	Smith	01/08/1987	07735638410	11-10-2018

First name to search for

Search

Add new user

Edit user

Delete user

Take register

User is no longer present in the table (i.e. it has been deleted from the system)

Form Filler

View Questions View Forms Create questions Create forms **Users** Statistics Import and Export

Username	Password	First Name	Last Name	Date of Birth	Phone Number	Sessions attended
gobololos	uZar 5Zar6	Grasima	De Cano	06/10/1950	4111194225	
nbiaggiottiz	qSdA0pYq3a	Nikoletta	Biaggiotti	17/08/2005	4841217883	2018-07-22
gjaycox10	INnctbRAF5	Garner	Jaycox	03/01/2007	8619603372	
agobolos11	LC0sl8qssSAg	Abbe	Gobolos	24/07/1989	3724626831	
efenty12	uQ7HFYv6h5	Eliza	Fenty	13/04/1978	2134951916	

First name to search for

Search

Add new user

Edit user

Delete user

Take register

Objectives met	40 (Ability to delete users)
# (from 4.1)	28
Area of System	Users - Take register
Output & Storage Requirements	UserDB
Data Used	current date, UserList, Sessions present at array (for each user) (a list containing the dates of the sessions that the user attended)

Evidence

Username	Password	First Name	Last Name	Date of Birth	Phone Number	Sessions attended
udecanoy	uZarSzaro	Drosima	De Caro	00/00/1900	4111199229	2018-07-22
nbiaggiottiz	qSdA0pYq3a	Nikoletta	Biaggiotti	17/08/2005	4841217883	
gjaycox10	lInctbRAF5	Garner	Jaycox	03/01/2007	8619603372	
agobolos11	LC0sl8qssSAg	Abbe	Gobolos	24/07/1989	3724626831	
efenty12	uQ7HFYv6h5	Eliza	Fenty	13/04/1978	2134951916	

First name to search for

Search

Add new user

Edit user

Delete user

Take register

Press take register

Check the checkboxes of the users that are present in the register that opens



Press save register

The users selected have had today's date added to their sessions attended array (i.e. they have been marked present for today)

Username	Password	First Name	Last Name	Date of Birth	Phone Number	Sessions attended
ddecarloy	oZaP92af0	Drusilla	De Carlo	08/10/1950	4111144229	
nbiaggiottiz	qSdA0pYq3a	Nikolelta	Biaggiotti	17/08/2005	4841217883	2018-07-22
gjaycox10	lNnctbRAF5	Garner	Jaycox	03/01/2007	8619603372	2018-10-11
agobolos11	LC0sl8qssSAg	Abbe	Gobolos	24/07/1989	3724626831	2018-10-11
efenty12	uQ7HFYv6h5	Eliza	Fenty	13/04/1978	2134951916	2018-10-11

First name to search for

Search

Add new user

Edit user

Delete user

Take register

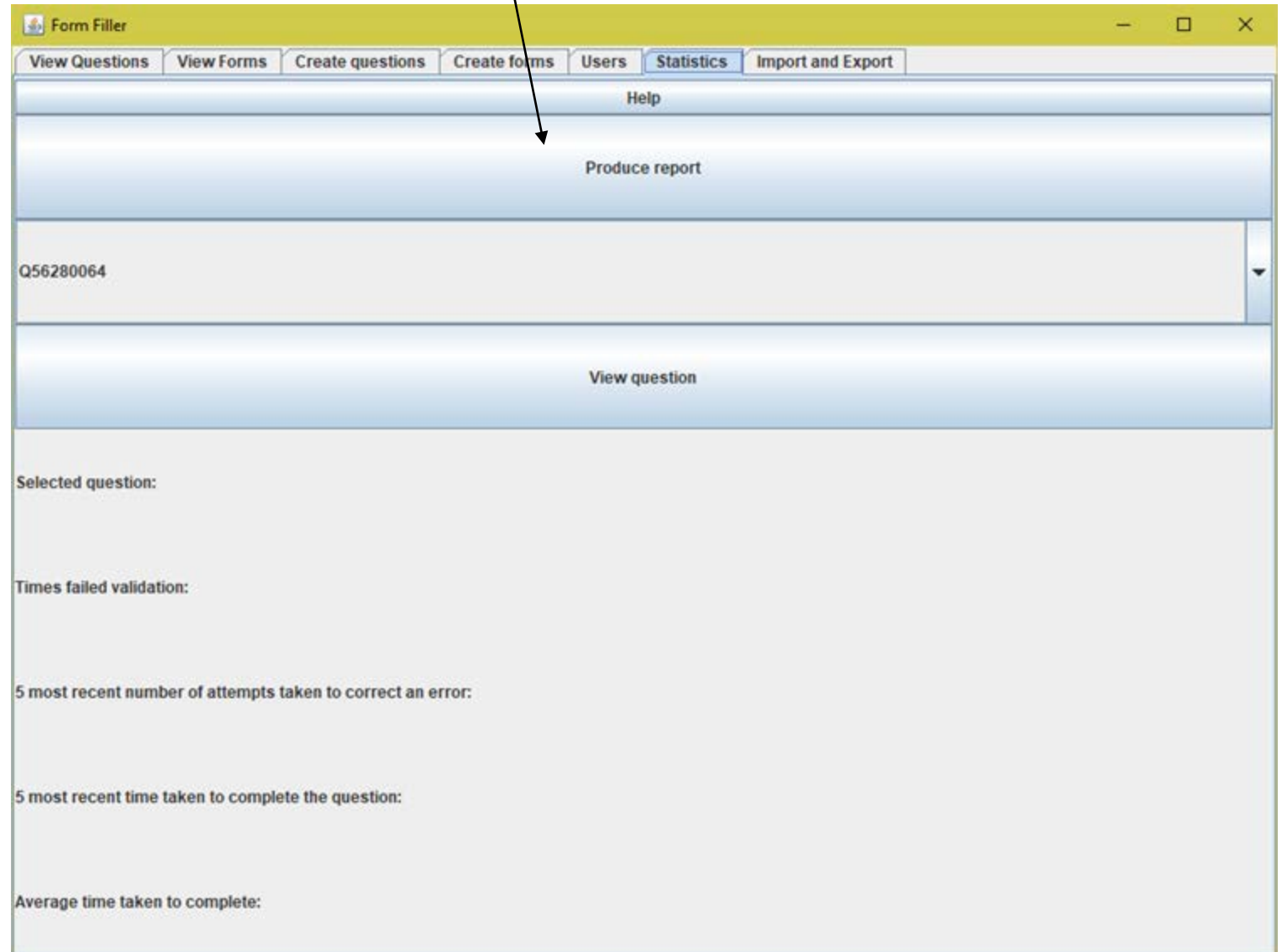
Objectives met	63 (The admin user should be able to take a register)
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Statistics

# (from 4.1)	30
Area of System	Statistics - Produce report
Output & Storage Requirements	Report (printable)
Data Used	user id (id of the user to produce a report of), question stat list (belonging to the user), time taken to complete array (for each question), number of attempts taken to correct an error array (for each question), total number of times failed validation (for each question)

Evidence

Press produce report



The screenshot shows the 'Form Filler' application window. The title bar is yellow and contains the text 'Form Filler' and standard window controls. Below the title bar is a menu bar with the following items: 'View Questions', 'View Forms', 'Create questions', 'Create forms', 'Users', 'Statistics' (which is highlighted), and 'Import and Export'. Below the menu bar is a toolbar with three buttons: 'Help', 'Produce report' (which is highlighted with a blue gradient), and 'View question'. Below the toolbar is a list box containing the text 'Q56280064'. Below the list box is a button labeled 'View question'. Below the button is a large text area containing the following text: 'Selected question:', 'Times failed validation:', '5 most recent number of attempts taken to correct an error:', '5 most recent time taken to complete the question:', and 'Average time taken to complete:'.

Form Filler

View Questions View Forms Create questions Create forms Users Statistics Import and Export

Help

Produce report

Q56280064

View question

Selected question:

Times failed validation:

5 most recent number of attempts taken to correct an error:

5 most recent time taken to complete the question:

Average time taken to complete:

Report opens in new window



The screenshot shows a web browser window with a yellow title bar containing the text 'bob123's report' and standard window controls. The main content area displays a table with the following data:

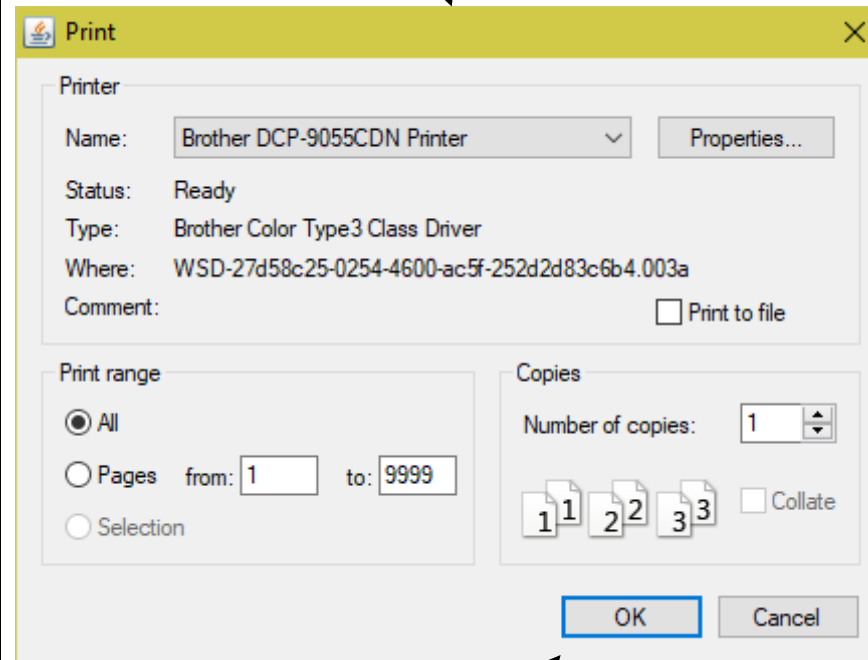
Question type	Number of times failed validation	Average number of attempts needed to correct	Average time to complete
Email	0	0	5
Name	0	0	8
Phone			
Gender			
File Upload	2	2	9
Password	2	2	1
Calendar			
Checkboxes			
Drop-down			
Radio Buttons			
Birthday	1	1	1
Address			
date of birth			
Country of Birth	2	2	6
Photo upload			

Below the table is a large, empty light gray rectangular area. At the bottom of the window is a blue button labeled 'Print'.

Press
print



Use system
dialog to print
the report



Press ok to
print

How the report will look when
printed



bob123's report

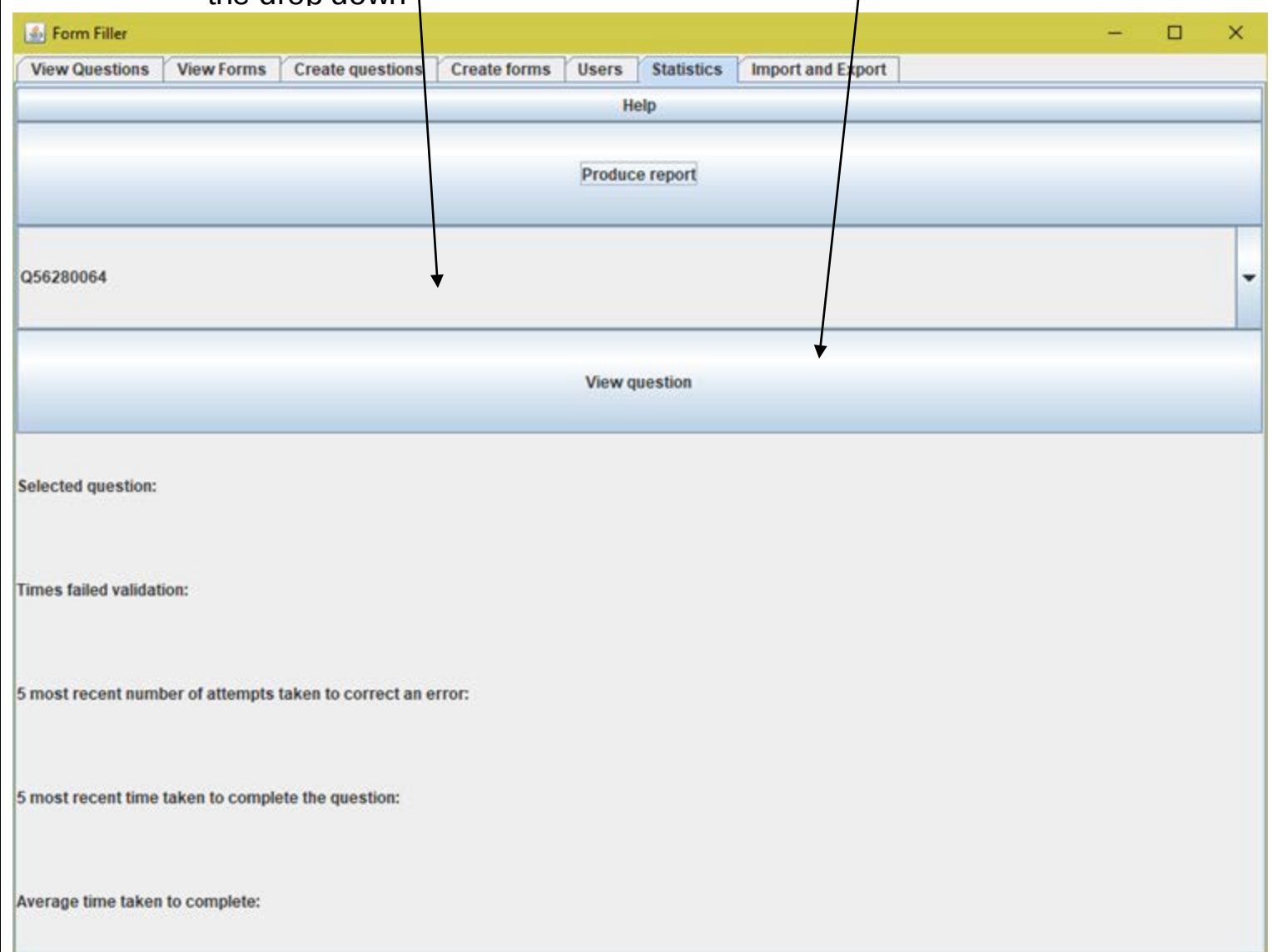
Question type	Number of times failed validation	Average number of attempts needed to correct	Average time to complete
Email			
Name	6	3	34
Phone			
Gender	2	2	3
File Upload			
Password			
Calendar			
Checkboxes			
Drop-down	1	1	15
Radio Buttons			
Birthday	1	1	1
Address	2	2	4

Objectives met	42 (Ability to produce a report on the user's weak areas/progress)
# (from 4.1)	32
Area of System	Statistics - View stats for individual question
Output & Storage Requirements	-
Data Used	id (of the question to view the statistics for), number of times failed validation, 5 most recent number of attempts to correct an error, 5 most recent times taken to complete the question, average time taken to complete

Evidence

Select a question using the drop down

Press view question



The screenshot shows the 'Form Filler' application window. The 'View Questions' tab is selected. A list of questions is displayed, with 'Q56280064' selected. A 'View question' button is visible below the list. The interface includes a menu bar with options: View Questions, View Forms, Create questions, Create forms, Users, Statistics, and Import and Export. A 'Produce report' button is also present.

Form Filler

View Questions View Forms Create questions Create forms Users Statistics Import and Export

Help

Produce report

Q56280064

View question

Selected question:

Times failed validation:

5 most recent number of attempts taken to correct an error:

5 most recent time taken to complete the question:

Average time taken to complete:

The statistics are for the selected question are displayed

The screenshot shows the 'Form Filler' application window. The 'Statistics' tab is selected in the top menu. Below the menu, there are buttons for 'Help', 'Produce report', and 'View question'. The 'View question' button is highlighted. Below this, the statistics for the selected question 'Q56280064' are displayed:

Selected question: Q56280064

Times failed validation: 2

5 most recent number of attempts taken to correct an error: 2,

5 most recent time taken to complete the question: 0:26,0:04,

Average time taken to complete: 0:15

Objectives met	-
-----------------------	---

Import

# (from 4.1)	34
Area of System	Import - Import questions
Output & Storage Requirements	questionDB, questionPanelDB
Data Used	id, difficulty, type, component list (list of components in the question)

Evidence

Press import question

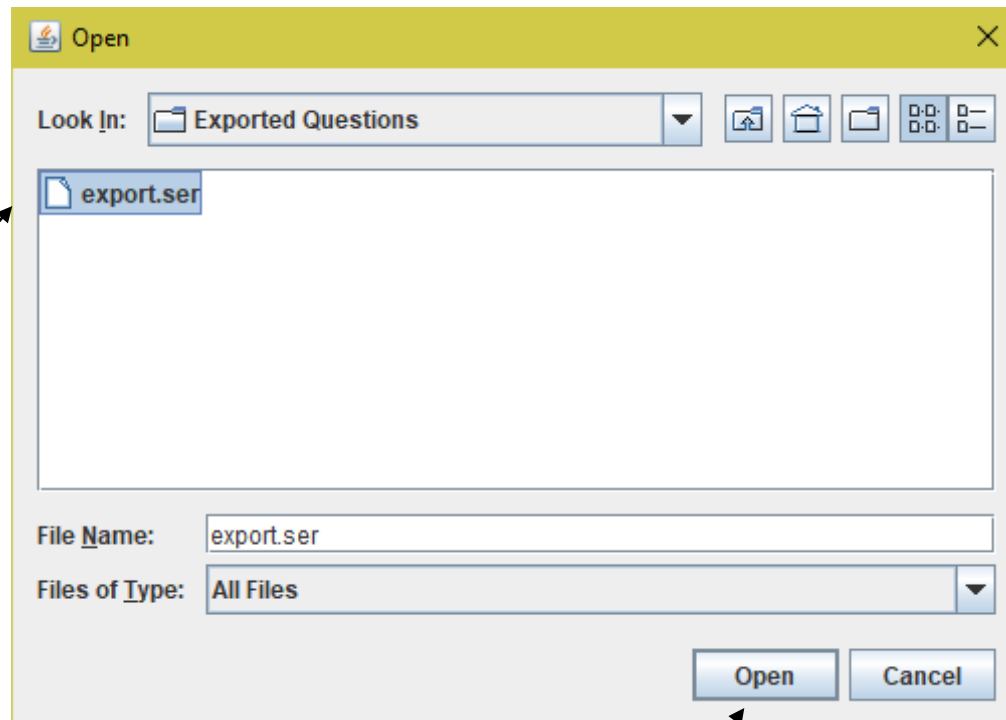
The screenshot shows the 'Form Filler' application window. The 'Import and Export' tab is active. On the left, there is a list of question types with their respective difficulties. The main area contains several buttons for managing questions and forms. An arrow points to the 'Import question' button.

Difficulty	Type
1	Gender
1	Name
3	Birthday
4	Country of Birth
4	Password
6	Email
8	File Upload

Questions	Description	Main Skill...	Difficulty
Q815587...	large am...	Name Ge...	6
Q815587...	Name an...	Name Fil...	5
Q562800...	Gender a...	Gender.C...	3
Q815587...	Name pa...	Name Pa...	2

Navigate to an exported question in
the file chooser that opens

Select the
question



Press open

Imported question

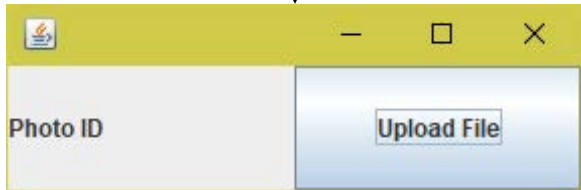
The screenshot shows the 'Form Filler' application with the 'Import and Export' tab selected. The interface is divided into several sections:

- Question Types Table:** A table with two columns: 'Difficulty' and 'Type'. It lists various question types and their associated difficulty levels.
- Imported Questions Table:** A table with four columns: 'Questions', 'Description', 'Main Skill', and 'Difficulty'. It lists questions that have been imported into the system.
- Action Buttons:** A series of buttons for managing the question types and imported questions, including 'Difficulty Sort', 'Type Sort', 'Preview Question', 'Export', 'Import question', 'Import form', 'Preview Form', and 'Export'.

A vertical arrow points from the 'Imported question' header to the 'Photo upload' question type in the 'Question Types' table. A horizontal arrow points from the 'Photo upload' question type to the 'Import question' button.

Difficulty	Type
3	Birthday
4	Country of Birth
6	Email
8	File Upload
1	Gender
1	Name
3	Photo upload
4	Password

Questions	Description	Main Skill	Difficulty
Q815587...	large am...	Name Ge...	6
Q815587...	Name an...	Name Fil...	5
Q562800...	Gender a...	Gender C...	3
Q815587...	Name pa...	Name Pa...	2

	The imported question 
Objectives met	22 (Ability to import a Question from file)
# (from 4.1)	35
Area of System	Import - Import Forms
Output & Storage Requirements	formDB
Data Used	id (of the form), question panel id list, description, difficulty

Evidence

Press import form

The screenshot shows the 'Form Filler' application window with the 'Import and Export' tab selected. The interface is divided into several sections:

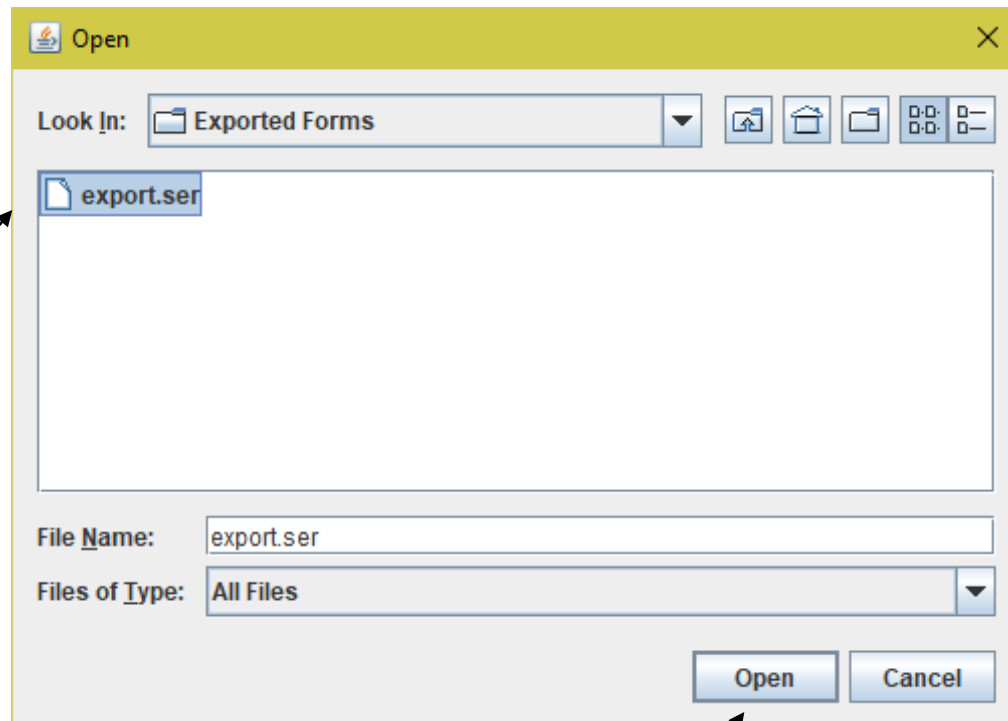
- Top Menu Bar:** View Questions, View Forms, Create questions, Create forms, Users, Statistics, Import and Export.
- Left Panel (Questions List):**

Difficulty	Type
1	Gender
1	Name
3	Birthday
4	Country of Birth
4	Password
6	Email
8	File Upload
3	Photo upload
- Right Panel (Import/Export Options):**
 - Import question:** A large blue area for importing individual questions.
 - Import form:** A large blue area for importing entire forms.
- Bottom Panel (Preview/Export Options):**
 - Preview Question:** A button to preview a selected question.
 - Export:** A button to export a selected question.
 - Preview Form:** A button to preview a selected form.
 - Export:** A button to export a selected form.

An arrow points from the text 'Press import form' to the 'Import form' button in the bottom right section.

Navigate to an exported form in the
file chooser that opens

Select the
form



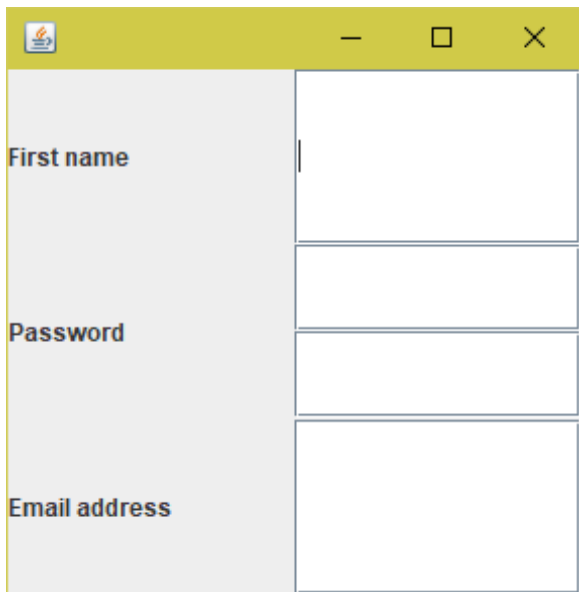
Press open

Imported form can be
seen the form table

The screenshot shows the 'Form Filler' application window. The 'Import and Export' tab is selected in the menu bar. The left sidebar contains a table of question types. The central panel has buttons for 'Difficulty Sort', 'Type Sort', 'Preview Question', and 'Export'. The right panel has 'Import question' and 'Import form' options. An arrow points from the text 'Imported form can be seen the form table' to the 'Import form' button.

Difficulty	Type
1	Gender
1	Name
3	Birthday
4	Country of Birth
4	Password
6	Email
8	File Upload
3	Photo upload

Questi...	Descripti...	Main ...	Difficu...
Q8155...	Name p...	Name...	2
Q5628...	Gender ...	Gend...	3
Q8155...	Name a...	Name...	5
Q8155...	large am...	Name...	6

	The imported form  The screenshot shows a web browser window with a yellow title bar. The page has a light gray background. On the left, there are three labels: 'First name', 'Password', and 'Email address'. To the right of each label is a white input field. The 'First name' field is a single line, while the 'Password' and 'Email address' fields are multi-line text areas. The window has standard minimize, maximize, and close buttons in the title bar.
Objectives met	33 (Ability to import forms from file)

Export

# (from 4.1)	36
Area of System	Export - Export Questions
Output & Storage	Exported question file

Requirements	
Data Used	id, difficulty, type, component list (list of components in the question)

Evidence

Select a question

Press export

The screenshot shows the 'Form Filler' application window. The 'Import and Export' tab is selected in the menu bar. The interface is divided into several sections:

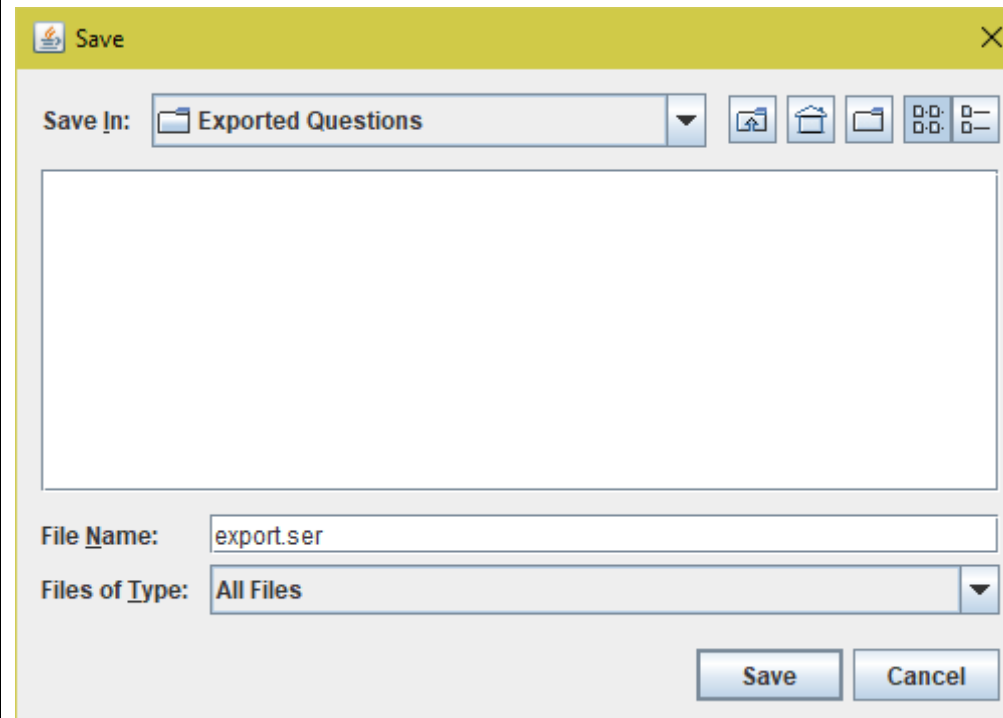
- Question List:** A table with columns 'Difficulty' and 'Type'. The first row is selected.
- Question Details:** A table with columns 'Questi...', 'Descripti...', 'Main ...', and 'Difficu...'. The first row is selected.
- Action Buttons:** Buttons for 'Difficulty Sort', 'Type Sort', 'Preview Question', 'Export', 'Import question', and 'Import form'.

The 'Export' button is highlighted with a red arrow pointing to it from the text 'Press export'.

Difficulty	Type
1	Gender
1	Name
3	Birthday
4	Country of Birth
4	Password
6	Email
8	File Upload
3	Photo upload

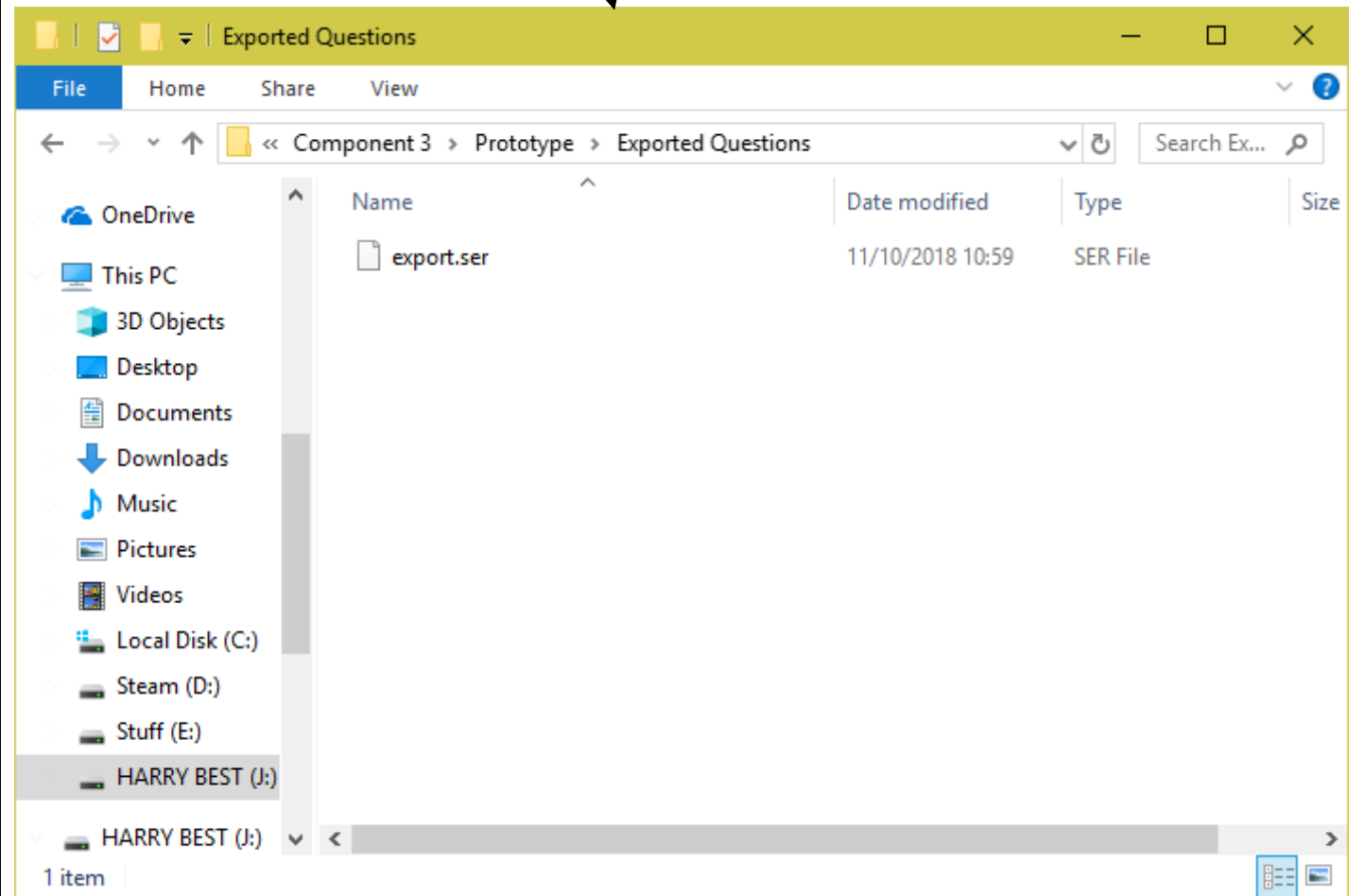
Questi...	Descripti...	Main ...	Difficu...
Q8155...	Name p...	Name...	2
Q5628...	Gender ...	Gend...	3
Q8155...	Name a...	Name...	5
Q8155...	large am...	Name...	6

Use the file chooser that opens to
navigate to a location to export the
question to



Press save

Question has
been exported



Objectives met	21 (Ability to export a question to file)
# <i>(from 4.1)</i>	37
Area of System	Export - Export Forms
Output & Storage Requirements	Exported Form file
Data Used	id (of the form), question panel id list, description, difficulty

Evidence

The screenshot shows the 'Form Filler' application window with the 'Import and Export' tab selected. The interface is divided into several sections:

- Top Menu Bar:** View Questions, View Forms, Create questions, Create forms, Users, Statistics, Import and Export.
- Left Panel (Form List):** A table with columns 'Difficulty' and 'Type'. It lists various form types and their associated difficulty levels.
- Right Panel (Import/Export Options):** A large area with buttons for 'Import question', 'Import form', 'Difficulty Sort', 'Type Sort', 'Preview Question', 'Export', 'Preview Form', and 'Export'.

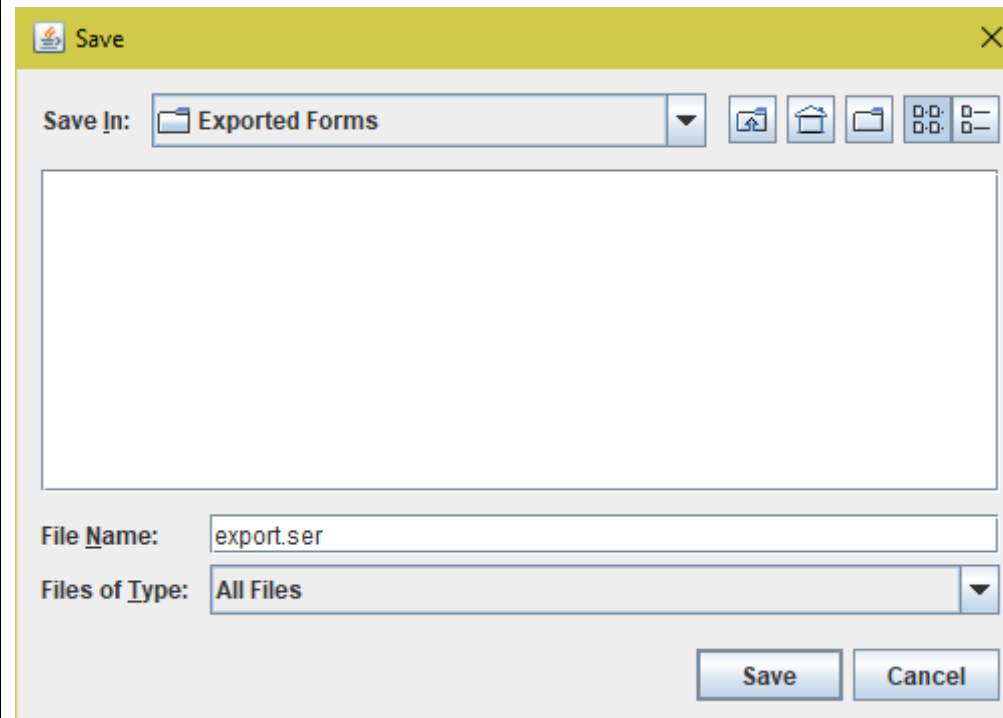
Below the screenshot, two arrows point to specific elements:

- An arrow points to the 'Export' button in the bottom right of the 'Import and Export' tab.
- An arrow points to the 'Export' button in the bottom right of the 'Import and Export' tab.

Select a form to
export

Press
export

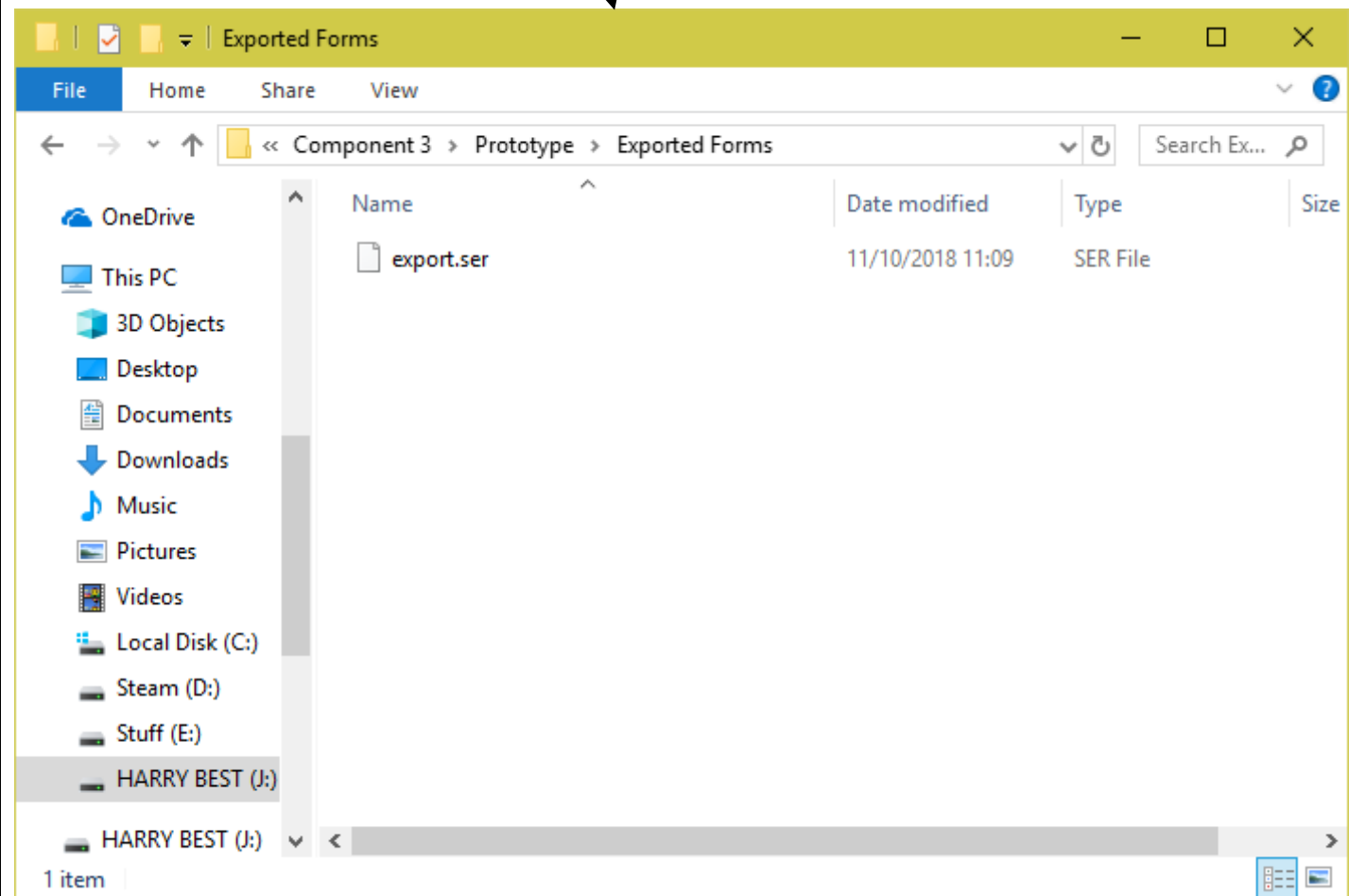
Use the file chooser that opens to
navigate to a location to export the
form to



Press save



Form has been
exported



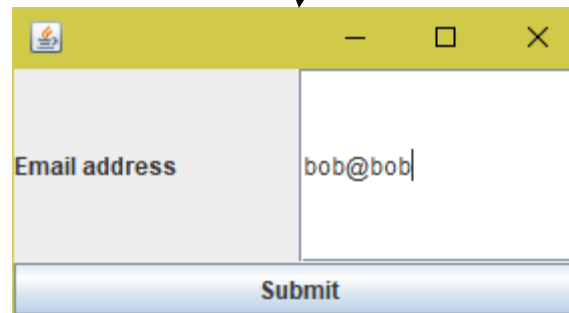
Objectives met	32 (Ability to export forms to file)
-----------------------	--------------------------------------

Component Validation

# (from 4.1)	38
Area of System	Component validation - Email address
Output & Storage Requirements	-
Data Used	Text (entered into the field)

Evidence

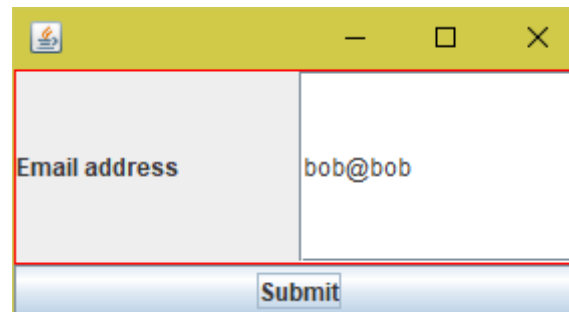
Invalid email entered



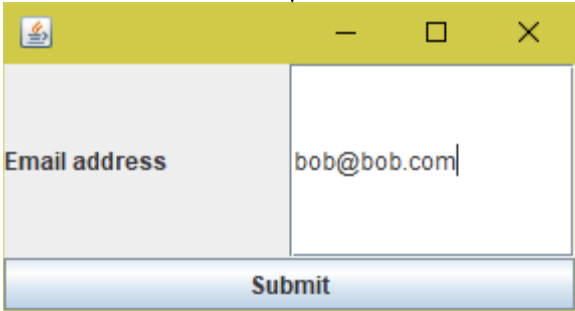
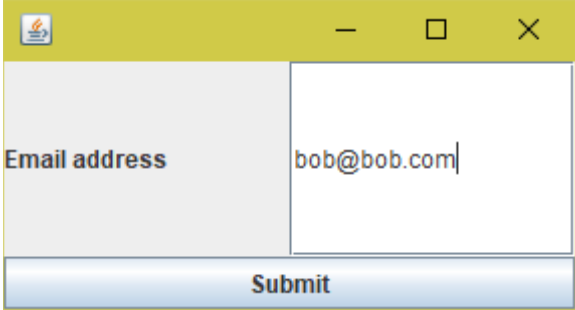
A screenshot of a web form window with a yellow title bar. The form contains a label "Email address" on the left and a text input field on the right containing the text "bob@bob". Below the input field is a blue "Submit" button. An arrow points from the text "Invalid email entered" to the input field.

Press submit

Red border indicating
it failed the validation
check



A screenshot of the same web form window as above, but with a red border around the entire form area, indicating a validation failure. The text "Email address" and "bob@bob" are still present, along with the "Submit" button. An arrow points from the text "Red border indicating it failed the validation check" to the red border.

	Valid email entered  Press submit No red border indicating it passed the validation check 
Objectives met	16 (Email addresses, dates, and phone numbers should have format checks)

# <i>(from 4.1)</i>	39
Area of System	Component Validation - Date
Output & Storage Requirements	-
Data Used	Date entered

Evidence

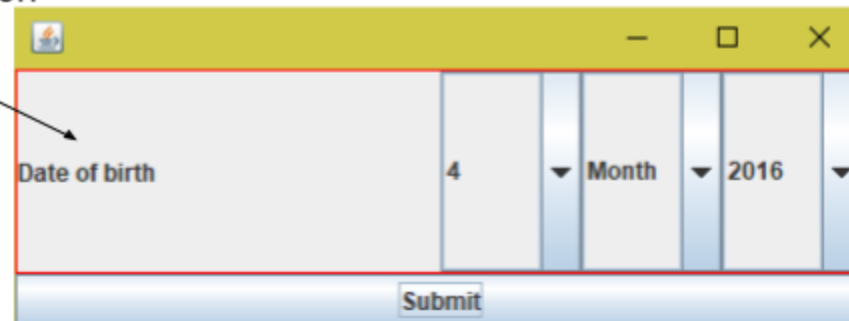
Invalid date entered



A screenshot of a web form titled "Date of birth" with a yellow header bar. The form contains three dropdown menus: the first shows "4", the second shows "Month", and the third shows "2016". Below the dropdowns is a "Submit" button. An arrow points from the text "Invalid date entered" to the first dropdown menu.

Press submit

Red border indicating
it failed the validation
check

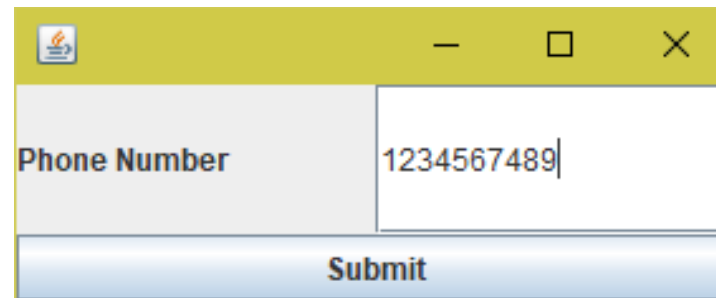


A screenshot of the same web form as above, but with a red border around the entire form area, indicating a validation failure. The "Date of birth" label, dropdowns, and "Submit" button are visible. An arrow points from the text "Red border indicating it failed the validation check" to the red border.

Objectives met	16 (Email addresses, dates, and phone numbers should have format checks)
# (from 4.1)	40
Area of System	Component Validation - phone number
Output & Storage Requirements	-
Data Used	Phone number entered

Evidence

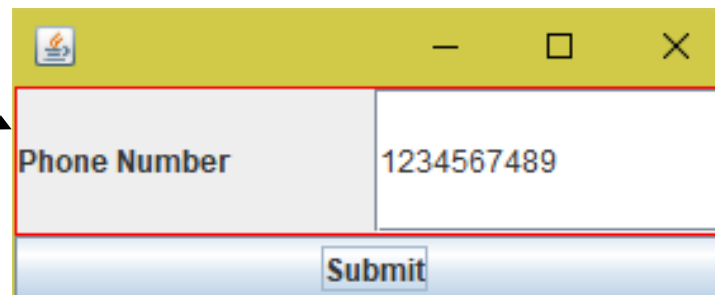
Invalid phone
number entered



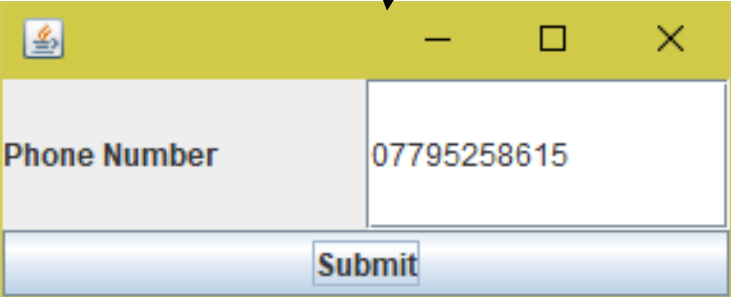
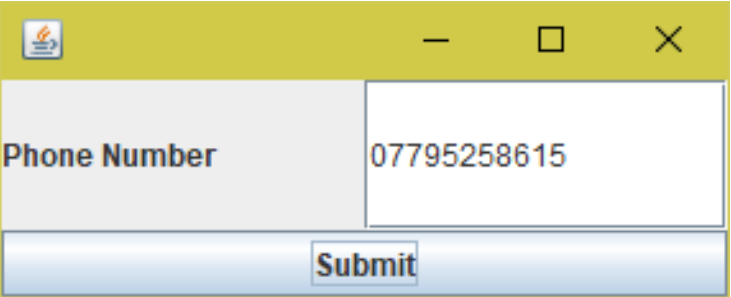
A screenshot of a web form with a yellow title bar. The form has a label "Phone Number" on the left and a text input field on the right containing the number "1234567489". Below the input field is a blue "Submit" button. The entire form is outlined with a thin red border, indicating a validation error.

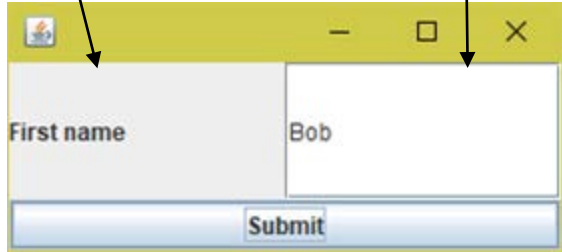
Press submit

Red border indicating
it failed the validation
check



A second screenshot of the same web form. The "Phone Number" label and the input field containing "1234567489" are still present. The "Submit" button is now disabled and appears lighter blue. The red border around the form is still visible.

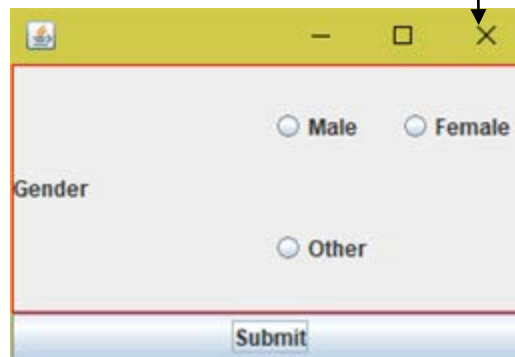
	Valid phone number entered  No red border indicating it passed the validation check  Press submit
Objectives met	16 (Email addresses, dates, and phone numbers should have format checks)
# (from 4.1)	42
Area of System	Component validation - presence
Output & Storage	-

Requirements	
Data Used	Data entered in each field
Evidence	Text field Red border indicating failed validation check  No data No red border indicating passed validation check  Valid data entered

Radio buttons

Red border
indicating failed
validation check

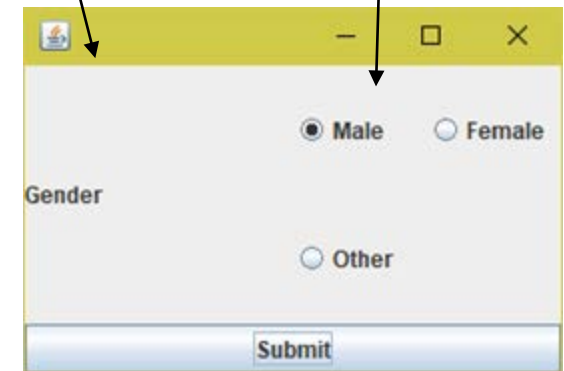
No option
selected



A screenshot of a web form titled "Gender" with a yellow header bar. The form contains three radio buttons: "Male", "Female", and "Other". The "Other" radio button is selected. A red border is drawn around the form area, indicating a failed validation check. A "Submit" button is at the bottom.

No red border
indicating passed
validation check

Option
selected

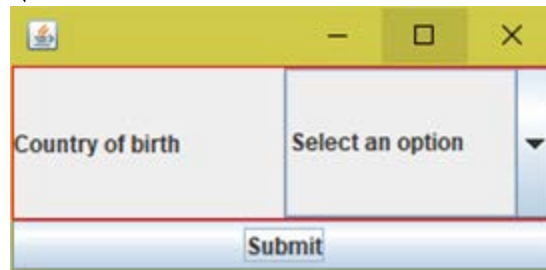


A screenshot of a web form titled "Gender" with a yellow header bar. The form contains three radio buttons: "Male", "Female", and "Other". The "Male" radio button is selected. There is no red border around the form area, indicating a passed validation check. A "Submit" button is at the bottom.

Drop-down

Red border indicating
failed validation check

No option
selected



A screenshot of a web form window with a yellow title bar and standard Windows window controls. The form contains a label 'Country of birth' on the left and a drop-down menu on the right. The drop-down menu displays 'Select an option' and has a downward arrow. A red border is drawn around the entire form area, indicating a failed validation check. Below the form is a 'Submit' button.

No red border
indicating passed
validation check

Option
selected

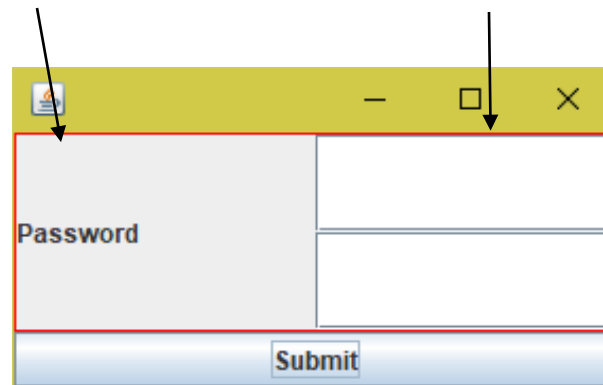


A screenshot of a web form window, identical in layout to the first one. The drop-down menu now displays 'England' instead of 'Select an option'. There is no red border around the form, indicating that the validation check has passed. A 'Submit' button is located at the bottom.

Password fields

Red border indicating failed validation check

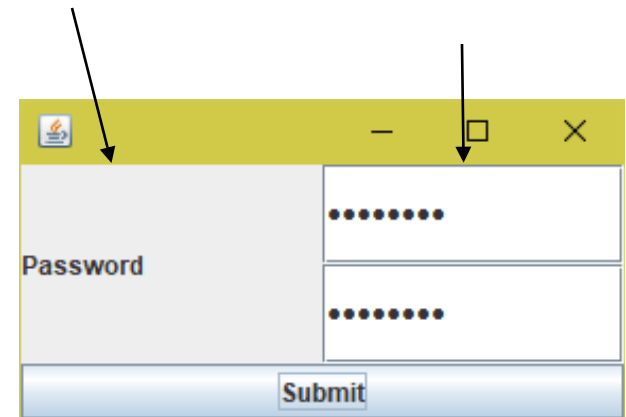
No data entered



A screenshot of a web form with a yellow title bar. The form contains a label 'Password' next to a text input field. The input field is empty and has a red border around it. Below the input field is a 'Submit' button. An arrow points from the text 'Red border indicating failed validation check' to the red border. Another arrow points from the text 'No data entered' to the empty input field.

No red border indicating passed validation check

Valid data entered



A screenshot of a web form with a yellow title bar. The form contains a label 'Password' next to a text input field. The input field contains ten dots, representing masked text. Below the input field is a 'Submit' button. An arrow points from the text 'No red border indicating passed validation check' to the input field. Another arrow points from the text 'Valid data entered' to the input field.

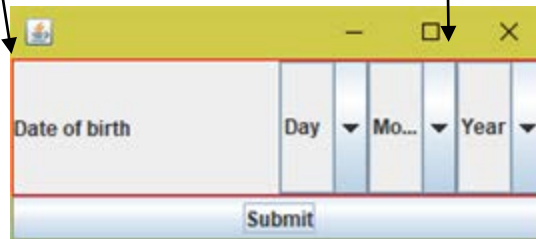
Calendar entry

Red border indicating failed validation check

No date entered

No red border indicating passed validation check

Valid date entered

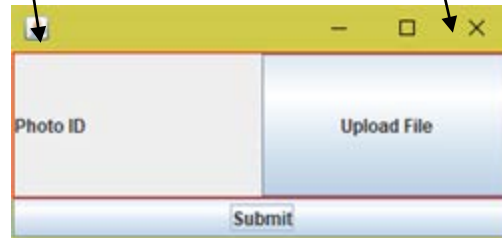
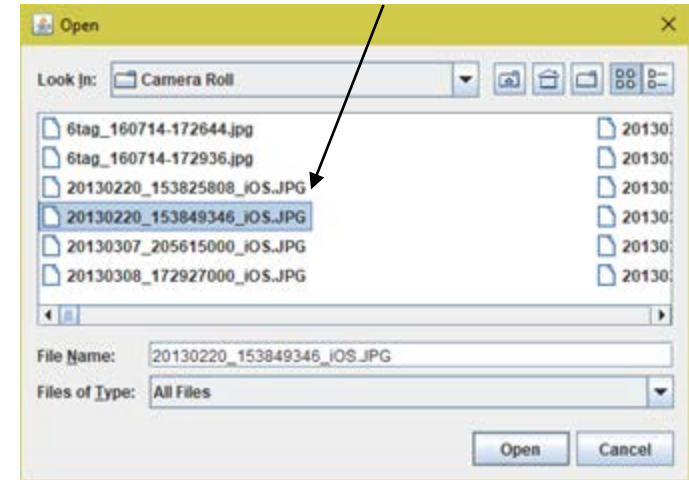
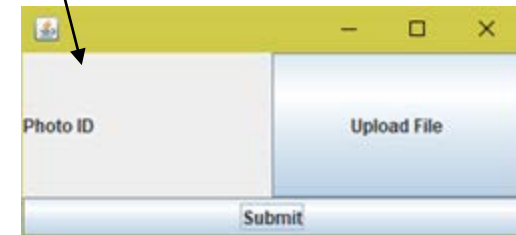


A web form titled "Date of birth" with a "Submit" button. The form has a red border, indicating a failed validation check. The date fields are empty. The form is part of a larger interface with a yellow header bar and a blue footer bar.



A web form titled "Date of birth" with a "Submit" button. The form has no red border, indicating a passed validation check. The date fields are filled with "1", "March", and "2017". The form is part of a larger interface with a yellow header bar and a blue footer bar.

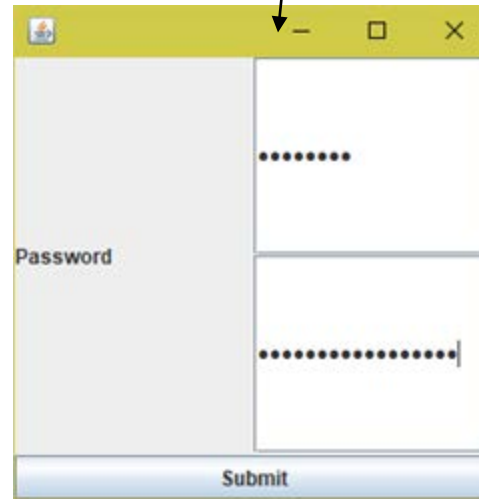
File chooser

	Red border indicating failed validation check No file uploaded Photo selected   No red border indicating passed validation check 
Objectives met	19 (All fields should have presence checks)

# (from 4.1)	43
Area of System	Component Validation - double entry verification
Output & Storage Requirements	-
Data Used	password1 and password2 (the passwords entered in the password fields)

Evidence

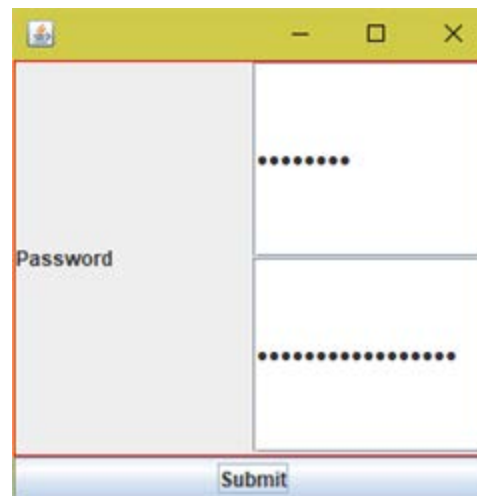
Mismatching passwords



A screenshot of a web application window with a yellow title bar. The window contains a form with two password input fields, each with a label 'Password' to its left. The first field contains 8 dots, and the second field contains 16 dots. A 'Submit' button is at the bottom. An arrow points from the text 'Mismatching passwords' to the top of the window. Another arrow points from the text 'Press submit' to the 'Submit' button.

Press
submit

Red border



A screenshot of the same web application window as above, but with a red border around the entire form area. An arrow points from the text 'Red border' to the left side of the red border.

Matching passwords entered

Form window showing two password input fields, both containing masked characters (dots). The label 'Password' is visible on the left. A 'Submit' button is at the bottom.

Press submit

No red border



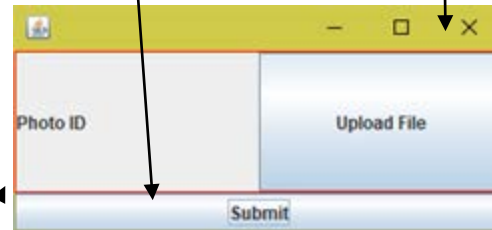
Form window showing two password input fields, both containing masked characters (dots). The label 'Password' is visible on the left. A 'Submit' button is at the bottom.

Objectives met	18 (Passwords should have double entry verification)
# (from 4.1)	44
Area of System	Component Validation - document upload fields
Output & Storage Requirements	-
Data Used	File path (of uploaded file)
Evidence	

	Photo
--	-------

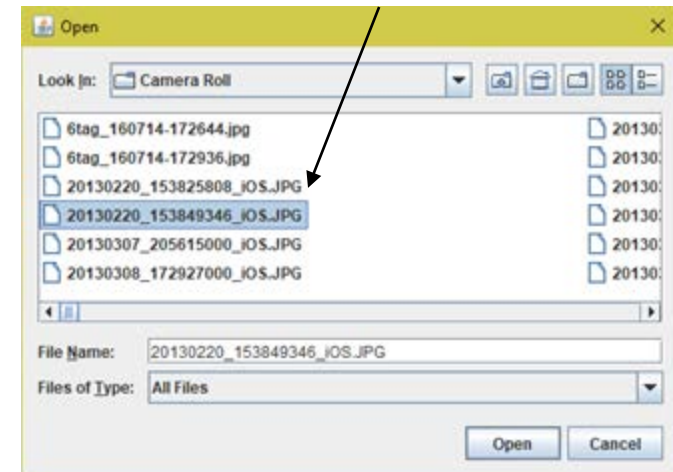
Press
submit

No file
uploaded



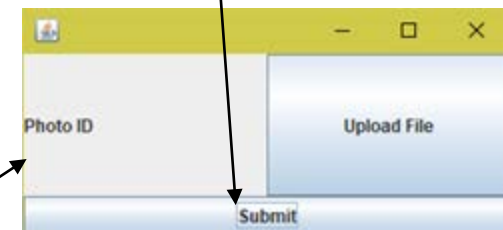
Red border
indicating failed
validation

Photo selected



Press
submit

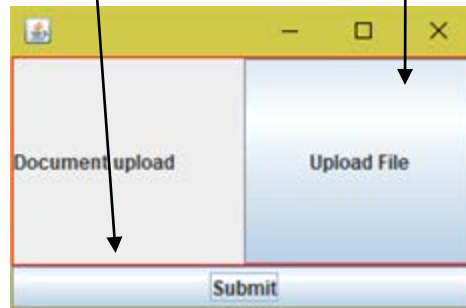
No red border
indicating passed
validation check



Document upload

Press
submit

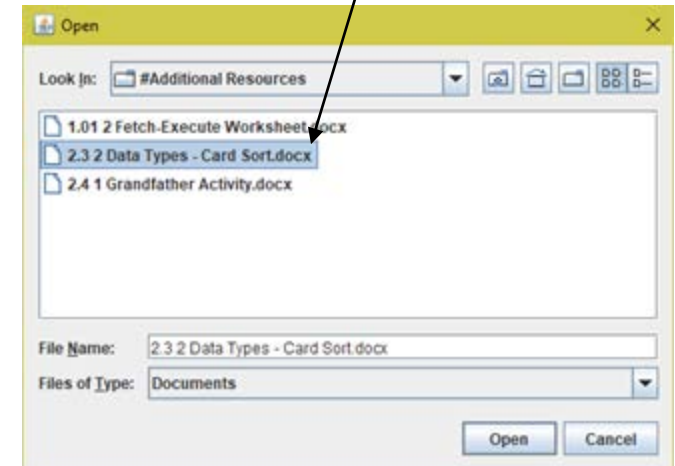
No file
uploaded



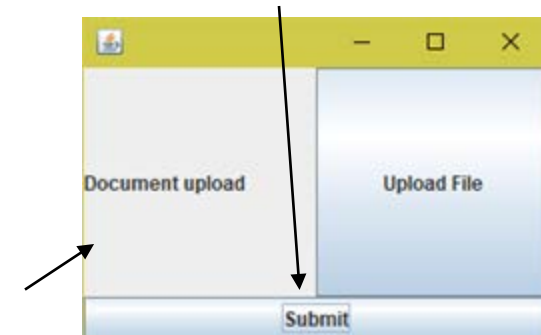
Red border
indicating failed
validation check

No red border
indicating passed
validation check

Document selected



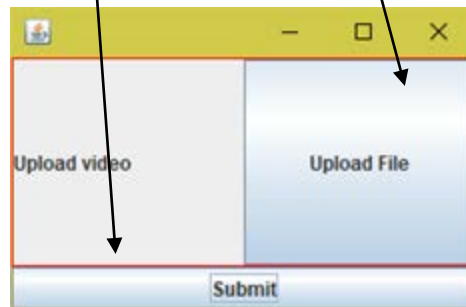
Press
submit



Video upload

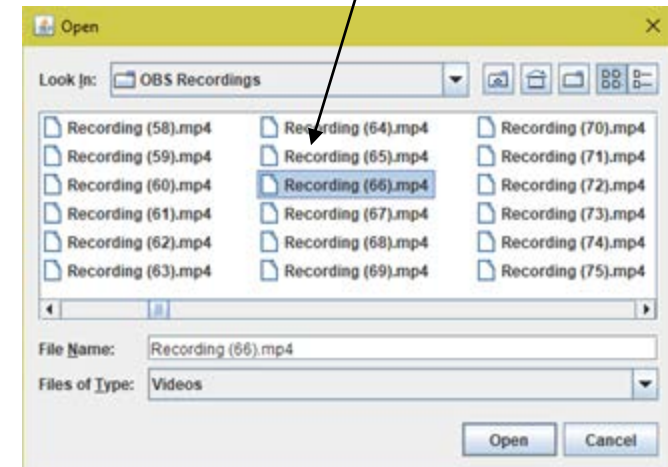
Press
submit

No file
uploaded



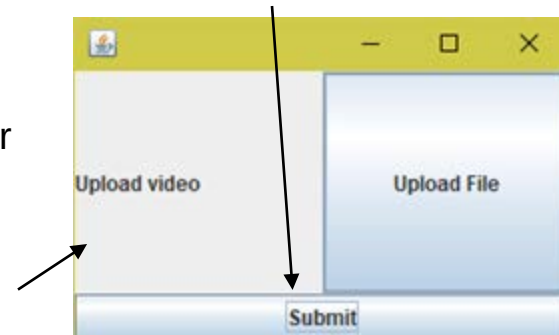
Red border
indicating failed
validation check

Video
selected



Press
submit

No red border
indicating
passed
validation
check



Objectives met	20 (Document upload fields should only allow certain types of files to be uploaded)
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Attempt form

# (from 4.1)	45
Area of System	Attempt form - save progress
Output & Storage Requirements	FormInProgressDB
Data Used	Data entered into the fields of each question

Evidence

Select a form

Press attempt

Questio...	Descript...	Main Ski...	Difficulty	Percent ...
Q81558...	Name a...	Name.F...	5	0%
Q81558...	large a...	Name....	6	0%
Q81558...	Name p...	Name.P...	2	0%

Difficulty Sort

Attempt Form

Delete Form

Attempt form based on weaknesses

Filter by difficulty

Filter by type

- ☐ Email
- ☐ Name
- ☐ Phone
- ☐ Gender
- ☐ File Up...
- ☐ Pass...
- ☐ Calen...
- ☐ Check...
- ☐ Drop-d...
- ☐ Radio ...
- ☐ Birthday
- ☐ Addre...
- ☐ date o...
- ☐ Countr...
- ☐ Photo ...
- ☐ Video ...

Fill in some fields

Press close

Form Fields	
First name	Bob
Password	
Email address	bob@bob.com
Submit	

Restart the system and
log in as the same user

Select the
same form

Percentage complete
updated

Press attempt

Form Filler

View Questions View Forms Create questions Create forms Users Statistics Import and Export

Help

Questi...	Descri...	Main S...	Difficulty	Percen...
Q8155...	Name ...	Name...	5	0%
Q8155...	large a...	Name...	5	0%
Q8155...	Name ...	Name...	2	66%

Difficulty Sort

Attempt Form

Delete Form

Attempt form based on weaknesses

Filter by difficulty

1 2 3 4 5 6 7 8 9 10

Filter by type

☐ Email ☐ Name ☐ Phone ☐ Gend...
☐ File U... ☐ Pass... ☐ Cale... ☐ Chec...
☐ Drop-... ☐ Radi... ☐ Birth... ☐ Addr...
☐ date ... ☐ Coun... ☐ Phot... ☐ Vide...

What was entered in the fields has been saved

A screenshot of a web form with a yellow title bar. The form contains three input fields: 'First name' with the value 'Bob', 'Password' (empty), and 'Email address' with the value 'bob@bob.com'. A 'Submit' button is at the bottom. A red rectangle highlights the password field. An arrow points to the top of the form.

Objectives met	43 (How much of each form a user has attempted should be tracked)
# (from 4.1)	46
Area of System	Attempt form - Validate answers

	See component validation section
# <i>(from 4.1)</i>	2
Area of System	Attempt form - Track statistics
	See statistics section. The statistics are gathered automatically the user doesn't have to do anything.

4.4 Self-Evaluation

4.41 Functioning of the Prototype

Overall, I think that the prototype functions well. It fulfils almost all of the requirements for the system.

Questions

The admin is able to create a wide variety of questions. Each component that can be added to a question has the correct and appropriate validation applied to it and this was one of the most important requirements of the system. However, the prototype doesn't include the facility to add a date picker than uses a calendar, a drop down date entry was used instead. In the final system, an actual calendar will be used to enter dates.

The admin is also not able to add components to a question using a drag and drop system and can only use buttons in the prototype.

The admin can enter all the required information about the question once they have finished creating it.

The user is able to find questions in the system and attempt them. This also satisfies another important requirement of the system.

All other requirements for questions are satisfied. E.g. sort filter, and attempt.

Forms

The admin user can create forms in the system by adding questions from the list of questions. They are able to sort and preview the questions before they had them to the form. There is also a facility to edit forms that are already created, and the admin can rearrange the questions in the form, and add and remove questions from the form.

The admin is able to enter a description of the form and its difficulty after creating it.

The user can view the forms that are in the system and select one to attempt. They can perform all the sorts and searches that are detailed in the requirements of the system. They can also attempt a form that is automatically created and includes questions that the system has identified as the ones that the user struggles with the most.

When the user attempts a form all of the components are correctly validated. When the user closes a form and reopens it (even after system restarts) their progress is saved.

The prototype satisfies all requirements for the forms section of the system.

Statistics

The prototype tracks all of the statistics that were specified in the requirements of the system. When the user attempts a question (on its own or as part of a form) the system tracks how many times that they have failed the validation checks, how long it took them to complete the question, and how many times it took them to correct an error.

There is also a screen that displays all of these statistics to the user. They can view statistics about individual questions or produce a report which they can print that shows their overall progress broken down by type of question. However, there are no graphs displayed to user which is a feature that is required for the final system. There is also no facility for the admin to produce reports for multiple users at once, this is also a feature that is required for the final system.

User

All data that was specified in the requirements to be stored about each user is stored in the prototype. The admin can also view this information in the users tab of the prototype. The admin can add, edit, and delete users. All requirements for user are fulfilled by the prototype.

4.42 Good Features

Questions

I think that the method of creating questions is simple and allows the admin to create questions very quickly because they only need to press a button to add a component. Although a drag and drop feature is planned for the final system, I will not implement it and I will use buttons instead to add components because of how quick and easy it makes adding components. I think that the preview of the question is helpful and works well.

I think that the wide range of components that I have successfully managed to implement allows the admin to create a wide range of questions for the system. The only component that I wasn't able to fully successfully implement was a calendar entry.

From a programming standpoint, I think that the way that validation for each component was implemented was a good way of doing it. This is because the validation is contained within each component object using an interface. This means that other parts of the system could call the validation method on each component and have a boolean returned based on whether the validation check was successful without having to know how the validation routine actually worked. This made it very easy to write the form display class as it just had to call the validate method of each component.

The use of a type list meant that there was no risk of repeated types and so meant that the type filters and sorts worked correctly because there were no redundant types. I will handle the types the same way in the final system.

I'm mostly happy with the progress saving features of the system. This is because it all happens behind the scenes and the user never has to consciously save their progress. Whenever the user closes a question or form that they are attempting the progress is automatically saved (what they've entered into each field) and the percentage that they have completed the form is updated. I think that the best part about the implementation of this feature is that it's user specific e.g. each user's progress is saved individually therefore if one user attempts a form and closes it, when a different user opens the same form it will be blank for them and save their own progress without interfering with the other user's progress. Another good part of this feature is that if the same question is in multiple forms the progress for each is kept separate also, i.e. if the user enters something into a text field of the question on one form when they open the same question in a different form it will be blank (or have what they entered into that form saved).

Forms

The admin is able to easily create forms in the system by selecting questions from a list, this was implemented exactly as specified in the requirements. I think that the form creation is well implemented because it meets these requirements. I think that the facility to rearrange the questions in the form and delete them is a very useful and easy to use feature. This is because all that the admin has to do is press the buttons that appear next to the questions in the form preview. I also think that it's useful for the admin to be able to edit past forms and I think that using a dropdown on the create forms screen to select a form to edit is very easy to use because they don't have to use a different interface to edit forms - they can use the one that they're familiar with from creating forms.

I think that the form creation screen is probably the best laid out screen in the application. This is because it's clear what each area of the screen is for because of the borders. There is also little wasted space.

Statistics

Automatic statistic tracking was difficult to implement however I think that it works well. I'm most happy with how the system determines how long the user has spent on a question. A mouse listener was used. When the mouse is over any component of a question a timer is started and times how long it was over that question. If the mouse moves to another question a timer is started for that question and the timer is stopped for the previous and the time is stored. This is repeated until the user stops attempting the form. Once the user has stopped attempting the form the total time that the mouse was on each question is saved.

I think that the method of using QuestionStat objects also worked very well it because it allows the statistics for each user and each question to be kept entirely separate. This means that the system works very well with multiple users and the data for one user doesn't affect the data of another user.

User

I am pleased that every requirement for the user section of the system was met (apart from encryption). I think that the process of adding or editing users is simple and well implemented. The automatic attendance tracking also works well and makes it easy for the attendance of a user to be tracked.

4.43 Shortcomings

Questions

The major shortcoming of the implementation of the questions is the actual implementation of how the graphical components are saved. Rather than just the data that is stored in each field being saved the actual JComponent object is serialised. This is ok when the system is running on a fast computer however when running on a slow computer, like those that are used at the computer club, the system grinds to a halt whenever a question has to be saved. This happens when a new question is created or when a user closes a form and their progress is saved. This method of serialising the JComponent also had negative consequences during the development, other than the slow down. Because the actual object was saved, whenever I changed the class file of each component object I would have to delete the entire question database and recreate all the questions as there would be a mismatch between the stored objects and the altered class. This made the prototype very temperamental as the question database was easy to corrupt. There was also no easy way of fixing a corrupt database. This would be unacceptable for the finished system because Steph will not have time to recreate the question database and so it must be reliable in the final system.

There was also no facility to edit questions. I didn't include the facility to edit questions because questions are only around 4 components at most therefore it would be just as fast to create a new question, however I think that the final system would benefit from the ability to edit questions. This is because if a question is recreated all of the user's progress on that question gets wiped because the new 'edited' question has a different id. It also means that the question has to be re-added to all of the forms that it is in. This would be very inconvenient for Steph if she ever had to edit a question.

When a question is deleted it is not deleted from all of the forms that it is in. This means that when one of the forms that it's in is loaded the system produces an error. This effectively breaks all of the forms that the question is in and so makes deleting a question useless. In the final system the system should also remove the question from all of the forms that it's in when the question is deleted.

Forms

When editing a form, the old description is not displayed to the user to be edited and the user has to enter the entire description again. This means that if a form is to be edited the admin has to remember what the old description was and retype it.

Statistics

There are no graphs displayed to the user on the view statistics screen. There is also no way for the user to view statistics about multiple questions at once. This limits the usefulness of the statistics screen because the data could be represented in a better way and there is only so much information that can be conveyed only by using labels.

The layout of the statistics panel is also quite poor as everything is in one column and it's not clear immediately what each part of the screen does. This is because the layout from the design wasn't able to be achieved because I omitted graphing and decided on an easier layout for the prototype.

User

The user data isn't encrypted in the prototype. This doesn't matter for the prototype because only 'dummy' data is used, however for the final system it's very important that encryption is implemented well because sensitive data will be stored about each user.

The user screen of the system is perhaps the worst laid out screen. The main purpose of the screen is to show the data stored about each user to the admin user, however this isn't effectively done in the prototype. This is because the majority of the space is taken up by the buttons at the bottom of the screen and this means that the table is very small and only a few users can be seen at a time.

The register is also poorly laid out because so many users are on it and so it's a bit too densely populated with names.

4.44 Suggestions for Improvements

Questions

Fixing the implementation of the saving of questions is the main improvement that needs to be made. The data stored in each field should be saved rather than directly saving the JComponent object. This will speed up the system a large amount and is essential for the system to run well on the slow computers of the computer club. It will also mean that the database is less likely to become corrupt and so the system will be more reliable.

Adding the facility to edit questions would also be a good improvement because it means that if a small error is made in a question it will be able to be fixed easily without having to recreate the entire question and update the forms that it's in. However, I don't think that I will be implementing it because this will encourage the admin to create longer questions, and really questions should only be 1 row of components in order to stop the admin simply creating an entire form as a question. Because questions are very few components it wouldn't be much of a benefit to be able to edit them. If a question was edited it also means that the statistics stored about the question would be invalid as the older version of the question could have been easier or harder. It makes more sense for the old question to just be deleted from the system and a new one created instead.

When a question is deleted it should be automatically removed from all of the forms that it's in so those forms aren't corrupted.

Forms

A more robust way of editing forms would be a good improvement. This would include the old description of the form being able to be edited rather than having to enter an entirely new description. The difficulty should also be able to be edited like this also.

It would be good if there was a way to enter a name for each form so that the forms can have another way to be identified rather than just the description (e.g. "Blue badge form"). This might make it easier for users to find forms in the system. It also means that when editing forms, the form to be edited can be selected by title rather than by ID which is easier and more user friendly.

The layout of forms could be improved. Making them take up the entire screen would be better, with realistic spacing between questions. It would also be good if the admin could add headings to different sections of the forms as this would mean that the forms would more closely mimic real world forms. The potential to add pages / have the system automatically add pages would be good for large forms too.

Statistics

A potentially better way of monitoring the time spent on each question could be implemented. This could be taking into account where the text cursor is in addition to where the mouse pointer is. However, this may not make much of an improvement because it would be difficult to tell which question the user was attempting if the mouse pointer and the text cursor were in different places.

It would be good if there was a way for the user to view the statistics for multiple questions at once. It would also be better if the user could select questions from the question table rather than just selecting questions by id. This is because the user wouldn't actually know the id of the question as it's hidden from them in the system. Using the question table means that the user would be able to search, sort, and preview the questions before selecting one.

User

A better layout of this screen should be implemented (like that detailed in the design). A more intelligent register could be used to make it easier for the admin to take a register and to reduce the clutter of displaying all of the users in the register. This more intelligent register system could only show the users that have attended at least once in the past two weeks and if the admin wanted to mark a user that wasn't present in the past two weeks they would have to search for them by first name. This would be a good way of reducing the clutter because it's most likely that the users that are present will be those that were present in at least one of the last two sessions. Populating the register this way means that those users that don't come anymore won't appear in the register without having to delete their accounts.